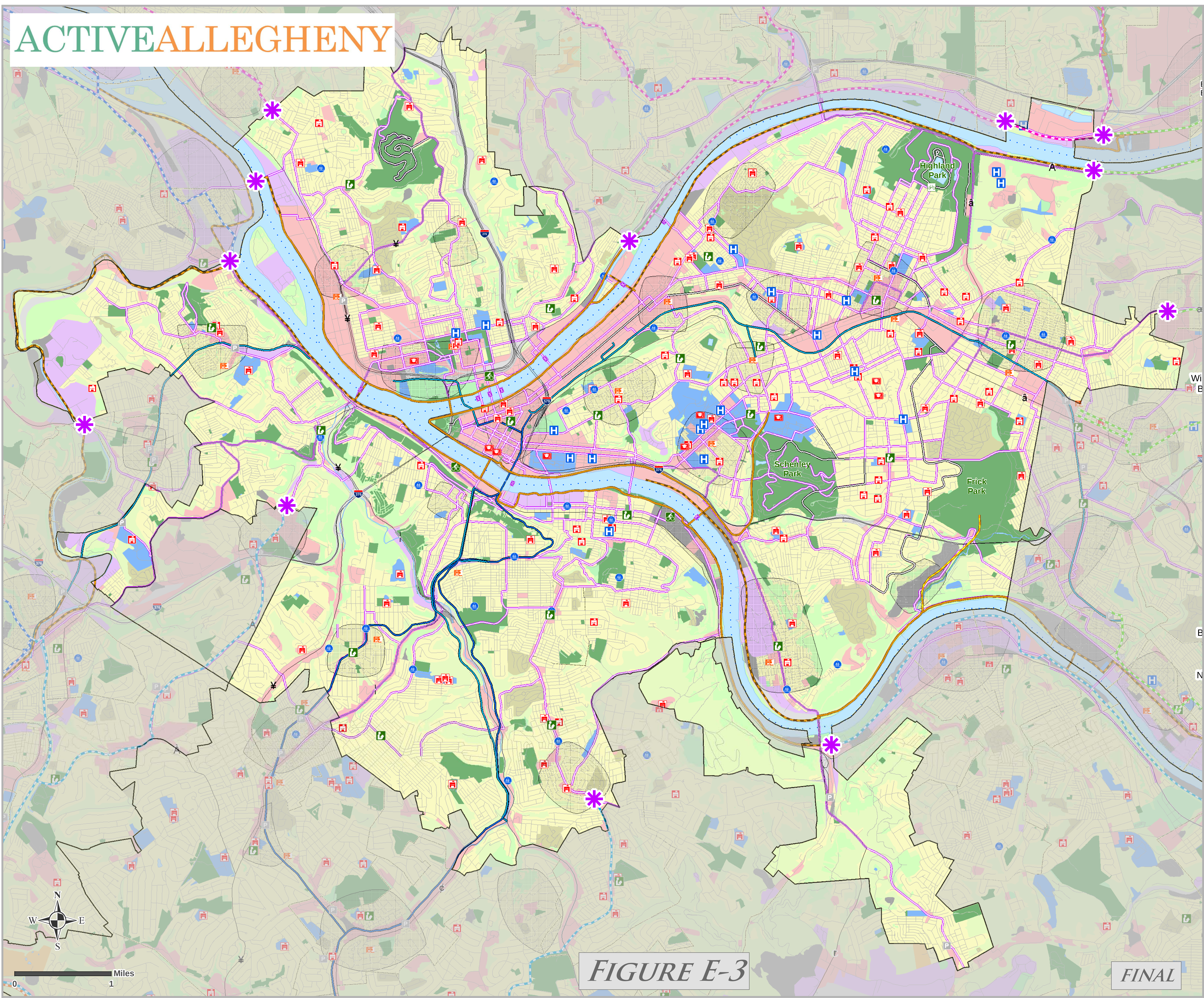




LEGEND

PROPOSED BICYCLE ROUTES

- NORTH BICYCLE ROUTES
- WEST BICYCLE ROUTES
- EAST BICYCLE ROUTES
- SOUTH BICYCLE ROUTES
- CONNECTION TO PITTSBURGH BICYCLE FACILITIES

EXISTING FACILITIES

- BIKE PGH BICYCLE ROUTES
- CITY OF PITTSBURGH MARKED BIKEWAYS

TRAILS

- EXISTING TRAILS
- PROPOSED TRAILS
- TRAIL UNDER DEVELOPMENT
- THREE RIVERS WATER TRAIL

TRANSPORTATION

- HIGHER ORDER ROADWAYS
- LOCAL ROADWAYS
- BRIDGES
- BUSWAYS AND STATIONS (EAST, NORTH SHORE, SOUTH, WEST)
- T AND STATIONS

FOCUS AREA ATTRACTORS

- FOCUS AREA BRIDGE
- BIKE RENTAL
- CYCLE TRACK
- SPORTS LEGACY
- HOSPITAL
- SENIOR CENTER
- LIBRARY
- PARK AND RIDE LOTS
- PRIVATE SCHOOL
- PUBLIC SCHOOL
- COLLEGE / UNIVERSITY
- POTENTIAL TOD SITES

PHYSICAL BOUNDARIES

- COUNTY BOUNDARY
- MUNICIPAL BOUNDARIES

WATER FEATURES

- RIVERS AND LAKES

LAND USES (ALLEGHENY PLACES)

- ECONOMIC GROWTH AREAS
- RESIDENTIAL
- COMMERCIAL
- INDUSTRIAL
- TRANSPORTATION
- RECREATION / CONSERVATION
- PARKS
- COMMUNITY FACILITIES
- AGRICULTURE
- CEMETERY
- UNDEVELOPED
- VACANT

AUGUST 2010
REV. 11/30/10
Public Transportation and Multimodal Open End II, Contract # 358R10

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Table E-1. Project List

Project	User Group Benefit	Time to Implement
<i>Bicycle Accommodation and/or Safety Improvements</i> (providing on-road bicycle facilities through the use of shared lane markings, parking restrictions, reduced speed limit, traffic calming volume and/or speed control measures, bicycle lanes, signing, striping, widening, and by other means identified in the Bicycle Facilities Toolbox in Chapter 2).		
Freeport Road (Aspinwall to Cheswick)	Bicyclists	Mid-term
Route 837/East Carson Street (Pittsburgh to West Elizabeth)	Bicyclists	Mid-term
Allegheny River Boulevard (Pittsburgh to Oakmont)	Bicyclists	Long-term
Bigelow Boulevard (Pittsburgh)	Bicyclists	Long-term
Fifth Avenue (Pittsburgh)	Bicyclists	Long-term
Liberty Avenue (Pittsburgh)	Bicyclists	Long-term
Penn Avenue (Pittsburgh)	Bicyclists	Long-term
Route 8/Butler Street (Etna to Richland)	Bicyclists	Long-term
Route 28 (Pittsburgh to Blawnox)	Bicyclists	Long-term
<i>Road Diets</i> (reducing four-lane cross-section to three lanes including a center left turn lane, addition of on-road bicycle facilities, reduced crossing distance for pedestrians, bus stops located at intersections or in dedicated pull offs).		
Route 48/Long Run Road (White Oak)	Bicyclists	Mid-term
Route 51 (Coraopolis)	Bicyclists, Pedestrians, Transit Users	Mid-term
Bigelow Boulevard (Pittsburgh)	Bicyclists, Pedestrians, Transit Users	Long-term
Negley Run Boulevard (Pittsburgh)	Bicyclists, Pedestrians	Long-term
Park Manor Boulevard (Robinson)	Bicyclists, Pedestrians, Transit Users	Long-term
Washington Boulevard (Pittsburgh)	Bicyclists, Pedestrians	Long-term

Project	User Group Benefit	Time to Implement
<u>Potential Partial Closures to Motor Vehicles for Active Transportation Accommodation</u>		
Roberto Clemente Bridge, Pittsburgh (Allegheny County)	Bicyclists, Pedestrians, In-Line Skaters, Skateboarders, Water Trail Users	Mid-term
Wabash Tunnel, Pittsburgh (Port Authority of Allegheny County or others)	Bicyclists	Long-term
<u>Dedicated County-Wide Bicycle Routes*</u> <i>(signing, striping, widening, incorporation into roadway projects, peak hour bicycle use of transit facilities, increased storage on transit facilities, bicycle parking provisions, trail completion, roadway maintenance, and additional planning/design)</i>		
East Route 1 Connector (Deer Lakes Park to Proposed East Route 1)	Bicyclists	Mid-term
East Route 4 (North Versailles to Swissvale Station)	Bicyclists	Mid-term
East Route 4 Connector (Wall to Turtle Creek)	Bicyclists	Mid-term
East Route 5 (Boyce Park to Pittsburgh)	Bicyclists	Mid-term
North Route 2 (North Park to Allegheny River Trail)	Bicyclists	Mid-term
North Route 3 Detour (Leetsdale to PA Bike Route A)	Bicyclists	Mid-term
North Route 4 (West Deer to Allegheny River Trail)	Bicyclists	Mid-term
South Route 1 (West Elizabeth to Steel Valley Trail)	Bicyclists	Mid-term
South Route 3 (Youghiogheny River Trail to Pittsburgh)	Bicyclists	Mid-term
South Route 5 (South Park to Pittsburgh)	Bicyclists	Mid-term
West Route 1 (Panhandle Trail to Carnegie PAAC Station)	Bicyclists	Mid-term
West Route 2 (Montour Trail to Pittsburgh)	Bicyclists	Mid-term
West Route 3 Connector (Cliff Mine Road to PA Bike Route A)	Bicyclists	Mid-term

***East, South, West, North, and circumferential route (Beltway Bicycle Route) connecting the regional parks and other area land uses.**

Project	User Group Benefit	Time to Implement
East Route 1 (Allegheny River Trail to Pittsburgh)	Bicyclists	Long-term
East Route 2 (Plum to Pittsburgh)	Bicyclists	Long-term
East Route 3 (Monroeville to Wilkinsburg Station)	Bicyclists	Long-term
North Route 1 (Hampton Township Park to Proposed North Route 2)	Bicyclists	Long-term
North Route 2 Connector (Proposed Beltway Bicycle Route to Proposed North Route 2)	Bicyclists	Long-term
North Route 3 (Leetsdale to Pittsburgh)	Bicyclists	Long-term
North Route 4 Connector (Proposed Recreational County Park Route to Proposed North Route 2)	Bicyclists	Long-term
Recreational County Park Route (Beltway Bicycle Route following Orange Belt and Detour Routes to Connect County Parks)	Bicyclists	Long-term
South Route 2 (Montour Trail to South Hills Village Station)	Bicyclists	Long-term
South Route 2 Connector (Library Road to Proposed South Route 2)	Bicyclists	Long-term
South Route 4 (Bethel Park to Proposed Chartiers Creek Trail)	Bicyclists	Long-term
South Route 4 Connector (Greentree Road to Pittsburgh)	Bicyclists	Long-term
West Route 2 Connector (Pittsburgh International Airport to Montour Trail)	Bicyclists	Long-term
West Route 3 (PA Bike Route A to Pittsburgh)	Bicyclists	Long-term
West Route 4 (Moon to Carnegie Station)	Bicyclists	Long-term

Project	User Group Benefit	Time to Implement
<i>Pedestrian Corridor Improvements (sidewalk maintenance, repair and/or installation; curb ramp installation; street furniture; lighting; and other corridor enhancements desired by a variety of pedestrians)</i>		
Beaver Street (Glen Osborne)	Pedestrian, Transit User	Short-term
Belmar Place (Swissvale)	Pedestrian, Transit User	Short-term
Braddock Avenue (Braddock)	Pedestrian	Short-term
Chartiers Avenue (McKees Rocks)	Pedestrian, Transit User	Short-term
Cochran Road & Washington Road (Mt. Lebanon)	Pedestrian	Short-term
McLaughlin Run Road (Bridgeville)	Pedestrian	Short-term
Route 837 (Clairton)	Pedestrian	Short-term
Route 837 (Duquesne)	Pedestrian, Transit User	Short-term
Beadling Road (Mt. Lebanon)	Pedestrians	Mid-term
Route 19 Truck/Washington Road (Mt. Lebanon)	Pedestrian, Transit User	Mid-term
Route 19 Truck/West Liberty Avenue (Dormont)	Pedestrian, Transit User	Mid-term
Route 50 (Heidelberg)	Pedestrian, Transit User	Mid-term
Steubenville Pike (Robinson)	Pedestrian, Transit User	Mid-term
Ardmore Boulevard (Forest Hills)	Pedestrians, Transit User	Long-term
Bates Street (Pittsburgh)	Pedestrians, Transit User	Long-term
Bigelow Boulevard (Pittsburgh)	Pedestrian, Transit User	Long-term

Project	User Group Benefit	Time to Implement
Campbells Run Road (Robinson)	Pedestrian	Long-term
Grove Road (Castle Shannon)	Pedestrian, Bicyclist, Transit User	Long-term
Island Avenue/Route 51 (McKees Rocks)	Pedestrian, Transit User	Long-term
Lincoln Way (White Oak)	Pedestrian	Long-term
Mayview Road (Upper St. Clair)	Pedestrian, Transit User	Long-term
McLaughlin Run Road (Upper St. Clair)	Pedestrian	Long-term
Park Manor Boulevard (Robinson)	Pedestrian, Trail User	Long-term
River Road (Haysville)	Pedestrian, Motorist	Long-term
Robinson Town Center Boulevard (Robinson)	Pedestrian, Transit User	Long-term
Intersection Improvements (curb ramp maintenance, ADA upgrades, curb ramp installation, pedestrian signal upgrades, push button installation, restriping crosswalks, installing warning signs, and installing crosswalks)		
Smithfield Street & Carson Street (Pittsburgh)	Pedestrian	Short-term
Route 28 & River Front Drive (Millvale)	Pedestrian	Short-term
Ardmore Boulevard & Yost Boulevard (Pittsburgh)	Pedestrian	Mid-term
Belmar Place & Woodstock Avenue (Rankin)	Pedestrian	Mid-term
Bethel Church Road & Broughton Road (Bethel Park)	Pedestrian	Mid-term
Boulevard of the Allies & Bates Street (Pittsburgh)	Pedestrian	Mid-term
Boulevard of the Allies & Halket Street (Pittsburgh)	Pedestrian	Mid-term
Braddock Avenue & 4 th Street (Braddock)	Pedestrian	Mid-term

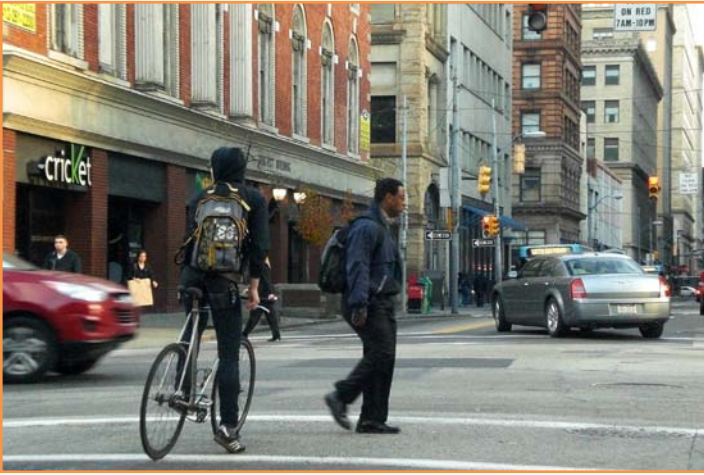
Project	User Group Benefit	Time to Implement
Braddock Avenue & 7 th Street (Braddock)	Pedestrian	Mid-term
Brinton Avenue & Ridge Avenue (East Pittsburgh)	Pedestrian	Mid-term
PJ McArdle Road & Liberty Bridge (Pittsburgh)	Pedestrian	Mid-term
Negley Run Boulevard & Washington Boulevard (Pittsburgh)	Pedestrian	Mid-term
Route 19 Truck & Brookline Boulevard (Dormont)	Pedestrian	Mid-term
South Bellefield Avenue & 5 th Avenue (Pittsburgh)	Pedestrian	Mid-term
Route 837 & Amity Street (Homestead)	Pedestrian	Mid-term
Beadling Road & Washington Road (Route 19) (Mt. Lebanon)	Pedestrian	Long-term
Campbells Run Road & Steubenville Pike (Robinson)	Pedestrian	Long-term
Castle Shannon Boulevard & Mt. Lebanon Boulevard (Castle Shannon)	Pedestrian	Long-term
Foster Street & 40 th Street (Pittsburgh)	Pedestrian	Long-term
Thorn Run Road & Route 51 (Coraopolis)	Pedestrian	Long-term
<u>Policy and Programmatic Recommendations</u> (education, enforcement, maintenance, coordination, plan making, and support)		
Support of Municipal Active Transportation Plans and Plans/Programs with Bicycle and Pedestrian Improvement Recommendations (e.g., City of Pittsburgh Bicycle Plan)	Bicycle and Pedestrian	Short-term
Support of County Active Transportation Plans and Plans/Programs with Bicycle and Pedestrian Improvement Recommendations (e.g., ALLEGHENY PLACES)	Bicycle and Pedestrian	Short-term

Project	User Group Benefit	Time to Implement
Support of Regional Active Transportation Plans and Plans/Programs with Bicycle and Pedestrian Improvement Recommendations (e.g., Move PGH, SPC Regional Traffic Signal Update Project, SPC Bike Count Program, Multi-modal Road Safety Audits, etc.)	Bicycle and Pedestrian	Short-term
Support of State Active Transportation Plans and Plans/Programs with Bicycle and Pedestrian Improvement (e.g., PCTI Program, Smart Transportation Guidebook, Bicycle and Pedestrian Project Checklist, Safe Routes to School (Federal Funded))	Bicycle and Pedestrian	Short-term
Encouragement and Technical Support in Completion of Regional Trail Network (e.g., Whitaker and Sandcastle Portions of Great Allegheny Passage in 2011, Ohio River Trail Development Feasibility Study)	Active Transportation Modes	Short-term
Assessment of Bicycle Parking Needs	Bicycle	Short-term
Installation of Bicycle Racks on Port Authority of Allegheny County Buses (expected in 2011)	Bicycle	Short-term
Continued utilization of PennDOT Bicycle and Pedestrian Checklist for Projects, consider making available to Public and conducting a Periodic Quality Audit	Bicycle and Pedestrian	Short-term
Coordination of Activities to Promote Safe Bicycling including Cyclovia and Bicycle Education Materials	Bicycle, In-line Skating, and Pedestrian	Short-term
Develop and Implement Plans for Snow Removal/Winter Maintenance for Trails and Bicycle Facilities	Bicycle and Pedestrian	Short-term
Continued Coordination with Allegheny County Public Works to Identify/Remediate Potentially Hazardous Conditions for Bicyclists on County Roadways, including Potholes, Sewer/Drain Grates, and/or Scuppers	Bicycle	Short-term
Exploration of Lighting Options for Trails that Experience Heavy Commuter Use	Bicycle and Pedestrian	Short-term
Continued Exploration of Funding Opportunities for Maintenance of Public Steps in Allegheny County	Pedestrian	Short-term
Expanded Crosswalk Needs Assessment and Marking Program	Pedestrian	Short-term
Consistent Enforcement of Winter Snow Removal Requirements for Sidewalks	Pedestrian	Short-term

CONCLUSION

ACTIVEALLEGHENY provides a roadmap for municipalities to integrate the active transportation improvements identified in the Plan into their own comprehensive and transportation plans. Allegheny County will coordinate with

neighboring counties with the goal of regional integration of active transportation and creation of active transportation byways that connect from county-to-county. Residents should be actively involved in assisting local governments and others to implement improvements identified in the plan. **Together we can integrate active transportation, an important component of everyday life in Allegheny County.**



*Bicyclist and Pedestrian, Downtown Pittsburgh
Photo: Lynn Heckman*

CHAPTER 1. An Active Transportation Plan

1.1 INTRODUCTION

“ACTIVEALLEGHENY integrates walking, biking, and other active, healthy travel modes into the existing transportation system. Planning and prioritizing investment in commuter bike routes and walking facilities will enhance our existing transportation network, and provide sustainable travel mode choices to move people to their destinations. This plan will be a blueprint for improved access to connect our communities, work sites, schools, attractions and homes. Active infrastructure will encourage investment and economic development.” – Dan Onorato, County Executive

What is Active Transportation?

Active Transportation is human-powered transportation including bicycling, walking, kayaking, and in-line skating. In the past decade, studies have detailed the potential benefits of active transportation for both users and society. *Active Transportation in Urban Areas: Exploring Health Benefits and Risk* released in June 2010 indicates that “people who use active transportation are, on average, more physically fit, less obese and have a reduced risk of cardiovascular disease compared to people who use only motorized transportation.”ⁱ Benefits to society include reduced air pollutants and greenhouse gas emissions.



Hot Metal Bridge Opening
Photo: Sara Walfoort

About this Plan

Allegheny County partnered with the Pennsylvania Department of Transportation’s (PennDOT) Bureau of Public Transportation to develop ACTIVEALLEGHENY. The plan was funded through a Pennsylvania Community Transportation Initiative (PCTI) “Smart Transportation” grant to enhance the ALLEGHENYPLACES (Allegheny County’s Comprehensive Plan) transportation element and prioritize active transportation action items. **The goal of ACTIVEALLEGHENY is to integrate non-vehicular modes of transportation, specifically walking and biking, into the transportation system through creation of a comprehensive active transportation plan.** The primary objective of ACTIVEALLEGHENY is to encourage and accommodate walking and biking as modes of commuting.

Plan Components

ACTIVEALLEGHENY focuses on connectivity, access, mobility, and health through specialized plan components: *Bike Allegheny, Walk and Roll Allegheny, Other Active Transportation Opportunities, Complete the Street, and Action for Active Transportation*. Each component provides resources, identified deficiencies, potential opportunities, system improvements, and other considerations such as policies and programs.

How to Use this Plan

Due to the size of the County and its evolving nature, the plan should not be treated as a static document, or one that identifies every desirable Active Transportation project in the County. Instead the plan should be used as a basis for the planning and development of pedestrian, bicycle, and active transportation modes of transportation in the County, and serve to provide examples of Active Transportation projects that others can follow. Further study and recommendations can and should develop from this document.

PCTI Program

In 2008, PennDOT introduced the Pennsylvania Community Transportation Initiative (PCTI) to advance Smart Transportation by “incentivizing collaborative decision-making, emphasizing regional, multi-municipal, and multi-agency cooperation, as well as advancing integrated land use and transportation decisions.”ⁱⁱ The PCTI program requires strong partnerships and extensive collaboration between PennDOT, Metropolitan Planning Organizations (MPO)/Rural Planning Organizations (RPO), counties, and municipalities. Selection criteria include a commitment to the principles of Smart Transportation and desire to enhance the transportation network in a proactive manner. PennDOT initially set aside \$60 million in State and Federal funding for projects that advanced Smart Transportation in the Commonwealth. **ACTIVEALLEGHENY** was funded in the initial phase. A second round of PCTI funding was initiated for the 2011-2014 Transportation Improvement Plan (TIP).

1.2 PERSPECTIVES

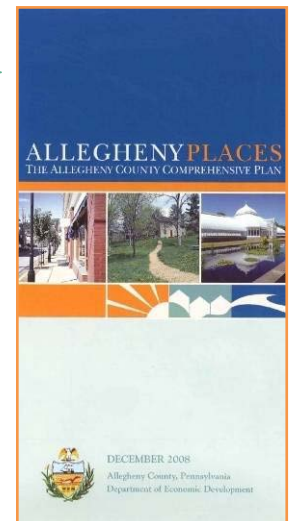
An Implementation Activity of ALLEGHENYPLACES

Allegheny County’s Comprehensive Plan establishes a vision and strategies to achieve that vision. It includes a transportation element with actions for commuter bicycle and pedestrian accommodation. **ACTIVEALLEGHENY** is the detailed plan for active transportation. It enhances and enriches **ALLEGHENYPLACES**.

Smart Transportation Guidebook Principles

The Pennsylvania Department of Transportation (PennDOT) and the New Jersey Department of Transportation (NJDOT) facilitated the development of the *Smart Transportation Guidebook* in 2008ⁱⁱⁱ with a goal of integrating planning and design aspects of roadways to advance development of livable and sustainable communities. The principles of Smart Transportation as detailed in the guidebook are:

- Tailor solutions to the context
- Tailor the approach
- Plan all projects in collaboration with the community
- Plan for alternative transportation modes
- Use sound professional judgment
- Scale the solution to the size of the problem



1.3 DEFINING THEMES

Through the course of **ACTIVEALLEGHENY**, four (4) defining themes were developed to guide future infrastructure improvements and assist in prioritizing identified projects. Through these four (4) themes, a quality transportation network can be developed and successfully implemented. The themes are access, connectivity, mobility, and health.

Access

Access for active transportation is the ability to safely enter destinations without barriers by multiple modes of transportation. Offices, commercial centers, neighborhoods, and other popular destinations should consider the needs of non-auto users and provide a safe and effective path of entry for non-motorized transportation.

Connectivity

Connectivity refers to the way which two features, destinations, or modes are *connected* to each other. In active transportation planning, connectivity links destinations (i.e., neighborhoods, commercial districts, offices, stadiums) through multiple modes of transportation and provides safe and reliable paths for active transportation users. Connectivity requires *access* and is ineffective without access to the destinations with which people wish to connect.

Mobility

When a transportation network has multi-modal access and substantial connectivity, an overall climate of mobility is created. A good transportation network puts an emphasis on the word *transportation*, creating an environment where many modes of transportation are given an opportunity to succeed in a safe and efficient manner. Mobility is at the root of good transportation networks by providing users the *choice* of several different modes of transportation to arrive at the same destination without feeling burdened by that *choice*.

Health

In a country that has seen a dramatic increase in obesity and chronic illness; healthy lifestyles are increasingly becoming an important life choice. Active transportation is a critical component of a healthy lifestyle, and it's important to offer a transportation network that allows for active options. Quite frequently commuters choose the automobile not out of desire but rather out of necessity. When networks are created that provide multi-modal access to destinations, exhibit high levels of connectivity, and have an overall character of mobility, healthy transportation options become viable and allow those who desire to improve their lifestyle to do so safely and effectively.



Bike Rack at Ingram Busway Station
Photo: David Wohlwill

1.4 SUSTAINABILITY

Merriam-Webster Dictionary^{iv} defines “sustainability” as “of, relating to, or being a method of harvesting or using a resource so that the resource is not depleted or permanently damaged.” This concept of sustainability has been forged throughout the nation’s history and in its most simplest of terms refers to “waste not, want not.” In order to provide a transportation network that is functional yet capable of safe trip conversion to alternative modes for the future requires investment in active transportation modes. The first step in active transportation investments is planning. This active plan outlines investments toward sustainable improvements for both short- and long-term community benefits.



Broad Street, Sewickley Borough

Sustainable Pittsburgh outlines the importance of transportation investment and livable communities on their website (www.sustainablepittsburgh.org) to include “Smart Growth and Integration of Transportation and Land Use” and “Public Transportation and Transportation for Livable Communities.”

Smart Transportation

PennDOT defines Smart Transportation as “partnering to build great communities for future generations of Pennsylvanians by linking transportation investments and land use planning and decision-making,” with a goal of creating transportation facilities “that are safe and affordable, responsive to the needs of all users, and support community planning goals.”^v They outline ten (10) themes for Smart Transportation:

- ✓ Money counts. Innovation for fiscal challenges.
- ✓ Leverage and preserve existing investments.
- ✓ Choose projects with high value/price ratio.
- ✓ Safety always and maybe safety only. Safety for all users is the most important consideration.
- ✓ Look beyond level-of-service. Evaluate other performance measures as well.

- ✓ Accommodate all modes of travel. Including walking, bicycling, and transit.
- ✓ Enhance local network. Highly connected local network with route options.
- ✓ Build towns not sprawl. Smart Growth!
- ✓ Understand the context; plan and design within the context. Context-sensitive design solutions.
- ✓ Develop local governments as strong land use partners.

Placemaking

ALLEGHENYPLACES is about PLACEMAKING for Allegheny County's bright and sustainable future. It integrates new and redevelopment activities into our established communities, while maintaining their character and respecting their history. **ACTIVEALLEGHENY** advances the **ALLEGHENYPLACES** vision within a placemaking context. It helps to create targeted "places" where residents can live, learn, work, invest, and play.

Measuring Performance

The National Bicycling and Walking Study: 15 Year Status Report^{vi} summarizes the numerous benefits associated with bicycling and walking as mode choices. Benefits to both the user and society as a whole are outlined and include: health, transportation, environmental and energy, economic, and quality of life. Each benefit can be measured either quantitatively or qualitatively prior to and after transportation system improvements.

Health

The National Collaborating Centre for Environmental Health^{vii} found that people who use active transportation are "more physically fit, less obese, and have a reduced risk of cardiovascular disease compared to people who use only motorized transportation."

Transportation

According to the *National Bicycle and Walking Study: 15 Year Status Report*, 72% of short trips (3 miles or less) are made in motor vehicles. This results in increased congestion on the roadway network and decreased level-of-service for motor vehicles. Trip conversion from motor vehicle to active transportation modes would reduce the amount of vehicle trips, thereby reducing congestion and delays.

Environmental and Energy

According to the Department of Energy, the transportation sector accounted for 29% of energy consumed in the U.S. in 2009.^{viii} Similarly, in 2006, 29% of total U.S. Greenhouse Gas Emissions (GHG) came from transportation sources.^{ix} The United States Environmental Protection Agency (U.S. EPA) notes that transportation is the fastest growing sector and U.S. emissions have increased nearly 47% over the past decade. Bicycles, kayaks, sneakers, and wheelchairs do not produce GHG and therefore trip conversion from motor vehicles to active transportation would reduce GHG caused by motor vehicles.

Economic

AAA estimates that commuting to work by motor vehicle costs approximately \$56 per 100 miles.^x They suggest that traveling by public or active transportation can save money.

Quality of Life

An increase in active transportation can benefit communities as well in creating livability through reduced vehicular traffic, reduced pollution, increased travel mode options, and revitalized business districts.

The U.S. EPA^{xi} encourages transportation agencies to integrate sustainability into planning, programming, and project development activities through performance measures to quantify the results. “Transportation performance measures predict, evaluate, and monitor the degree to which the transportation system accomplishes adopted public objectives. They can be applied at all stages of transportation decision making.” The U.S. EPA lists the following ten (10) performance measures:

- Transit Accessibility
- Bicycle and Pedestrian Mode Share
- Vehicle Miles Traveled (VMT) per Capita
- Carbon Intensity
- Mixed Land Uses
- Transportation Affordability
- Distribution of Benefits by Income Group
- Land Consumption
- Bicycle and Pedestrian Activity and Safety
- Bicycle and Pedestrian Level of Service



Bicycle Parking, Carnegie Mellon University

Photo: Lynn Heckman

Guidance on utilizing the performance measures is provided in the Draft *Guide to Sustainable Transportation Performance Measures*.

1.5 COMMITMENT, COORDINATION & OUTREACH

Stakeholder coordination and public participation was an integral part of the ACTIVEALLEGHENY planning process and several different methods were utilized to encourage involvement including a website, online survey, Facebook page, coordination with a Core Committee, participation by a Study Advisory Committee, and interaction with the general public.

Allegheny County Economic Development

The ACTIVEALLEGHENY study was administered in coordination with Allegheny County departments and officials. Allegheny County Economic Development (ACED) managed the study.

Core Committee

The Core Committee for the study consisted of county selected stakeholders to provide vision and guidance on active transportation in Allegheny County. Core Committee members included:

- City of Pittsburgh
- Southwestern Pennsylvania Commission (SPC)
- PennDOT District 11-0
- Allegheny County Parks Foundation
- Allegheny County Executive’s Office
- Pennsylvania Environmental Council
- Allegheny County Economic Development
- 10,000 Friends of PA
- Oakland Transportation Management Association
- Montour Trail Council
- Upper St. Clair Township
- Bike Pittsburgh
- Friends of the Riverfront
- CityLAB



Active Allegheny Core Committee

A Comprehensive Commuter Bicycle and Pedestrian Transportation Plan for Allegheny County

- Pittsburgh Downtown Partnership Transportation Management Association
- Allegheny County Public Works, Airport Corridor Transportation Management Association
- Port Authority of Allegheny County (PAAC)
- Allegheny County Task Force on Disabilities

Four (4) Core Committee Meetings were held during **ACTIVEALLEGHENY**. Meeting Memoranda from those meetings is contained in **Appendix B** (included in a separate document).

Study Advisory Committee

The Study Advisory Committee for the plan consisted of numerous county and state agency representatives, local municipal stakeholders, SPC members, sustainable advocacy groups, commuters, businesses, schools, and others, identified by the Core Committee, to present input on potential deficiencies, opportunities, and enhancements. Study Advisory Committee members included:

- | | |
|---|---|
| • Allegheny County | • Three Rivers Center for Independent Living |
| • SPC | • United Cerebral Palsy of Pittsburgh |
| • Sustainable Pittsburgh | • Mullen Advertising |
| • Venture Outdoors | • Rothschild Doyno Collaborative |
| • GoBurgh Initiative | • URS Corporation |
| • Pennsylvania Department of Conservation and Natural Resources (DCNR) | • Pennoni Associates |
| • RAND | • Gateway Engineers |
| • Airport Corridor Transportation Association | • Civil & Environmental Consultants, Inc. |
| • Oakland Transportation Management Association | • Carnegie Mellon University's Traffic 21 |
| • Pittsburgh Downtown Partnership Transportation Management Association | • Mon Valley Initiative |
| • Point Park University | • ACCESS Transportation |
| • Community College of Allegheny County (CCAC) | • Montour Trail Council |
| • Twin Rivers Council of Government (COG) | • Picadio, Sneath, Miller & Norton, P.C. |
| • Steel Valley COG | • 3 Rivers Wet Weather Inc. |
| • Quaker Valley COG | • Blind Leisure Outdoor Development |
| | • Manchester Bidwell Corporation |
| | • City of Pittsburgh, and a number local municipalities |

Three (3) Study Advisory Committee Meetings were held during **ACTIVEALLEGHENY**. Meeting Memoranda from those meetings is contained in **Appendix C** (included in a separate document).

Allegheny Municipalities

Allegheny County municipalities and other organizations (e.g., Mon Valley Initiative, Allegheny Together, Bike Pittsburgh, Port Authority of Allegheny County, and others) were contacted via email, fax, and phone to ascertain if they had existing plans and documents related to active transportation improvements in their municipality or their region. **Table 1-1** summarizes the plans and documents obtained as a result of the outreach effort.

These plans and documents were reviewed for identified issues as well as recommended improvements. In addition, Allegheny County municipalities and organizations were asked if they had other concerns, or if improvement opportunities exist, within their municipality. **Table 1-2** summarizes the issues and opportunities identified through input and plans collected.

Public Outreach

Website

www.activeallegheny.com serves as the study's website. The website contains background information on active transportation, updates on project progress, and served as a resource for obtaining study materials.

Online Survey

An online survey was designed and administered for ACTIVEALLEGHENY. The purpose of the survey was to gather public input and assist the Study Team in identifying active transportation deficiencies, opportunities, and locations for potential improvement.

The survey was available online from June 23, 2010 through July 26, 2010 through www.surveymonkey.com. During that time, a total of 738 responses were received from the online survey. Nearly half (48%) of the survey respondents were residents of the City of Pittsburgh. A second tier survey analysis was performed in August to filter residents by municipality and identify additional locations for improvements in the County. The Survey Results Summary and Filter by Municipality Addendum are contained in **Appendix D** (included in a separate document).

Social Networking on Facebook

ACTIVEALLEGHENY joined social networking in June with its very own Facebook page. It provided an opportunity for posting comments and blogs with the Project Manager, while also providing information on upcoming events, photos and other information about the study.

Public Presentations and Community Events

Public Meetings were held in June to present findings from the existing conditions analysis including crash overrepresentations and preliminary countywide bicycle routing. One Public Meeting was held in Carnegie, while the other was held in the City of Pittsburgh at Point Park University. Comments received at the Public Meetings are contained in **Appendix E** (included in a separate document).

Presentations were made to several entities as well to inform them about ACTIVEALLEGHENY and receive feedback on issues and opportunities relevant to their groups. Presentations were made to the following organizations, committees, and agencies: SPC Pedestrian-Bicycle Advisory Committee, Airport Corridor Transportation Association, 10,000 Friends of PA, and the Committee for Accessible Transportation (ACCESS Transportation).

ACTIVEALLEGHENY staffed a booth at the Allegheny Green + Innovation Festival on August 14, 2010 to provide residents with information about the study. Interactive games and a raffle were available for entertainment at the festival. The word find and the memory game developed for use at the Allegheny Green + Innovation Festival are contained in **Appendix F** (included in a separate document).

Table 1-1. Collected Plans and Documents

Municipality/ Organization	Plans and/or Documents Obtained
Bridgeville Borough	Design Guidelines, Strategic Plan, and Traffic and Parking Analysis
Elizabeth Borough	Design Guidelines, Streetscape Design Plan, and Strategic Plan
Stowe Township	Design Guidelines and Traffic and Parking Analysis
Edgewood Borough	Edgewood Avenue Revitalization Plan
Etna Borough	Etna's Walking Trails
Heidelberg Borough	Heidelberg Trail Journal
Moon Township	Sidewalk and Trails Master Plan
Oakmont Borough	Transportation Plan and Action Plan from the Comprehensive Plan
Pittsburgh, City of	Final Corridor Phasing Plan for Penn Avenue (between 34 th Street and Negley Avenue), Bicycle Plan, Pittsburgh Downtown Plan, City of Pittsburgh Bicycle Facility Guidelines and Policies, Bicycle Parking Policy and Guidelines, Bicycling Resources and Contacts, Employee Survey 2009, 2009 Sharrows Survey
Swissvale Borough	Design Guidelines, Strategic Plan, and Streetscape Master Plan
Tarentum Borough	Design Guidelines, Strategic Plan, and Traffic and Parking Analysis
Verona Borough	Design Guidelines and Traffic and Parking Analysis, Sustainable Verona
Southwestern Pennsylvania Commission (SPC)	Liberty Avenue from 12 th Street to Herron Avenue Road Safety Audit (RSA) Report, Census Block Maps for minorities, low income, disabled, and older adult populations, Bicycle Counts: Master Record by Location, Pedestrian and Bicycle Public Participation Report, Report on Environmental Justice, Allegheny County Transportation Improvement Program (TIP), Bike Suitability Maps
Allegheny County	Greenways Plan, ALLEGHENY PLACES Comprehensive Plan, Allegheny County: Early History and Subsequent Development, West Busway Transit Oriented Development Assessment and Plan, South Hills Transit Revitalization Investment District Study
Mon Valley Initiative	Mon Valley Corridor Vision Plan
Airport Corridor Transportation Association (ACTA)	Commercial Center Mobility Study, A Planner's Notebook, Intersections 06'-08' Annual Reports, Rethinking the Suburban Bus Stop, Study of Improved Shared Ride Transportation Services, Suburban Transportation Solutions, Transportation 101 Brochure
Oakland Transportation Management Association (OTMA)	Crosswalk/Stop Bar Line Striping Priority List, Engineering Applications for Society Final Report for Intersection Analysis Project (Bigelow Boulevard, Hamlet Street and Bates Street)
Pittsburgh Downtown Partnership Transportation Management Association (PDP)	Downtown Pittsburgh Made Easy Guide, Street Smarts Pedestrian Resource Guide, Street Smarts Cycling Guidelines, 2010 Downtown Resident Survey, Pedestrian Traffic Study (2008)
Bike Pittsburgh (BikePGH)	Bike to Work Guide (Bike Commuting 101), Rack 'N Roll Routes (also received from the Port Authority of Allegheny County), Pittsburgh Bicycle Map
Port Authority of Allegheny County	Transit Development Plan and proposed route changes, Wabash Tunnel input from various Port Authority Departments, Rack 'N Roll Program

Table 1-2. Identified Issues, Opportunities, and Improvements

Municipality	Identified Issues, Opportunities, and Improvements
Braddock Borough	• Extend Maple Way and develop into a greenway.
Bridgeville Borough	• Pedestrian Improvements and Traffic Calming for Washington Avenue. • Connect Washington Avenue to Triangle Park/Railroad Street via a trail or open space. • Connect Bower Hill Road to Railroad Street through an extension of existing sidewalk. • Connect Triangle Park/Railroad Street to the Bank Street Extension parking lot and lower Washington Avenue via pedestrian walkway. • Sidewalk buffer improvements, median improvements, and potential roundabout at Washington Avenue and Bower Hill Road.
Carnegie Borough	• Does not have a formal bicycle and pedestrian plan but does encourage biking and walking as part of the West Busway TOD Assessment and Plan completed in 2010 (West Busway TOD Assessment and Plan was an implementation activity of ALLEGHENY PLACES and the SPC 2035 Transportation and Development Plan). • Participant in multi-municipal Third Street Improvements including active components. • Has many projects moving forward that would encourage biking and pedestrian plan.
Collier Township	• Bike/ped/active transportation is not addressed in the Township's current comprehensive plan (expires in 2011). In the upcoming plan update, bike/ped/active transportation is an emerging topic which will be addressed and incorporated into the new plan. • The recently adopted conservation subdivision ordinance, based in part on the desire to create a network of interconnected greenways that could be used for trails, could provide additional pedestrian access and may have potential commuter bicycle connectivity. • Pedestrian concerns are an ongoing topic in Collier. Sidewalks are not required for all new developments. There is some difficulty in overcoming perception issues since much of Collier is semi-rural with steep terrain. Those required to construct sidewalks, at times, have difficulty understanding who will use them and what destinations they will connect. • Other important concerns are some hazardous road issues, cited by bicyclists, during public input for this plan. • The Township is home to a long stretch of Panhandle Trail. The Trail is a recreational asset and often functions as a community park. However, the 3-mile disconnect between the Walkers Mill Trailhead and Carnegie Borough limits the trail's use as a transportation asset. The trail has great potential for bike commuters to the West Busway park-and-ride and other Carnegie-and-beyond destinations; once this key connection is complete. • The Township, nonetheless, is a very popular destination for bicyclists, on-road and on the trail. Several opportunities exist for important bike/ped connections with Settlers Cabin County Park and the adjoining Botanic Garden site, currently under construction. • Another specific connection in the planning stage is on-road bike lane markings along Nike Site Road, which currently has extremely wide lanes. This is part of the Park Commission's plan for transition of the former Kelly Army Reserve facility into recreation facilities and a community center. The bike lane will connect the Panhandle Trail to these new park and recreation facilities. The connection length between the facilities is about a mile. Pedestrian access to the new park, from nearby residential developments, is another key element in the new park design. • Along the Route 50 Corridor in the Township, a smaller scale streetscaping and sidewalk improvement initiative, supplemented by transit service, is under study with the intent of linking Carnegie, Scott, Heidelberg, Collier, and Bridgeville. This initiative takes resource constraints into account that preclude major roadway improvements along the corridor.

Municipality	Identified Issues, Opportunities, and Improvements
Edgewood Borough	• Gaps in the sidewalk network exist and sections of sidewalk are in poor condition. • ADA compatibility issues noted for sidewalks. • Inadequate pedestrian lighting cited. • Install a new pedestrian bridge over Race Street with new linear park/pedestrian walkway. • Improve crosswalk and sidewalk at the corner of Swissvale Avenue and Race Street. • Install new crosswalks and sidewalks at the corner of Swissvale Avenue, Edgewood Avenue, and Maple Street. • Renovate pedestrian tunnel connecting Pennwood Avenue to busway station. • Stairs and crosswalk improvements for busway station platform at Pennwood Avenue. • Sidewalk and crosswalk improvements on Edgewood Avenue from Lacrosse Street to Maple Avenue. • Sidewalk and crosswalk improvements at Edgewood Towne Center to improve access for Park and Ride lot.
Elizabeth Borough	• Increase bicycle and pedestrian access to Duke and Wiley Parks. • Market and Plum Streets should be developed with pedestrians as the primary focus. • Develop Pedestrian Oriented Corridors perpendicular to the Riverfront. • Develop a repair and maintenance program that addresses pedestrian signals, sidewalks, and curbs. • Strengthen the connection between 2nd Avenue and the river by improving pedestrian accommodations along Market and Plum Streets. • Add curb extensions to assist in pedestrian crossing at 2nd and 3rd Avenues.
Etna Borough	• Identified five (5) walking routes that start in downtown Etna and then continue out into the rest of the Borough. Routes use sections of the following streets: Grant Street, Butler Street, Maplewood Street, Grandview Avenue, High Street, Bridge Street, Locust Street, Prospect Street, Vine Street, Walnut Street, Morelock Street, Bottomfield Street, and Freeport Street.
Greentree Borough	• Hazardous roads cited as issue affecting bicyclists.
Heidelberg Borough	• Developed 1.5 mile walking route for the Borough that uses the following roadways: W. Railroad Street, 4th Street, Garfield Street, Jackson Street, Industry, and Zero Street. • Participating in a multi-municipal streetscape/sidewalk project with Carnegie Borough and Scott Township. Completed pedestrian and transit stop improvements on Route 50 near the borough building.
Ingram Borough	• Local funding and budget cited as issue affecting ability to develop bicycle/ped plan.
Kennedy Township	• Infrastructure and topography challenges in the township affecting ability to develop bicycle and pedestrian plans.
McCandless Township	• Developing sidewalk plan for township. • Developing a pedestrian plan in conjunction with the Rachel Carson Trail. • Intersections of concern on McKnight Road include: Blazier Drive, Peebles Road, Perrymont Road, Cumberland Road, and Arcadia Drive.

Municipality	Identified Issues, Opportunities, and Improvements
Moon Township	• Nine (9) of the 26 pedestrian crashes in the Township between 2001 and 2006 were on University Boulevard. • The University Boulevard plan of improvements includes connectivity between the Airside Business Park, business strip development, local schools, and Robert Morris University. • New sidewalks are proposed to connect between and within residential neighborhoods. • Hiking trails are proposed to connect between residential neighborhoods. • Multi-use trails proposed along roadway corridors including: Flaugherty Run Road, Beaver Grade Road, Ewing Road, McCormick Road, Hookstown Grade Road, and Hershinger Road.
Mt. Lebanon, Municipality of	• Bike lanes through South Hills to Great Allegheny Passage (GAP) that link at Peter's Woods or Bethel Park. • Bike lanes through Mt. Lebanon to connect schools and municipal areas. • Develop way to allow bicycles on T service. • Other initiatives identified by Mt. Lebanon and the school board include a potential Safe Routes to Schools Study and improvements to Washington Road in response to school children needing access.
North Fayette Township	• Worn footpaths, or desire lines, were noted on both sides of Summit Park Drive. • Recommends sidewalk connections and new sidewalk where worn footpaths are present. • Recommends a Montour Trail spur connecting to Summit Park Drive which would connect to new shared use paths to nearby retail. • Montour Trail runs through the township and they do assist with maintenance. • There are plans to install a one (1)-mile walking trail at the new Donaldson Community Park. • Property acquisition and "Not In My Back Yard" (NIMBY) issues affecting the development of additional facilities. • Recommends improving crosswalks at bus stop locations.
Pennsbury Village Borough	• No plans to develop a bicycle and pedestrian plan.
Pittsburgh, City of	• Develop consistent cross-section along the Boulevard of the Allies, including bike lanes and wide sidewalks, develop a greenway connection from Boulevard of the Allies to Panther Hollow, and install new pedestrian connections across Boulevard of the Allies including a link to the Second Avenue Area Technology developments. • Extend greenway along hillside to Bates Hollow with new connection to Eliza Furnace Trail. • Intersections along Boulevard of Allies, Bates Street, and Dawson Street need re-striping/re-marking of pavement. • Enhance multi-modal connections along 2nd Avenue to Panther Hollow. • Extend Eliza Furnace Trail eastward along the river. • Sections of sidewalk along Penn Avenue found to be in poor or very poor condition. • Penn Avenue corridor in need of street lighting improvements. • Crosswalk and pedestrian signal improvement needed at intersections along Penn Avenue. • Bicycle issues along Penn Avenue include: speeding, lack of adequate lighting, uneven pavement and potholes, and lack of bike friendly facilities. • Penn Avenue is not a goal for a future bicycle route but as a destination, a recommended parallel bicycle route to Penn Avenue would utilize Coral Street, Comrie Way, and Woolslayer Way.

Municipality	Identified Issues, Opportunities, and Improvements
Pittsburgh, City of (continued)	• More bicycle racks needed on Penn Avenue, especially at key destinations. • Factors impacting bicycling in Pittsburgh include: topography, bridge crossings, narrow streets, high levels of road user conflicts, roadway surface issues, inadequate level of existing bicycle routes, and scarcity of bicycle parking. • Goals for the city bicycle route system include: connect downtown with surrounding neighborhoods via commuter routes and riverfront trails, connect riverfront trails to each other, connect downtown to Oakland via on-street and off-road routes, connect Oakland universities to surrounding neighborhoods and trip attractors, connect regional parks to riverfront trails, connect neighborhoods to local business districts, connect bicycle routes to transit facilities, cross-river connections, and connections that circumvent major traffic or topographic obstacles. • Improve pedestrian connections between the North Shore and North Side and Golden Triangle areas. Improve pedestrian connections between the Strip area and the Golden Triangle. • Establish a continuous pedestrian-friendly corridor along Federal Street across the Sixth Street Bridge to Market Street. • Reconstruct Fort Pitt Boulevard, Fort Duquesne Boulevard, Wood Street, and Forbes Avenue to provide a more conducive pedestrian street environment. • Improve the pedestrian link between the Lower Hill District, Civic Arena, and Grant Street. • Provide new pedestrian link between Station Square and the Golden Triangle via a new lower Triangle Monongahela River Crossing. • A Transit Revitalization Investment District (TRID) Planning Study was initiated in October 2010 for Broadway Avenue in Beechview, Pittsburgh. The study is expected to engage the community and provide conceptual design for Complete Streets and redevelopment. Streetscape improvements, neighborhood connectivity, regional transit commuter routes and regional bicycle and pedestrian facilities will be highlighted as part of the study, which is expected to be complete in June 2011.
Robinson Township	• Local funding and lack of personnel cited as issue affecting ability to develop bicycle plan. • Worn footpaths, or desire lines, were noted on south side of Park Manor Boulevard, and at existing gaps in the sidewalk network along Robinson Center Drive and Park Manor Boulevard. • Pedestrians observed transcending hillsides, which led to recommended locations for steps or potentially ADA compliant facility. Recommendations for sidewalk connections and new sidewalk where worn footpaths are present. • Recommends a second Montour Trail spur connection from the YMCA on Montour Run Road up the hill to the Mall at Robinson. • Recommends shifting and narrowing travel lanes on bridge over I-376 to create a pedestrian walkway/sidewalk. • Recommends improving crosswalks at bus stop locations.
Sewickley Heights Borough	• Blackburn Road in need of sidewalk from Sewickley Valley Hospital School of Nursing to Sewickley Borough War Memorial Park where there is parking lot for students.
Stowe Township	• Pedestrian crosswalks are in very poor condition.

Municipality	Identified Issues, Opportunities, and Improvements
Swissvale Borough	• Improve pedestrian connection between the Central Business District (CBD) and the residential neighborhood that lies to the east across Braddock Avenue and up the hill. • Improve pedestrian connection between the CBD and the residential area to the west of Monongahela River and Noble Street. • Pedestrian crosswalk pavement markings are worn. • Pedestrian countdown signal heads recommended at multiple intersections. • Add bulb-outs, concrete curbs, sidewalks, ADA compatible handicap ramps at pedestrian crossings, and pedestrian level street lighting along roadways, including Washington and Monongahela Streets.
Rankin Borough/ Swissvale Borough	• Create ped connection from Memorial Park and Schley Avenue to Carrie Furnace site. • Create greenway system along hillside and with pathway connections to Rankin ball fields.
Tarentum Borough	• Find opportunities to draw pedestrians from 6th Avenue down into the Corbet Street corridor and from Riverview Park back into Corbet Street and the CBD. • Sidewalk issues at Blackburn's Pharmacy, located at Corbet Street and 4th Avenue.
Turtle Creek Borough	• Build pedestrian connection between Keystone Commons and business district. • Develop greenway parallel to Monroeville Avenue/Lynn Avenue and Turtle Creek. • Create new pedestrian bridge over Tri-boro Expressway along Monroeville /Lynn Avenues.
Verona Borough	• Riverfront access for Allegheny River. • Existing rail line acts as a barrier to riverfront. • Trail connection from Verona to Oakmont. • Development of a riverfront trail. • Public dock on waterfront with bicycle and pedestrian access. • Trail using land adjacent to Allegheny River Railroad line. • Connection to Pittsburgh Heritage Trail to provide access to downtown Pittsburgh.
Wilkinsburg, Borough of	• Completed a Comprehensive Plan and Business District Revitalization Plan, as well as an initial sustainability assessment, which was performed by Sustainable Pittsburgh. Each plan has transportation components that set goals and objectives for transportation including bicycle and pedestrian traffic, as well as improvements in the business district.
Allegheny County	• Develop trails along greenways, including rail to trail conversions. • Develop commuter bikeways both along established roadways and through greenways. • Implement and promote recommendations in this Plan.

Organization	Identified Issues, Opportunities, and Improvements
Southwestern Pennsylvania Commission (SPC) (continued)	• High level of transit, pedestrian, and bicycle activity on Liberty Avenue between Lawrenceville and Downtown. Issues include: ○ Pedestrian crossings needed to access bus stops. ○ Bicycles from Lawrenceville to the Strip District need to be better accommodated. ○ Poor maintenance of sidewalk and crosswalk markings. ○ Vehicles park on sidewalk and impede pedestrians. ○ Few pedestrian refuge areas. ○ Pedestrian often crossing at unmarked mid-block crossings to access bus stops. ○ Crosswalks missing at some intersections. • There is a need for pedestrian countdown signals. There is an opportunity for needs-based planning based on observed behavior (e.g., bicycle activity counts). • There is an opportunity for residents wishing to bike to work through Bike Pools as part of SPC's CommuteInfo Program. • SPC's Pedestrian and Bicycle Program is advancing Bike Suitability Maps for roadways in Allegheny County.
Mon Valley Initiative	• Reclaim the Mon River as a "Front Door" for communities along the corridor. • Reconnect fragmented neighborhoods and communities to each other. • Create vibrant centers and gathering places along the corridor. • Create viable/interim uses for land in the path of the proposed Mon-Fayette Expressway. • Leverage investment pressures, particularly the Almono and Carrie Furnace sites as well as Oakland, to jump start development in all communities along the corridor. • Restore and increase stock and variety of high quality housing and public amenities. • Coordinate plans and initiatives among communities including Oakland, Hazelwood, Swissvale, Rankin, Braddock, and Turtle Creek.
Airport Corridor Transportation Association (ACTA)	• Over the past twenty years, retail, hotel, restaurant, and office development in the Robinson and North Fayette Township areas has been very successful. So successful, in fact, that Robinson Town Centre, The Mall at Robinson, and The Pointe at North Fayette are, by far, the largest concentration of retail in the western suburbs. With this success has come increased movement of all types: vehicular as well as pedestrian and bicycle. The Montour Trail is adjacent to the commercial area. The hub of Port Authority of Allegheny County (PAAC) service in the airport corridor is also in this commercial area. Pedestrian amenities are few. Most areas do not have sidewalks, steps, handicap ramps, etc. A commercial area developed for the automobile now hosts hundreds of pedestrians each day. Many of them are employees going to and from work. With the help of "Walkable Communities, Inc." and Southwestern Pennsylvania Commission (SPC), ACTA held a community workshop/audit to discuss mobility issues in the commercial area. As part of the workshop, ACTA documented a significant increase in pedestrian traffic through a series of photographs showing the "desire paths" in the unimproved grassy/earth areas, many on steep earth slopes. ACTA subsequently developed a walking tour of the area for local elected officials to illustrate mobility concerns.

Organization	Identified Issues, Opportunities, and Improvements
Airport Corridor Transportation Association (ACTA) (continued)	Over the past few years, ACTA conducted three major studies to look at commuting and mobility issues in the airport corridor. The studies were based on user surveys and focus groups of workers, shoppers, business owners, local residents, bus riders, bicyclists and pedestrians. The first, <i>Study of Improved Shared Ride Transportation Services in the Robinson/North Fayette Employment Center</i>, looked at where jobs are located and the current barriers and future opportunities for commuters getting to work. The second, <i>ACTA's Commercial Center Mobility Study</i>, took a much broader look at mobility issues in the same study area in order to develop a community and user-focused plan of action to improve mobility, enhance intermodal connectivity and create a sense of place in the commercial area which serves as the downtown for the community. Specific study recommendations include pedestrian crosswalks and a connection between the Montour Trail and the retail area. The third study, <i>Rethinking the Suburban Bus Stop</i>, uses existing conditions in the study area to design a set of replicable prototype bus stops and bus stop placements that address mobility and accessibility challenges faced by bus riders in an area with limited pedestrian amenities. Prototypes address typical suburban bus stop placements including a hub stop, a stop along a busy roadway, a stop in a retail area, and an intermodal transfer stop. The bus shelter designs incorporate technology related to real-time bus information and pedestrian amenities. The designs also address pedestrian access to and around the bus stops so that pedestrian circulation is raised to a level more equal to vehicular travel. In addition to the three technical studies, three other publications were produced as part of the studies: <i>Commuting in the Corridor</i>, <i>A Planner's Notebook</i> and <i>Suburban Transportation Solutions</i>. All of the publications are available at www.acta-pgh.org.
Oakland Transportation Management Association (OTMA)	- Focused on improving 11 intersections for pedestrians as part of the 5th/Forbes Pedestrian Safety and Mobility Improvements Study. OTMA developed a brochure "Hometown Streets, Re-imagining Safe Streets – Transforming an Urban Center" to highlight pedestrian safety elements (e.g., countdown signals) for Oakland. - OTMA would like to improve crossings for pedestrians at the following intersections: - o Halket Street and Boulevard of the Allies. - o Bates Street and Boulevard of the Allies. - o Fifth Avenue and Bellefield Avenue.
Pittsburgh Downtown Partnership (PDP)	Planning to implement a pilot project downtown in Market Square for an ADA Wayfinding System (www.clickandgomaps.com). The goal is to make Downtown Pittsburgh more accessible for the blind and deaf/blind. The City of Pittsburgh would be the first US Downtown to implement this program according to PDP. According to the website for Click and Go Wayfinding Maps, the maps "render public facilities...accessible to blind and deaf/blind travelers down to a level of detail that is unparalleled by any other service or technology...[and it also provides] customized narrative walking directions for outdoor landmark-to-landmark route travel."

Organization	Identified Issues, Opportunities, and Improvements
Allegheny Together (continued)	SPECIAL NEW PROGRAMS FOR ALLEGHENY TOGETHER COMMUNITIES The following new programs are being launched in Allegheny Together communities this summer: 1. Healthy Downtown Business Program - Allegheny County Economic Development, Allegheny County Health Department, and Town Center Associates (TCA) are currently implementing a new “Healthy Downtown Business Program” in the Allegheny Together communities. This new program has been created to make it easy for downtown business owners and their employees to implement regular exercise into their daily routine. Each downtown business owner and all of their employees are eligible to participate in this program, which is the first phase in creating healthier business districts in Allegheny County.* In August, TCA visited all of the Allegheny Together downtown businesses to let them know about the program and to provide them with: • A Free “Allegheny Together Pedometer” • A “Walking Route Map” of their downtown to make it easy for them to gauge the distance they walk each day, and achieve their mileage goals • A Downtown Business Directory to encourage downtown businesses to support one another’s businesses • A Walking Log to keep track of their progress (which will be collected by TCA in October) • Other handouts provided to encourage and support healthier downtown businesses 2. All of the above documents will also be available at DowntownFirst.net, making it easy for businesses to access additional copies, etc. During the month of September, downtown business owners and employees will track their steps walked using a free Allegheny Together pedometer. Each week of the program, TCA will be selecting participants to receive prizes to encourage their continued participation. An award, based on the average number of steps walked per employee, will be given to the “Healthiest Downtown Business” and the “Healthiest Downtown Business Employee” in each Allegheny Together Community. Local participating communities include: • Bellevue • Bridgeville • Coraopolis • Elizabeth • Stowe • Swissvale • Tarentum • Verona <i>*NOTE: The next phase of “Healthy Downtowns” program, pending available funding, will consist of downtown business owners providing their customers with free pedometers and walking logs. The pedometers will be given out during an established week in all of the Allegheny Together communities, and a similar contest will ensue. Additional features will be added to the Walking Route Map during this phase of the program, to encourage more residents to make walking in their business district a regular part of their daily routine. Having more people walking in the business districts will not only make them more vibrant and safe, but will also help to make residents more aware of their downtown businesses. The Allegheny Together Business District Advisory Committees have already expressed interest in this program.</i> Source: Allegheny Together, 09/2010

CHAPTER 2. *Bike Allegheny*

ALLEGHENY PLACES identified key challenges in increasing bicycle travel as a mode share. Some of those challenges included a “lack of a bicycle route signage program,” “lack of available, safe bicycle parking facilities,” and incorporation of bicycle facilities into roadway projects. **ACTIVE ALLEGHENY** details the issues and constraints for bicyclists in Allegheny County and offers solutions from engineering to education.

2.1 ALLEGHENY’S BICYCLE NETWORK

Bicyclists have identified the need for safe and convenient access to destinations in the County. Although some bicycle routes are available, most notably PA State Bicycle Routes and shared use trails including the Great Allegheny Passage (GAP) and Montour Trail, more designated commuter and recreational routes are still strongly desired. PennDOT’s Design Manual^{xii} states that although “most highways have not been designed with bicycle travel in mind...there are many methods to safely improve most roadways to accommodate bicycle traffic while also improving safety for motorized road users and pedestrians.”



Shoulder on Hulton Road, Oakmont

Desired Access

Through discussions with stakeholders and the public and analysis of the online survey results, bicycle access is desired primarily along spokes from the north, east, south, and west suburbs into the City of Pittsburgh. Secondary access desired includes employers and schools, new and proposed development sites, trail network and parks, transit stops and stations, and connecting adjacent communities. **Table 2-1** details desired access for bicyclists that were expressed through these discussions and surveys.

Table 2-1. Desired Access for Bicyclists

To	From
Downtown Pittsburgh	• North Hills • Oakland • Mt. Lebanon
Kennywood Park	• East End
The Waterworks Mall	• Downtown
The Waterfront	• Sandcastle/South Side Trail • Duck Hollow Trail
Carrie Furnace Site	• East End
Montour Trail	• South Park • IKEA • Mt. Lebanon • Bethel Park
Great Allegheny Passage	• Rankin Bridge • Glenwood Bridge • Montour Trail • Round Hill Park • Frick Park • Mt. Lebanon

To	From
West Busway Carnegie Station	• Panhandle Trail • Heidelberg
Bidwell Technical Institute	• Penn Hills
Pittsburgh International Airport	• Montour Trail • Clinton Road
Millvale Riverfront Park	• Millvale • Freeport
Duck Hollow Trail	• Frick Park • Eliza Furnace Trail • Glen Hazel
Eliza Furnace Trail (Jail Trail)	• Riverview Park • Schenley Park • Duck Hollow Trail • Glenwood Bridge
Schenley Park	• West End • Eliza Furnace Trail
Sandcastle Waterpark	• Route 885 • East End
New Kensington	• Dorseyville

Identified Deficiencies

A review and analysis of existing conditions, survey results, bicycle and pedestrian crashes, and public feedback was performed to identify locations where bicycle facility deficiencies exist. Although there are a number of locations throughout Allegheny County that could benefit from the installation or enhancement of bicycle facilities, for this plan ten (10) locations are identified for further study and improvement. The ten (10) locations would require additional study prior to design and are therefore classified as long term improvements (5+ years). **Table 2-2** details the location and the identified deficiency. The crash summary and bicycle crash map are contained in **Appendix G** (included in a separate document).



Bridge Street, Etna Borough

Table 2-2. Top 10 Bicycle Facility Deficiencies

Corridor	Limits	Identified Deficiency
Penn Avenue	City of Pittsburgh	10 reported bicycle crashes in 5 years (2005 to 2009). - Actively used by bicycle commuters. - Limited cartway width (36') with multiple demands (parking, transit, pedestrians, bicyclists, vehicles). 10' travel lanes, no shoulders, 7-8' parking provisions and bus stops. #1 listed roadway needing bicycle facility improvements by survey users.
Liberty Avenue	City of Pittsburgh	7 reported bicycle crashes in 5 years (2005-2009). - Identified as a top 5 roadway needing bicycle facility improvements by survey users.
Bigelow Boulevard	City of Pittsburgh	- Identified as not compatible for bicycle traffic from Oakland to Downtown based on PennDOT design guidelines for lane/shoulder widths and observed average operating speed. - Public desires safe and convenient access to Frank Curto Park and Downtown. - Speed limit posted at 35 mph, observed average speed at 55 mph. - Existing cross section is 4 travel lanes with less than 1' shoulder. - For pedestrians, sidewalk is not continuous and ends at the merge with the I-579 Ramp.
Allegheny River Boulevard	City of Pittsburgh to Oakmont	- Identified as not compatible for bicycle traffic as it would need a consistent 4'-6' shoulder based on PennDOT design guidelines. - Existing cross section is 30' with 11' and 12' lanes and 3'-4' shoulders.
Route 8/ Butler Street	Etna to Richland (Orange Belt)	6 reported bicycle crashes in 5 years (2005-2009). - Identified as a top 5 roadway needing bicycle facility improvements by non-Pittsburgh residents in the online survey. <i>In Etna</i> 38' pavement width, 12' lanes, 7' parking on-street. <i>At Saxonburg Boulevard</i> 50' pavement width, 4 lanes at a width of 11.5' and center median. - SPC Bicycle Compatibility Below Average.
Route 28 Corridor	City of Pittsburgh to Blawnox	- Identified as a top 5 roadway needing bicycle facility improvements by non-Pittsburgh residents in the online survey. - Three Rivers Heritage Trail extends only to Millvale. The trail could be extended to Blawnox and access across Route 28 for bicycles and pedestrians investigated.
Route 19/ Washington Road/ West Liberty Avenue	City of Pittsburgh to Upper St. Clair	- Identified as a top 5 roadway needing bicycle facility improvements by non-Pittsburgh residents in the online survey. 25 to 35 mph. 4 lanes with <1' shoulder. - SPC Bicycle Compatibility Above Average to Below Average with Significant Grade.
Route 837/ East Carson Street	City of Pittsburgh	5 reported bicycle crashes in 5 years (2005-2009). - Identified as a top 5 roadway needing bicycle facility improvements by non-Pittsburgh residents in the online survey.

Corridor	Limits	Identified Deficiency
Freeport Road	Aspinwall to Cheswick	8 reported bicycle crashes in 5 years (2005-2009). <i>In Etna (25 mph – 35 mph)</i> 2 lanes, no parking at 24' pavement width. <i>Aspinwall (35 mph)</i> 2 lanes with parking westbound and shoulder eastbound. <i>Blawnox (25 mph)</i> 2 lanes with parking. - East of Blawnox Business District, speed limit increases to 45 mph and the cross section becomes 3 lanes, then 4 lanes (under construction). - SPC Bicycle Compatibility Average to Below Average.
Fifth Avenue	City of Pittsburgh	7 reported bicycle crashes in 5 years (2005-2009). - Identified as a top 5 roadway needing bicycle facility improvements by Pittsburgh residents in the online survey. - Connects Downtown to Shadyside through Oakland.

Potential Opportunities

Several innovative opportunities for active transportation were identified by the stakeholders, study team, and public during the course of the study. They are described in this section.

Wabash Tunnel

The Wabash Tunnel is open daily for one-directional motor vehicle travel with HOV restrictions during the weekday morning and evening peak periods and without restrictions at other times. Traffic is allowed inbound during the weekday mornings and early afternoons and outbound during the weekday evenings and early mornings. The tunnel operates with outbound traffic during the weekends. In all of these instances, only one of the two travel lanes is being used at any one time by vehicular traffic.



Wabash Tunnel, South Side
Photo: Port Authority of Allegheny County

Users of the tunnel include the Fayette Area Coordinated Transit, which currently operates scheduled weekday bus service through the tunnel. Special event buses (e.g., incline shuttles) utilize the tunnel as well as significant traffic volumes outbound from Station Square and vicinity after sporting and other entertainment events.

Bicycles are restricted from the Wabash Tunnel for many valid reasons. Stakeholders and the public suggested during the course of the study that the community consider how the Wabash Tunnel could potentially accommodate bicycle traffic sometime in the future. The Port Authority of Allegheny County has considered and investigated the feasibility of the suggestion in the past. Based on PennDOT Design Guidelines (Publication 13M, Design Manual 2), emergency vehicle access must be maintained

through the tunnel. The existing tunnel provides this access per design guidelines. As travel modes, bike access, and federal and state policy evolve over the next decade to accommodate and more fully utilize all modes of transportation, it may be advantageous to revisit Wabash Tunnel bicycle access, in the context of other changes in mode shift. Port Authority cannot allow bicycles in the Wabash Tunnel due to safety, design, liability, and operational concerns. However, it may be possible if there is significant demand, to pursue an alternative ownership scenario, where bicycle access to the tunnel can be considered, although that would be a major undertaking and cost.

Roberto Clemente Bridge

It was suggested during the study that the Roberto Clemente Bridge (6th Street) in Pittsburgh be permanently closed to motor vehicle traffic. The bridge is currently closed to motor vehicles before sporting events at PNC Park and Heinz Field. The Roberto Clemente Bridge is one of the Three Sister's Bridges owned by Allegheny County serving as a connection between Downtown and the North Side. According to the book "The Bridges of Pittsburgh", they are the only three identical side-by-side bridges in the world.^{xiii} The other two bridges are the Andy Warhol Bridge (7th Street) and Rachel Carson Bridge (9th Street). While comparing activity on the three bridges, the Roberto Clemente Bridge was observed to have the most bicycle and pedestrian activity on its 10-foot sidewalks both northbound and southbound. A feasibility study to determine traffic impacts of permanently restricting motor vehicles from the Roberto Clemente Bridge can be performed to determine if the bridge could be a bicycle and pedestrian only facility in the future. Performing a "road diet" on the bridge to narrow it to two vehicular travel lanes plus bicycle lanes could also be considered in this feasibility study.



Roberto Clemente Bridge, Pittsburgh

Road Diet Feasibility Studies

During the course of field investigations for the study, several roadways were observed to be under capacity in terms of vehicle-to-capacity ratio. Those roadways, with four (4) lane cross-sections and without shoulders, are:



East Liberty Boulevard, Pittsburgh

- Park Manor Boulevard in Robinson between Montour Run Road and Robinson Town Center Boulevard
- Negley Run Boulevard in East Liberty between E. Liberty Avenue and Washington Boulevard
- Washington Boulevard in Highland Park between Negley Run Boulevard and Allegheny River Boulevard
- Route 51 in Coraopolis (study to include appropriate use of one-way pair system)
- Bigelow Boulevard between Oakland and Downtown, currently a limited access divided highway
- Long Run Road/Route 48 in White Oak

The six (6) roadways are recommended for study to accommodate a road diet. A road diet is the conversion of a four lane cross-section to a configuration with two through lanes, one two-way left turn lane, and two bike lanes. The *Smart Transportation Guidebook* outlines the benefits of a road diet to include:

- Creates a designated facility for bicyclists,
- Reduces crossing distance in which pedestrians are exposed to vehicular traffic,
- Provides a refuge for crossing pedestrians if physical medians are created,
- Can reduce the incidence of left turn crashes for motorists, and;
- Can reduce vehicular speeds by 1 to 5 mph on roadways where speeding is common.^{xiv}

The Borough of Carnegie has submitted a PCTI grant application to perform a road diet assessment for Main Street near the Carnegie Station of the West Busway.

Washington Road (Route 19) in Mt. Lebanon was recommended for a Road Diet as well by residents of Mt. Lebanon. They cited high turning movement volumes and a safety concern for children walking and biking to school. It is recommended that Mt. Lebanon pursue several funding opportunities for Washington Road as well.

A Road Diet Before...



And After



Photos: FHWA University Course on Bicycle and Pedestrian Improvement, Lesson 15, Bicycle Lanes Publication No. FHWA-HRT-05-114

It should be noted that as part of a road diet assessment, traffic analysis should be performed to determine potential impacts to vehicle level-of-service.

2.2 SYSTEM IMPROVEMENTS

Designated County Commuter Bicycle Routes

It is recommended that Allegheny County designate bicycle routes to serve bicycle commuters from north, south, east, and west suburbs to the City of Pittsburgh to connect with the City of Pittsburgh Bicycle Network. The proposed routes in this plan were reviewed and compared to existing data sources (e.g., traffic volumes, jurisdiction, SPC bicycle suitability rating) for potential designation as a bicycle route. Prior to the designation of a bicycle route, facilities should be evaluated for compatibility per PennDOT design guidelines and brought up to bicycle standards where needed. The SPC Region's neighboring counties may also wish to connect to these routes to extend the spokes from the entire region to the regional hubs in Pittsburgh and Oakland, as well as to access points on the Great Allegheny Passage.



East Carson Street, Pittsburgh

The bicycle routes proposed in this Plan may make use of existing or proposed off-road trail segments, roadways, and/or transit routes. In the case of off-road trail segments which are under construction or proposed, user connectivity can often be provided in the interim through the use of local streets and roadways. While formal route designation should not occur until a facility is complete, it is recommended that segments be constructed and existing roadways be brought up to applicable bicycle standards where feasible in support of on-going bicycle route development. Constructing portions of a route and utilizing interim sections may ultimately help to achieve the goal of route completion and formal designation. Proposed routes could be designated as future bicycle routes which can help promote usage.



Rack and Roll Bus in Pittsburgh
Photo: Sara Walfoort

Proposed bicycle routes that involve utilizing transit would also need to be evaluated and coordinated with the Port Authority, as current policy allows folding bicycles on the rail system and inclines, but currently does not permit regular bicycles on light rail and incline vehicles during peak periods (folding bicycles are allowed inside all Port Authority vehicles at all times of the day). The Port Authority has indicated that in the year 2011, all PAAC buses will be equipped with racks as part of the “Rack ‘N Roll” Program.^{xv} **Table 2-3** details the proposed countywide designated commuter bicycle routes.

Appendix H (included in a separate document) contains the System Improvements Map for Countywide Bicycle Routes. This map is also included in the Executive Summary. **Appendix A** contains cue sheets (detailed maps and templates which list specific information for a pre-determined list of attributes) for each route illustrating the utilized roadways, bridges, bike routes, trails, and transit, as well as providing information for that section.

Table 2-3. Designated County Commuter Bicycle Routes

Route	Description
N1	Mc Cully Road (<i>Hampton Township Park</i>), Craighead Road, Mt. Royal Road, Grant Avenue, Crescent Avenue, Butler Ave, <i>Connection to Proposed N2 Route</i>
N2	South Ridge Drive (<i>North Park</i>), Peebles School Road, Babcock Boulevard, Evergreen Avenue, Seavey Road, Parker Street, Butler Street, Allegheny River Trail (existing), <i>Connection to City Facilities</i>
N2 Connector	Nicholson Road, Rochester Road, <i>Connection to Proposed N2 Route</i>
N3	Beaver Street, Ohio River Boulevard, Allegheny Avenue, Center Avenue, Church Avenue, California Avenue, Lincoln Avenue, <i>Connection to City Facilities</i>
N3 Detour of Ohio River Boulevard	Broad Street, Sewickley Bridge, PA Bike Route A (existing), <i>Connection to Proposed W3 route @ Bike Route A</i>
N4	Little Deer Creek Road (<i>@ Proposed Beltway Bicycle Route</i>), Michael Road, Russelton-Dorseyville Road, Saxonburg Boulevard, Harts Run Road, Dorseyville Road, Riding Meadow Trail, Old Squaw Trail, Salamander Trail, Fox Chapel Road, Allegheny River Trail (existing), <i>Connection to Proposed N2 Route</i>
W1	Panhandle Trail (existing), Walkers Mill Road, Noblestown Road, Scotts Run Road, Ewing Road, Cabbage Road, Logan Road, <i>Connection to West Busway</i>
W2	Montour Trail (existing), Bike Route A (SR 51), McKees Rocks Bridge, Helen Street, River Avenue, Three Rivers Heritage Trail (existing), <i>Connection to City Facilities</i>
W2 Connector	Moon Clinton Road (<i>Pittsburgh International Airport</i>), McCaslin Road, Clinton-Enlow Road, Montour Trail (existing)
W3	PA Bike Route A (existing), Grand Avenue Bridge, Ohio River Trail (proposed), Neville Island Bridge, <i>Connection to Proposed W2 Route</i>
W3 Connector	Cliff Mine Road, Thorn Run Road, <i>Connection to Proposed W3 Route</i>
W4	Beaver Grade Road (<i>Connection to Proposed W2 Route</i>), Steubenville Pike, Tidball Road, McMichael Road, Campbells Run Road, Chartiers Avenue, 5 th Avenue, Dick Street, Carnegie Station (West Busway), <i>Connection to Proposed S4 Route</i>

Route	Description
E1	Allegheny River Trail (existing) (<i>Harrison Hills Park</i>), E. 1 st Avenue, Worth Avenue, E. 7 th Avenue, Freeport Road (<i>Connection to Proposed E1 Route Connector</i>), Pittsburgh Street, Freeport Road, Blockdale Street, Allegheny River Trail (proposed), <i>Connection to Proposed N4 Route</i>
E1 Connector	Baileys Run Road (<i>Deer Lakes Park</i>), Freeport Road, <i>Connection to Proposed N1 Route</i>
E2	Cox Comb Hill Road (<i>@ Proposed Beltway Bicycle Route</i>), Hulton Road, Three Rivers Heritage Trail (proposed), <i>Connection to City Facilities</i>
E3	Old William Penn Highway, Rodi Road, Nottingham Drive, Homer Road, Churchill Road, Beulah Road, William Penn Highway (Penn Avenue), Montier Street, North Avenue (Eastbound) / Wallace Avenue (Westbound), Wilkinsburg Station, <i>Connection to East Busway</i>
E4	East Pittsburgh-McKeesport Boulevard, Versailles Avenue, First Street, Greensburg Avenue, Penn Avenue, Braddock Avenue, Electric Avenue, Linden Avenue Pedestrian Plaza, Linden Avenue, Bessemer Avenue, Western Avenue, Center Street/Bell Avenue, Jones Avenue, Braddock Avenue, Kenmawr Avenue, Belmar Place, Woodstock Avenue, Swissvale Station <i>Connection to East Busway</i>
E4 Connector	Wall Avenue (<i>@ Proposed Beltway Bicycle Route</i>), Patton Street, Airbrake Avenue (Eastbound) / Middle Avenue (Westbound), Penn Avenue, <i>Connection to Proposed E4 Route</i>
E5	Pierson Run Road, Saltsburg Road, Frankstown Road, <i>Connection to City Facilities</i>
S1	Route 837, (<i>Connection to Proposed Beltway Bicycle Route</i>), Steel Valley Trail (existing), <i>Connection to Proposed S3 Route</i>
S2	Montour Trail, Logan Road, Bethel Church Road, Fort Couch Road, Village Road, <i>Connection to South Hills Village T Station</i>
S2 Connector	Library Road, Logan Road, <i>Connection to Proposed S2 Route</i>
S3	Youghiogheny River Trail (existing), Steel Valley Trail (existing), Riverton Railroad Bridge, <i>Connection to City Facilities</i>
S4	McMurray Road, McLaughlin Run Road, Ridge Road, Bower Hill Road, Chartiers Creek Trail (proposed), <i>Connections to Proposed W1, W4, S4 Connector Routes, and West Busway</i> with connection to Panhandle Trail
S4 Connector	Greentree Road, <i>Connection to City Facilities</i>
S5	Brownsville Road, Curry Road, Brownsville Road, <i>Connection to City Facilities</i>

Designated County Beltway Bicycle Route

Members of the Core and Study Advisory Committee, as well as the public, expressed desire for a circular bicycle route to connect County Parks and nearby land uses. The Orange Belt, which is comprised of 91.7 miles of miscellaneous county and state owned roads and color coded for navigational purposes^{xvi}, was a logical starting point. Spurs and parallel routes were then added to or substituted for the existing Orange Belt to avoid high volume cross sections. Prior to designating the recommended route, existing roadways need to be evaluated for compatibility per PennDOT design guidelines. Study Team observations noted that the proposed beltway route, utilizing a majority of Orange Belt roadways, is scenic with light truck traffic and relatively low volumes and motor vehicle operating speeds. A typical cross section is comprised of 10' travel lanes with 0' - 4' shoulders. **Table 2-4** details the proposed beltway bicycle route. **Appendix H** (included in a separate document) contains the Preliminary System Improvements Map for Countywide Bicycle Routes, which includes this route. This map is also included in the Executive Summary.

Table 2-4. Designated County Beltway Bicycle Route

Route	Description
Beltway Bicycle Route	Montour Trail (existing), Library Road, Clifton Road, McMurray Road, McLaughlin Run Road, Ridge Road, Baldwin Street, Railroad Street, Bower Hill Road, Washington Avenue, Prestley Road, Thoms Run Road, Battle Ridge Road, Boys Home Road, Union Avenue (Route 978), W. State Street, Clinton Avenue, McKee Road, Steubenville Pike, Enlow Road, Montour Trail (existing), Beaver Grade Road, University Boulevard, Sewickley Bridge, Broad Street, Hill Street, Blackburn Road, Fern Hollow Road, Camp Meeting Road, Rochester Road, Wexford–Bayne Road, Wexford Road, Gibsonia Road, Oak Road, Bairdford Road, Saxonburg Boulevard, East Union Road, Starr Road, Little Deer Creek Road, Creighton-Russellton Road, Butler Logan Road, Crawford Run Road, Freeport Road, New Kensington Bridge (C.L. Schmitt Bridge), Industrial Boulevard, 3 rd Avenue, 2 nd Street, Logans Ferry Road, Leechburg Road, New Texas Road, Saltsburg Road (Route 380), Center Road, Haymaker Road, Mosside Boulevard (Route 48), Jacks Run Road (Route 48), Long Run Road (Route 48), Walnut Street (Route 48), Boston Bridge, Boston Hollow Road (Route 48), Scenery Drive (Route 48), Lovedale Road, McKeesport Road, Hayden Boulevard (Route 51), State Street (Route 837), Montour Trail (existing)

City of Pittsburgh Bicycle Network

The City of Pittsburgh, BikePittsburgh (BikePGH), and Friends of the Riverfront have been working diligently to enhance the bicycle network in the City of Pittsburgh. Many roadways have been retrofitted for bicycle facilities and many more are proposed. In 2009, the Mayor of Pittsburgh created a list of bicycle and pedestrian specific initiatives to be advanced in the categories of engineering, education, enforcement, and events (**Appendix I**, included in a separate document). As a result of that initiative, a Bicycle Route and Signage Plan is currently under development, as well as other efforts to enhance the bicycle and pedestrian network within the City. A map illustrating the current City of Pittsburgh Bicycle Network, and routes proposed as part of this plan, is contained in **Appendix J**, which is included in a separate document as well as the Executive Summary. The City will further its Bicycle and Pedestrian Network as part of its upcoming MovePGH Comprehensive Transportation Plan and Bicycle Master Plan update. The City and the County should coordinate any future routes to promote system connectivity.

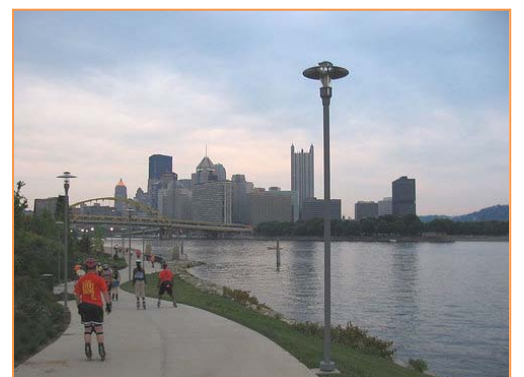


Approaching Roberto Clemente Bridge, North Shore

Three Rivers Heritage Trail

Friends of the Riverfront is performing planning, development and construction to expand the Three Rivers Heritage Trail along the Allegheny County Riverfront. They are one of many groups working to support beneficial and community focused riverfront development through the Pittsburgh Metropolitan Area. The following projects are key regional priorities:

- **Community Initiatives – Extending the trail/commuter bike facility up the Allegheny River:** A public/private initiative to complete a trail through 17 municipalities along the North Shore of the Allegheny River to connect the Three Rivers Heritage Trail with the Armstrong Trail.



North Shore Trail, North Shore

- **Allegheny Riverfront – Convention Center to Highland Park:** As part of the Allegheny Riverfront Vision Plan, create an urban design vision and implementation plan for the south bank of the Allegheny River between the Convention Center and Highland Park.
- **Allegheny Riverfront – Allegheny River Boulevard to Rail Trail:** Planned trail connection between Penn Hills and Verona along Allegheny River Boulevard.
- **Verona Borough Sustainability Needs Assessment:** Includes recommendation for a multipurpose walkway along the railroad tracks and activities at Riverbank Park.
- **Carrie Furnace:** Preliminary engineering design for a trail to connect Duck Hollow through Carrie Furnace to Braddock.
- **East End Loop:** Planning to connect the Duck Hollow Trail to the Pittsburgh Zoo.
- **Ohio Riverfront Montour Connection:** Planning to connect the Three Rivers Heritage Trail downstream to the Montour Trail.
- **Carnegie Science Center Development Plan:** A plan to expand the Science Center to include a redesigned riverfront park and trail access to the Three Rivers Heritage Trail.
- **Convention Center Riverfront Park:** Development of a riverfront park on the Allegheny River adjacent to the Convention Center with connections to the Three Rivers Heritage Trail.
- **Mon Wharf Landing:** Development of the Mon Wharf for riverfront open space to the public including a linear park and trail.
- **South Shore Riverfront Park:** Extension of the South Side Riverfront Park to include the Three Rivers Heritage Trail and South Side Trail.

2.3 BICYCLE FACILITIES TOOLBOX



The Bicycle Facilities Toolbox is a resource for County and local officials, staff, residents, and stakeholders that will assist in planning and developing bicycle facilities as part of the **ACTIVEALLEGHENY** Plan. The toolbox is composed of the following four (4) sections:

- 🚲 **Bicycle Facility Users:** An overview of the types of cyclists that are a focus of this study.
- 🚲 **Bicycle Facility Types and Design Guidelines:** A review of common bicycle facilities and relevant design guidance from PennDOT, Allegheny County, and the City of Pittsburgh, as well as national guidelines and standards.
- 🚲 **Order of Magnitude Costs:** A guide to typical costs for design and installation of bicycle facilities.
- 🚲 **Innovative Bicycle Facilities:** A presentation of innovative bicycle facilities that are being developed and evaluated, both nationally and internationally.



*Bicyclists in the Strip District
Photo: Kevin Smay*

Bicycle Facility Users

There are multiple ways to categorize cyclists. They can be separated into groups by purpose (e.g., commuter, recreational, etc.), by skill level (e.g., experienced, inexperienced, etc.), or by age (adult, child, etc.). For the purposes of **ACTIVEALLEGHENY**, the focus is the following types of bicyclists: Experienced Commuter, Casual Commuter, Utilitarian, and Children, which are described briefly in this section:

- **Experienced Commuter:** This type of user is comfortable in mixed traffic and using shared travel lanes or on-road bicycle facilities. Experienced commuters are adults who have been bicycling over a long period of time, and who may ride by choice or because bicycling is their only means for getting to work (e.g., no personal vehicle, no transit access, etc.). These users are focused on direct trips, which usually occur during peak travel periods. Trip lengths for the experienced commuter vary from short trips of 3 to 5 miles to long trips of 10 to 20+ miles.
- **Casual Commuter:** This type of user has limited experience with bicycling and may lack confidence with on-road bicycle travel, but has interest in commuting by bicycle. Casual commuters are adults who may have experimented with bicycling to work, but typically rely on another means for commuting, such as driving alone or walking to transit. These users are focused on direct trips, but feel more comfortable on: off-road facilities; low-volume roadways; or exclusive on-road bicycle facilities. The casual commuter has a trip length of 1 to 5 miles, and a preference for transit connections to traverse difficult locations, such as congested roadways and bridges.
- **Utilitarian:** Although commute trips can be utilitarian, this descriptor is used for bicyclists who are making off-peak trips or daytime work trips, shopping, medical visits, or other non-commute purposes. These cyclists can be experienced or casual, adults or adolescents, and are often prepared to make trips in mixed traffic. Trip lengths for the utilitarian likely range between 1 to 3 miles.
- **Children:** Children tend to not travel as fast as adult cyclists, but still desire and require access to destinations such as schools, stores, recreational areas, and neighboring residences. According to AASHTO, “residential streets with low motor vehicle speeds, linked with shared use paths and busier streets with well-defined pavement markings between bicycles and motor vehicles can accommodate children without encouraging them to ride in the travel lane of major arterials.” Ideally children on bicycles should be accommodated with barrier-separated bicycle lanes and accompanied by adults until they are an appropriate age when they have an understanding of traffic movements and signage and they are able to operate a bicycle on-street.



*Bicyclists in Pittsburgh
Photo: Sara Walford*

KIDS

Bicycle Facility Types and Design Guidelines

There are a common set of on-road and off-road bicycle facilities that are used to enhance and accommodate bicycle travel. These facilities, which are grouped as “Bikeways” by PennDOT, have recommended physical dimensions and characteristics, as well as typical pavement striping, markings, and signs. Bicycle facility guidelines are outlined in Chapter 16 and Chapter 19 of PennDOT Design Manual 2. Roadways and intersections should be compliant with PennDOT and MUTCD guidelines and standards. The current approved version of the MUTCD in Pennsylvania is the 2003 edition. PennDOT is currently in the process of evaluating the 2009 edition for potential adoption. States are given two years from the publication date (January 15, 2010) to conduct their review process.

On-Road Bicycle Facilities

The common on road bicycle facilities are: Shared Lanes/Shared Roadways, Paved Shoulders, and Bicycle Lanes. Specific roadway attributes (e.g., pavement widths, cross-sections, parking provisions, traffic volumes, posted speed limit, etc.) determine the applicability of each facility. In addition to these common facilities, there are approaches, including signed bicycle routes and bicycle boulevards, that can be used to integrate the roadway improvements into discrete corridors and an overall on road network. **Table 2-5** describes these facilities in detail. Striping for bicycle facilities should be thermoplastic for longevity.

Table 2-5. On-Road Bicycle Facilities

Shared Lane/ Shared Roadway	A Shared Lane, or Shared Roadway per the PennDOT Design Manual 2, accommodates bicyclists and motorists in the same travel lane. Shared lanes can be located on urban or rural roadways with low vehicular traffic volumes and low posted speeds, and are often supplemented with 'Share the Road' warning signs. Wide outside travel lanes, which have widths of 12' to 15' depending on the roadway context (e.g., rural or urban), are desired for shared lane facilities. A new pavement marking used to guide bicyclists with lateral positioning in a shared travel lane, especially in locations with on-street parking, is the shared lane marking (informally referred to as 'Sharrows'). Sharrows are included in the MUTCD 2009 Edition.	 <i>Shared Lane with Shared Lane Marking</i>  <i>'Shared The Road' Used in Pennsylvania</i>
Paved Shoulder	A paved shoulder provides accommodation for bicyclists adjacent to vehicle travel lanes. Paved shoulders can be located on urban or rural roadways with moderate to high vehicular traffic volumes and moderate to high posted speeds. Paved shoulders for bicyclists range in width from 4' to 6'+ depending on the available pavement width, and can be supplemented with 'Share the Road' warning signs. Chapter 16 of the PennDOT Design Manual (Publication 10A) does not identify paved shoulders, but paved shoulders are listed in the Bicycle and Pedestrian Checklist (Appendix K, included in a separate document). In the checklist, paved shoulders are grouped with bicycle lanes, and appropriate widths are cited as 6' standard and 5' adjacent to curb.	 <i>Paved Shoulder</i>

Bicycle Lane

Bicycle lanes are designated travel lanes for exclusive or preferential use by bicyclists. Bicycle lanes are typically located on roadways in urban settings with moderate to high vehicular traffic volumes, moderate to high posted speeds and permitted or designated on-street parking.

According to the PennDOT Design Manual, bicycle lanes include the application of pavement striping, markings, and regulatory signage. Bicycle lane facilities should be one-way facilities that carry traffic in the same direction as motor vehicles, and utilize configurations that encourage merging to occur in advance of intersections. Following the Bicycle and Pedestrian Checklist, bicycle lane widths are cited as 6' standard and 5' adjacent to curb, and are to include marking as per AASHTO guidance.



Bicycle Lane

Signed Bicycle Route

Signed bicycle routes are treatments used to provide wayfinding guidance to cyclists. Route signs can be used to provide directional, distance, and destination information to assist bicyclists in navigation. Signed routes can also be used to direct cyclists to corridors that have existing on-road facilities, or access locations for off road facilities. The PennDOT Design Manual 2 does not provide specific guidance on signed bicycle routes, but AASHTO does provide guidance on the design and application of signed bicycle routes.



Bicycle Route Sign

Source: MUTCD 2009 Edition

Bicycle Boulevard

A bicycle boulevard is a corridor treatment that prioritizes bicycle travel. Bicycle boulevards accommodate shared travel for bicyclists and motorists and utilize traffic calming measures, signs, pavement markings, and crossing improvements to enhance bicycle travel. Corridors identified for bicycle boulevards are typically characterized by low volumes and low speeds. Bicycle boulevards are not included in the PennDOT Design Manual; however, a Bicycle Boulevard Guidebook was recently released by the Initiative for Bicycle and Pedestrian Innovation at the Center for Transportation Studies (Portland State University).^{xvii} The guidebook provides direction on selecting routes and application of design elements (e.g., priority, intersection, traffic calming, and traffic reduction treatments).



Chronicle / Paul Chinn

Bicycle Boulevard in Berkley, CA

Photo: sfgate.com (Paul Chinn)



Three Rivers Heritage Trail
Photo: Tom Baxter

Off-Road Bicycle Facilities

The typical facility used to accommodate off-road bicycle travel is the Shared Use Path, or Bicycle Path (PennDOT). The Shared Use Path has specific physical attributes which are described in **Table 2-6**. At one point in time, people considered “trail development” to be a recreational accommodation. Not any longer. Trail development within Allegheny County provides connections to our communities, opportunities for economic development, and offers transportation alternatives through the use of the corridors. Trails are used for recreational purposes, but also for commuting from one place to another.

The Allegheny County comprehensive plan outlines the goals for trails, greenways, and bike routes. **ACTIVE ALLEGHENY** will integrate existing trail corridors with planned on-road and off-road (trail) segments to connect to the destinations across our region.

Table 2-6. Shared Use Path or Bicycle Path

Shared Use Path	A shared use path or bicycle path is a facility that is physically separated from the roadway and typically accommodates bi-directional travel by both bicyclists and pedestrians. The path can be located either within publicly owned right-of-way, within an exclusive right-of-way, or on an easement. Shared use paths typically have a hard surface (e.g., asphalt, concrete, compacted gravel, etc.) and have a recommended width of 10', although a minimum width of 8' may be used where space is constrained or in environmentally sensitive areas. Wider paths are also recommended if there is a high volume of existing or anticipated bicycle and pedestrian traffic. Sidepaths are a subset of shared use paths that denote paths that run adjacent to a parallel roadway. Sidepaths can assist in providing bicycle connections between on- and off-road facilities, but often require a more in-depth operational and safety analysis.	 Shared Use Path  Sidepath at an intersection
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Order of Magnitude Costs

Costs associated with implementing bicycle facility improvements will vary. Interim improvements (e.g., shoulder striping, bicycle route signage) will have less design requirements and will therefore be less expensive than an improvement that would need to go through feasibility assessment and design before obtaining funding for construction (e.g., shared use path/bicycle path construction, accommodating bicycle lanes through signalized intersections). Current typical costs are contained in **Table 2-7**.

Table 2-7. Typical Average Costs for Bicycle Facilities

Item	Cost Per Unit	Quantity	Cost	Source
Cost Per Lane Mile				
Bicycle Lane Striping	\$1.00 / LF	10,560 LF	\$10,560.00	(2) 4" White Hot Thermoplastic Pavement Markings (1 MILE) (ECMS)
	\$110.00 EA	22 Legends	\$2,420.00	(22) White Waterborne Pavement Legend, "Bicycle With Rider", 8'-0" X 4'-0" (1 Every ~250 FT) (ECMS)
	--	--	\$12,980.00	--
Sharrow Decals	\$135.00 EA	22 Decals	\$2,970.00	(22) White Waterborne Pavement Legend, "Bicycle with Rider", 8'-0" X 4'-0" With 2 Chevrons (1 Every ~250FT) (ECMS + Estimate)
Shoulder Striping	\$1.00 / LF	5,280 LF	\$5,280.00	4" White Hot Thermoplastic Pavement Markings (1 MILE) (ECMS)
Cost Per Item				
Share the Road Sign	\$25.00 / SF	6.25 SF	\$156.25	30" X 30" Share The Road Sign, Type B Post Mounted (W16-1)(ECMS)
Bicycle Warning Sign	\$25.00 / SF	6.25 SF	\$156.25	30" X 30" Bicycle Warning Sign, Type B Post Mounted (W11-1) (ECMS)
Bicycle Route Sign	\$25.00 / SF	3 SF	\$75.00	24" X 18" Bicycle Route Sign, Type B Post Mounted (D11-1)(ECMS)
Bicycle Lane Sign	\$25.00 / SF	5 SF	\$125.00	30"X 24" Bicycle Lane Sign, Type B Post Mounted (R3-17) (ECMS)
Bicycle Rack (Hoop)	\$175.00 EA	1 Rack	\$175.00	BikePGH (\$99.00 without installation cost)
Bus Mounted Bicycle Racks	\$1,030.00 EA	1 Rack	\$1,030.00	Port Authority of Allegheny County, Summer 2010

Source: PennDOT, <http://www.dot14.state.pa.us/ECMS/> and 2010 RS Means Site Work and Landscape Cost Data, 29th Edition.

Innovative Bicycle Facilities

In certain situations, traditional bicycle facilities (e.g., bicycle lanes) may not achieve desired results due to the nature of the existing roadway network. For this reason, the application of innovative facilities can be utilized to make important connections that would otherwise be unavailable through traditional means. Examples of innovative facilities are presented in **Table 2-8**. These facilities may be applicable in the future to bicycle compatibility improvements in Allegheny County. These facilities have been evaluated by the Institute of Transportation Engineers (ITE) and have successfully been implemented in many cities throughout the United States.

Table 2-8. Innovative Bicycle Facilities

Cycle Track	A cycle track is a bicycle facility that is adjacent to the roadway but separated by a physical barrier. Physical barriers can include the addition of concrete islands or the movement of the parking lane away from the curb, where space permits. Cycle tracks often require right-of-way of up to 14' but can be constructed in situations with as little as 9' of additional right-of-way. Cycle tracks would be applied where significant demand for bicycling exists, and often permit bi-directional travel, eliminating the need for accommodations on both sides of the roadway.	 <i>Cycle Track in Montreal, Quebec</i> Photo: BikePortland.org
Contra-flow Bicycle Lanes	Contra-flow bicycle lanes are similar to traditional bicycle lanes, except they provide for travel down a one-way street against the flow of motor vehicle traffic. This application is best utilized in extraordinary circumstances when vital connections are excluded from a bicycle route network. Prior to application, significant study should be performed to identify alternate routes which follow existing travel lane directions. In many cases, alternate routing through the use of shared use paths and parallel roadways will exist. Applications of contra-flow bicycle lanes often include the use of bollards or permanent physical barriers as a means of physical separation from oncoming vehicular traffic.	 <i>Contra-flow Bike Lane in Washington D.C.</i> Photo: DCist.org
High Visibility Bicycle Lanes	High visibility bicycle lanes are similar to traditional bicycle lanes with the exception that the entire lane is painted to differentiate it from vehicular travel lanes. This application provides an additional layer of visibility which will alert motorists to the presence of cyclists. Prominent examples include New York City's bicycle lanes which utilize the color green, and Portland, Oregon, which use blue markings at merging locations, such as highway ramps. Despite this difference, the application of the high visibility bicycle lanes have produced favorable results by way of bringing attention to the presence of cyclists and additional traffic calming effects to the roadway.	 <i>High Visibility Bicycle Lane in NYC</i> Photo: NYC Street Design Manual

Advance Stop Line/ "Bicycle Box"	The advance stop line or "bicycle box" is a roadway treatment developed to provide the bicyclist with a space to position themselves for turning movements at signalized intersections. This treatment marks an area for bicyclists in front of stopped vehicles at signalized intersections. Current applications use a contrasting surface color to mark the entire area occupied by the bicycle box and to enhance visibility. A prominent example of this treatment currently in use and under evaluation is Portland, Oregon.	 <i>Bicycle Box in Portland, Oregon.</i> Photo: BikePortland.org
Buffered Bicycle Lane	Similar to a striped bicycle lane, the buffered bicycle lane provides a dedicated travel lane for bicycle travel. The difference is that the buffered lane is marked with a typical 2' – 4' "shy zone" that creates a wider physical separation between vehicles and bicycles. Buffered bicycle lanes have been built adjacent to travel lanes, as well as adjacent to parking lanes.	 <i>Buffered Bicycle Lane in NYC</i> Photo: economyleague.org

2.4 POLICY AND PROGRAMMATIC CONSIDERATIONS

There are policy and programmatic factors in developing and maintaining a convenient, attractive, and accessible bicycle network. Described below are policies and programs that will impact and influence the development of a bicycle network in Allegheny County. These factors reference existing local guidelines, laws, and requirements, and identify best practice policies and programs that should be considered as Allegheny County, its municipal partners, and other stakeholders develop the bicycle network in the County.

Bicycle Parking



Bicycle Racks at Capacity in Downtown Pittsburgh

Bicycle parking is a necessary amenity at trip destinations for bicyclists. Parking accommodations allow for secure placement of bicycles, and in some locations, protection from the elements. Bicycle parking encompasses racks that can accommodate an individual bicycle or multiple bicycles, lockers to secure and store bicycles, and bike stations where bicycles are locked up at indoor locations that offer additional amenities such a repair and maintenance services.

Bicycle parking can be separated in short- and long-term parking. Short-term parking accommodates bicycles used for utilitarian trips where cyclists will only need to secure the bicycle temporarily. Short-term parking facilities include bicycle racks (both covered and uncovered) and

on-street bicycle parking stations conveniently located at civic buildings and in commercial areas. Long-term parking would be used for commuter trips, where the bicycle can be secured and left unattended over a lengthier period of time. The length of time parking is required impacts the location and type of parking used. Short-term parking should be provided in highly visible and easily accessible locations, whereas long-term parking tends to be placed in low traffic and low visibility locations that offer exclusive access for bicyclists. Long-term parking facilities include lockers, cages or designated rooms within buildings to offer a higher degree of security and weather protection.



*BikePGH in Pittsburgh
Photo by Kevin Smay*

In Allegheny County, the City of Pittsburgh has Bicycle Parking Guidelines for the installation, location, and design of bicycle racks. In conjunction with the guidelines, the City also recently passed a bicycle parking ordinance that requires and provides incentives for the installation of bicycle parking when a building is being built or altered. Other municipalities in Allegheny County can reference and/or utilize the city's guidelines as well as guidelines, from the Association of Pedestrian and Bicycle Professionals (APBP) to develop local guidance and ordinances.

Traffic Calming

Traffic calming involves the installation of volume or speed control measures on a roadway to modify driver behavior and improve conditions for non-motorized transportation users. Volume control measures range from full street closures to diagonal diverters to reducing the width of vehicle travel lanes, while speed control measures include enforcement equipment (e.g., speed trailers), speed humps, bulb-outs, chicanes, roundabouts, raised crosswalks and intersections, medians and gateway features (e.g., welcome signs, speed limit reduction, landscaping, archway, etc.). According to Chapter 9 of the Smart Transportation Guidebook, traffic calming may apply to many different roadway classifications, however prior to choosing a measure for installation, factors such as design speed, right-of-way, pedestrian and bicycle accommodation and ample warning for motor vehicles should be evaluated. In the Smart Transportation Guidebook, Table 9.1. "Traffic Calming Measures Appropriate to Roadway Classifications" provides planning guidance regarding the installation of traffic calming measures.^{xviii}

Policies

State Bicycle and Pedestrian Coordinator

The Federal Highway Administration (FHWA) indicates that each state should have a Bicycle and Pedestrian Coordinator "to promote and facilitate the increased use of non-motorized transportation, including developing facilities for the use of pedestrians and bicyclists and public educational, promotional, and safety programs for using such facilities." For Pennsylvania, that person is based out of PennDOT's Bureau of Design, Highway Quality Assurance Division in Harrisburg. The Bicycle and Pedestrian Coordinator position facilitates the following:

- ✓ "Increased use of non-motorized transportation, including developing facilities for the use of pedestrians and bicyclists and public education, promotional and safety program for using such facilities."^{xix}
- ✓ Implementation of the goals and objectives set forth in the 2007 PennDOT Bicycle and Pedestrian Plan.
- ✓ Information exchange among public agencies with regards to bicycle and pedestrian travel.

Given this ambitious program, the implementation of this plan may require additional staff resources to achieve the program goals.

Roadway Maintenance and Repairs

Regular maintenance and repair of roadways is an essential activity for locations where there are on-road bicycle facilities. Maintenance activities, which include clearance of obstacles in bicycle lanes, sweeping shoulder areas, clearing overgrown vegetation, and keeping drainage inlets clear, ensure accessible bicycle facilities and, importantly prevent bicyclists from needing to merge into traffic in order to avoid roadside debris and other impediments. A standard and regular repair schedule for roadway surfaces is very critical for corridors with bicycle facilities. Potholes, cracks, and heaved pavement disrupt smooth pavement surfaces and can contribute to loss of control of a bicycle. PennDOT has a number of regular maintenance and repair approaches that it employs. These include:

- **Seal Coat (Tar and Chip)** – A maintenance activity utilized to extend the performance and condition of low traffic volume roads. This activity involves the use of liquid asphalt and fine stones in order to seal the road surface, keep water out, and restore surface friction. Although it is a low-cost highly effective means of maintaining the road surface, this technique is problematic for bicyclists. It was noted by several online survey users as a constraint to bicycle riding on-road. Efforts should be made to clear the shoulders of the treated roadway, as soon as possible after applying the seal coat (tar and chip), to minimize disruption to bicycle travel and the process should be clearly communicated to the public and bicycle organizations in advance and once excess material has been cleared from the roadway and the shoulder.
- **Joint/Crack Sealing** – A maintenance activity used to close joints and prevent water from seeping into paved roadway. When water seeps in the paved surface, it can lead to potholes and breaks in the pavement.
- **Vegetation Management** – A maintenance activity that includes mowing, thinning of trees, and other processes to control encroachment into the roadway and prevent visibility issues. This should include removing overgrown weeds from road shoulders on designated bike routes at regular intervals during growth season.
- **Line Painting** – A maintenance activity to improve visibility of roadway dividing lines and markings.
- **Mechanized and Manual Patching** – A maintenance activity performed both by machines and individual workers to fill in sections of roadways with extensive potholes and cracking. Patching is used to fill in these gaps and restore pavement smoothness.

It is recommended that local municipalities and other responsible organizations in Allegheny County coordinate with PennDOT and the County to identify and set a regular maintenance and repair schedule for roadways with bicycle facilities. The use of seal coat (tar and chip) should be minimized on roadways with high bicycle usage, or the process should be clearly communicated in advance. PennDOT alerts should be distributed to include biking organizations. In addition, localities should look how to integrate these activities into local capital programs with the aim of keeping bicycle facilities on their roadways clear of impediments.

Railroad Crossings

When bicyclists cross over a rail line, bicycle tires may become trapped in the openings adjacent to rail line where they cross roadways (referred to as the flangeway). Bicyclists may also have trouble maintaining friction over surface materials around the flangeway. PennDOT's Statewide Bicycle and Pedestrian Master Plan^{xx} emphasizes that bicycle facilities crossing rail lines should ideally be at a 90-degree angle to reduce the potential for the wheels to get trapped. If a crossing cannot be close to 90-degrees, and especially if it cannot be maintained greater the 45-degrees, consideration should be give to providing a wider path so that bicyclists can angle their approach over the rails.

Recommended bicycle facility treatments that include angling of crossings, signing, and striping are included in the AASHTO Bicycle Guide.



Railroad Crossing on Bower Hill Road in Bridgeville

With the presence of many rail lines in Allegheny County, it is recommended that all entities involved with advancing bicycle improvements follow AASHTO guidance and adopt them in locally developed guidelines.

Bikeways on State Highways

To construct a bicycle facility on a state highway, there are two policies to be considered: 1) Procedures for Processing Bikeway Construction Projects, and 2) Bikeway Occupancy Permits.^{xxi} The procedures primarily apply to independent bikeway projects, which include the construction of just a bicycle facility¹, and follow a prescribed set of procedures for document submissions and reviews by PennDOT and FHWA. The Bikeway Occupancy Permit is also required for the development of a bicycle facility and is issued by PennDOT. The permit, which is in the form of resolution, includes stipulations for local ordinances addressing enforcement, maintenance, and design of the bikeway. To obtain a permit, the entire route (beginning to end) must be submitted and the local municipality is responsible for the maintenance of the bikeway, including snow removal.

It is recommended that as part of the ongoing development of the bicycle network in Allegheny County, proposed changes that will utilize state highways be grouped to provide a more efficient and comprehensive review. A joint proposal should illustrate the network connections proposed for development and allow for a more informed coordination between project sponsors and PennDOT.

Rumble Strips (Placement and Need)

PennDOT's Publication 46, Chapter 11, Section 11.11 establishes the criteria for the use of rumble strips in PennDOT projects. A rumble strip is a pavement treatment that creates noise and vibration in order to alert motorists to changes in travel, such as crossing over shoulders or centerlines, or the need to slow down. Textured or grooved pavement treatments are used for the rumble strips, and they can be placed parallel to travel (e.g., along shoulders or centerlines) or perpendicular to travel (e.g., across travel lanes). Although the strips can provide safety benefits for motorized vehicle travel, when rumble strips are placed on shoulders, they impact the mobility of bicyclists. The vibrations experienced while traveling over the strips can force bicyclists to merge into non-compatible travel lanes and even make bicyclists avoid an otherwise compatible bicycle roadway.



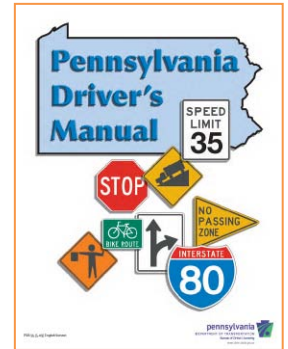
Rumble Strips on SR 837 in West Mifflin

For the future development of the bicycle network, it is recommended that rumble strips not be included on shoulder bicycle facilities unless there is a minimum clear passage of 4' from edge of shoulder strip to edge of pavement or 5' clearance from strip if a curb is present. If these dimensions are not obtainable and rumble strips are necessary, it is recommended that the rumble strip be placed under the shoulder stripe. With this treatment, it is recommended that intermittent gaps in the rumble strip be provided to allow a bicyclist to move off the shoulder when needed (e.g., avoiding obstacles in shoulder, for turning movements, etc.). These recommendations should be integrated into design manuals at the local, county, and state levels.

¹ The other category of bikeway projects is incidental. These are constructed as part of a more comprehensive highway project and are reviewed along with other elements of the project.

Pennsylvania (PA) Department of Motor Vehicles (DMV) – Driver’s Manual

The PA DMV driver’s manual is applicable to the three (3) legally-recognized types of vehicles in the state: bicycles, motor vehicles, and horse-drawn vehicles. It provides guidance for operations according to the PA Vehicle Code and notes that the rules of the road are applicable to each of these vehicles. Specifically relevant to bicycle interactions with the other vehicle types and rules of the road are: compliance with regulatory signs (e.g., R5-6, ‘No Bicycle’ where bicycles may not use the roadway), observance of warning signage (e.g., “Share the Road”, which is used to warn motorists to provide adequate space for bicyclists to share the roadway), and required behavior for motorists travel in the presence of bicyclists (e.g., allow a longer following distance, do not sound horn in close proximity to bicyclists unless needed, being alert when making turns, etc.). In addition to the existing aspects of the Pennsylvania vehicle code, there has been interest in a ‘Safe Passage Law’, which requires motorists to allow at least 3’ – 4’ when passing a bicyclist. There are 16 states that currently have enacted this law and organizations such as BikePGH and PA Walk & Bikes are actively working to have a similar bill considered in Pennsylvania.



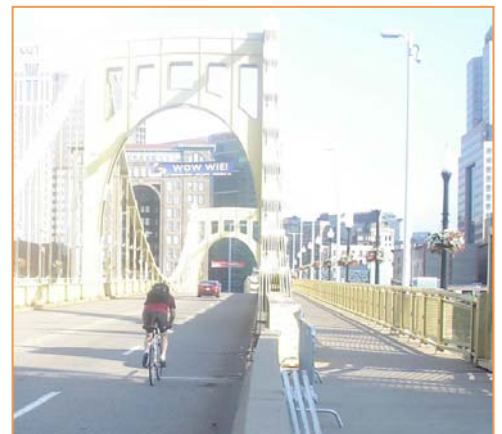
It is recommended that partners and stakeholders in **ACTIVEALLEGHENY** initiate and continue outreach efforts to highlight traffic operations and rules of the road guidance, especially sections relevant to bicycle travel. This could include a regular schedule of announcements and promotions at the county level with support from more locally focused campaigns by municipalities. A goal should be consistency of messages that can reinforced at multiple levels and with various audiences. Furthermore, it is recommended that support be provided for the passing a ‘Safe Passage Law’ in Pennsylvania.

Pennsylvania (PA) Department of Motor Vehicles (DMV) – Bicycle Driver’s Manual

Pennsylvania has developed a driver’s manual specifically for bicycles. The manual provides information about recommended bicycle travel behavior, with specific sections addressing: preparations before riding, traveling with motorists along roadways and through intersections, responding to poor driving behavior by motorists, traveling at night or in the rain, and riding with groups of bicyclists. The manual also identifies elements of the PA Vehicle Code that specifically apply to bicyclists. It is recommended that the Bicycle Driver’s Manual be excerpted or referenced and included in outreach efforts to increase knowledge of rules of the road and recommended travel behavior for bicyclists and motorists.

Bikes on Bridges

Bridges can present bicyclists with mobility and accessibility issues due to changes in cross-sections from approaching lanes (e.g., narrower lanes, shoulder drops, etc.) to features on the bridge’s roadway surface like expansion joints. In many cases, bridges are currently being addressed for reconstruction or rehabilitation, and that presents an opportunity to retrofit the structure with bicycle accommodations. One (1) approach for improving bicycle compatibility across bridges is to re-allocate available roadway space to enable the installation of a bicycle lane or shoulder facility. If space is constrained on the roadway, the addition of a shared use path or the widening of existing sidewalk to accommodate bi-directional bicycle and pedestrian travel should be investigated. For both approaches, it is important to include applicable signing and striping, to thoughtfully address locations where transitions will occur between facilities (e.g., markings shoulder to bicycle lanes, ramps to accommodate on-road to off-road travel, etc.), and to provide lighting, where needed, to improve visibility for roadway users.



Cyclist on the Roberto Clemente Bridge

Based on AASHTO guidelines, if bicyclists will be operating directly adjacent to railings or barriers, these elements are recommended to be a minimum height of 3.5'. Taller railings are recommended for consideration in locations where there are major changes in vertical or horizontal roadway elements (e.g., sharp curves, steep hills, etc.). Local emphasis relative to bicycles on bridges has been focused on bicycle friendly accommodations including compatible bridge scuppers/grates and expansion joint design, and bicycle lane striping in the vicinity of ramps (e.g., Birmingham Bridge).

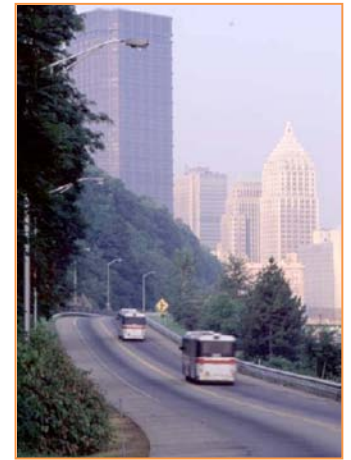
Bikes on Busways

Busways are exclusive rapid transit rights-of-way provided for bus travel. Busways in Allegheny County and other urbanized areas are designed to be higher speed and high capacity public transportation facilities and are not designed for bicycle traffic. Port Authority's three busways link outlying areas in the South, West and East to downtown Pittsburgh and Oakland.



In-line skater on Baldwin Borough Trail (CSX)

Another arrangement, with active railroad lines and bicyclists in close proximity, is rails-with-trails. This approach features the development of a bicycle and pedestrian shared use path adjacent to an active railroad line or rapid transit line. The *AASHTO Guide for the Development of Bicycle Facilities* and the *Rails-to-Trails Conservancy*^{xxii} both provide guidance and lessons learned on the practice of rails-with-trails. Pittsburgh has two examples of active freight railroad lines with adjacent trails. The North Shore and Millvale trails are located along the Norfolk Southern's Conemaugh Line. The Station Square, South Side and Baldwin Borough trails are located next to CSX Transportation's Pittsburgh Subdivision.



East Busway

Although some bicyclists desire to use the busways as commuter routes, Port Authority policy does not allow bicycles on busways due to design, safety liability and operational issues. Additionally, there is not sufficient right-of-way in the busway corridors to allow placement of bicycle lanes adjacent to any of the three busways. However, establishment of bike lanes on streets near the busways can be explored. As more buses become equipped with bicycle racks, it becomes increasingly feasible for bicyclists to take advantage of the rapid service offered by buses on busways to get to downtown Pittsburgh, Oakland and other destinations in Allegheny County.

Lighting on Road for Bicycles

On roadways, street lights are mostly provided to assist motorists. These lights, which are typically set 25' to 30' above the road, may provide adequate illumination for motorists to see the roadway, but for bicyclists, typical street lights may not provide the best visibility. *BIKESAFE*^{xxiii}, a publication by FHWA, provides strategies to improve lighting on road for bicyclists.

Bike Lanes and On-Street Parking (Parallel and Angled)

Bicycle lanes are recommended to be a minimum of 5' wide, which is the preferred operating width of cyclists as noted in the AASHTO Bicycle Guide. In the presence of on-street parking, bicycle lanes are recommended to be 6' to 7' wide, with the wider option used in locations with a high rate of parking turnover. The additional width is recommended to allow bicycles room to maneuver and respond to motorists behavior during parking or while parked.

In the case of parallel parking, a common hazard for bicyclists is being ‘doored’. Dooring occurs when a motorist opens the door of their parked vehicle and strikes a traveling bicyclist. Dooring can seriously injure a bicyclist. Dooring incidents can potentially be reduced by installing striping and pavement markings, such as the bicycle lane symbol further to the left in a lane and using parking stall ‘T’ stripe that extends into the bicycle lane.



Bicycle Lanes adjacent to parallel parking in New York City
Photo: nycbikemaps.com

Angled parking can pose a different kind of hazard to bicyclists. It is not recommended that bicycle lanes be striped adjacent to front-in angled parking due the restricted visibility of motorists as they back out of spaces. An approach that can be utilized in locations where proposed bicycle lanes are adjacent to angled parking is back-in angled parking. This approach leads to improved sight distances and access to rear of vehicles outside of travel lanes.^{xxiv}

When bicycle lanes are proposed for locations with on-street parking, it is recommended that they be provided at a width of 6’ to 7’. In locations where a bicycle facility is proposed adjacent to angled parking, a study exploring back-in angled parking should be performed.

Bicyclists using Stop Signs as Yield Signs

Bicycling requires physical efforts and most bicyclists prefer to maintain a level of momentum while traveling. Routes where bicyclists have to continually stop and start again can serve as an impediment to bicycle travel. To this end, there has been recent interest in a legal change that would allow bicyclists to treat stop signs as yield signs under specific conditions. This would mean that cyclists would still have to slowly approach the stop sign, but if the intersection is clear of vehicles, the bicyclist could proceed without stopping. A law to this effect has been legal in Idaho^{xxv} since the 1980’s and has recently been considered in Oregon and California.

The PA Vehicle Code does not allow bicyclists to treat stop signs as yield signs. It is recommended that Allegheny County and its municipalities monitor this law and studies of its results on bicycle travel and crashes. There is merit to this proposal given the travel preferences of bicyclists, but it will likely have impacts on vehicle operations so it should be continued to be studied. Topography in Allegheny County should be a prime consideration when deciding if this type of change could be safe for bikes and motorized vehicle in this area.

Programs



B-Cycle Automated Rental Kiosk in Denver, Colorado
Photo: votewithyourfeetchicago.blogspot.com

Bike Sharing and Rental Stations

Programs to support bicycle sharing and rentals have been growing both nationally and internationally. Examples of popular programs such as the ‘Velib’ in Paris, France, which has deployed 20,000 bicycles and over 1,600 stations throughout the city. Another is the SmartBike DC (www.smartbikedc.com) program^{xxvi} in Washington D.C., a public-private collaboration between Clear Channel Outdoor and the District Department of Transportation with over approximately 100 bikes at ten (10) stations. Bike rental stations can be either full-service or self-service, with many recent programs using a self-service approach. Bicycles are placed in locked racks which release a bike once the rental payment is made. The bicycle can then be returned at the original location or at other stations at

different locations. The bike rental station can work well with transit to provide a non-motorized connection at the end of a transit trip.

Bike sharing and bike rental programs should be considered for downtown Pittsburgh, Oakland, and other town center in Allegheny County, especially those near existing bicycle and shared use facilities. There are bicycle rental facilities at the end of the Eliza Furnace Trail (across from the First Avenue LRT Station) and at Station Square along the South Side Trail. Also of note, The Friends of the Riverfront currently have a bicycle loan program on some of their trails near downtown Pittsburgh.

Cyclovía

Cyclovías are events where a street or a set of streets are shut down to motorized traffic for a certain period of time and are made available for non-motorized use. Cyclovía, or *ciclovía* in Spanish, started in Columbia^{xxvii} where streets were closed to cars and opened for bicyclists, pedestrians, skaters, and various other individuals. The Cyclovía becomes a celebrated event and brings diverse members of the community out to interact around active living. This commonly occurs on Sunday afternoons, but many options could be explored.



Pittsburgh BikeFest Poster
Source: BikePGH

The cyclovía movement has come to North America and several locations now host cyclovías, sometimes under different names such as 'Sunday Streets' (San Francisco), 'Sunday Parkways' (Portland, Oregon, and Summer Streets ('New York, New York)). These events provide car free space for people to experiment with bicycle tripmaking, take recreational rides, and have space where parents may feel comfortable having children and teenagers learn how to ride on-street.



Car Free Fridays Poster
Source: BikePGH

As partners in the **ACTIVEALLEGHENY** look to encourage more bicycling, a program of cyclovía events could be scheduled to provide opportunities and support for bicycle riders, especially casual riders who are looking to become more comfortable on bikes. These events could also be coordinated with Car Free Fridays and the annual BikeFest that are sponsored by BikePGH. Each month, Car Free Fridays highlights a different neighborhood or municipality in our region to promote active transportation. It's a great chance to learn about the neighborhood, check out the business district, and figure out how to travel car free.

A strategic program of rotating Cyclovía opportunities each Sunday, spring through fall, (especially on scenic routes such as Grandview Avenue in Mt. Washington, Bigelow Boulevard from Downtown to Oakland, etc.), with each road closed one Sunday afternoon per year, could become a major amenity/attraction for our region.

Bicycle to Work Month

May is bicycle to work month^{xxviii}, and the annual bike to work day is usually scheduled for the third Thursday of the month. Bike to work month provides a tremendous opportunity to focus efforts and promote bicycle commuting both due to the extra attention given to bicycling at this time of year and the support that bicyclists have as more people ride to work. SPC, BikePGH, and other local groups participate in and promote bike to work week in Allegheny County. This activity helps to raise awareness which ultimately helps to increase safety for all roadway users.

Bicycle Ambassador

The Bicycle Ambassador Program is a method of providing person-to-person outreach regarding bicycle travel to a community. Comprised of experienced and trained bicyclists, the ambassadors perform educational and encouragement activities to assist with route planning for bicycle trips. This increases knowledge of rules-of-the-road for newer bicycle commuters and motorists and assists in gathering feedback for needed bicycle facility improvements. In some cases, bicycle ambassadors are paid staff, but more often they are volunteers who want to be a resource for bicyclists in their community, especially to those who are newly exploring bicycle travel.

Cities like Chicago^{xxix}, Philadelphia^{xxx}, and Minneapolis/St. Paul^{xxxi} have successful bicycle ambassador programs. The ambassadors attend community events, reach out to schools, and are frequently just present in neighborhoods or high bicycle traffic locations as an available resource. Chicago's program also has a junior ambassador element where teenagers are trained through a 10-week course. Those individuals are then able to provide outreach to fellow teenagers and other young riders about recommended bicycle travel behavior.



*Bicycle Ambassadors in Philadelphia
Photo: Bicycle Coalition of Philadelphia*

A bicycle ambassador program for Allegheny County can be explored. In cooperation with local municipalities and organization like BikePGH, the County, the City, and others, ambassador teams could provide assistance in various locations or employment centers. This could increase the amount of outreach being provided to both adults and children. The Bike Pool Program, offered through the CommuteInfo Program and BikePGH, offers a Bike Mentor Program for new bicycle riders.

Bicycle Commuter Incentives

There are approaches to incentivize bicycle commuting that are used on a national level as well as by many states. The Internal Revenue Service (IRS) allows a monthly tax credit of up to \$20 that can be applied to the maintenance and purchase of bicycle equipment. Many states offer Guaranteed Ride Home Programs (GRH) through commuter organizations such as Transportation Management Associations. The GRH assists bicyclists, as well as other commuters without access to a personal motor vehicle, with a ride home in response to emergencies. In our region, SPC provides this service. Employers can provide incentives for bicycling commuting as well. Such

incentives are: pre-tax deductions for bicycle commuting expenses, parking cash-out to provide bicyclists a cash equivalent for not using subsidized parking, and shower and changing facilities at employment locations.

Bethesda's 2009 Bicycle Commuter Spirit Awards

In conjunction with the regional Bike to Work Day to be held Friday, May 15, 2009, Bethesda Transportation Solutions is sponsoring the **Bicycle Commuter Spirit Awards**. The awards recognize downtown Bethesda employees who are dedicated to biking to work.

Bike to Work Day 2009

We need your help in finding two commuters devoted to two-wheeled bicycle transportation. The first award is for the **Most Committed Bicycle Commuter**, an employee who bikes to work on a consistent basis. The second award is for the **Longest Distance Commuted by Bicycle** – for that dedicated bike commuter who goes the distance!

Nominate a co-worker or yourself. The deadline for nominations is **Friday, April 17, 2009**. Awards will be distributed during Bethesda's Bike to Work Day pit stop event on Friday, May 15, 2009 between 6:30-8:30am at Reed Street (corner of Woodmont and Bethesda Avenues).

It is recommended that local municipalities, Allegheny County, and other agencies explore the possibility of implementing bicycle commute incentives in places of employment. As a first step, these incentives would provide the foundation for encouraging the use of bicycle commute incentives at private employment locations. Promotion of existing programs, such as the GRH (guaranteed ride home), can help those interested in commuting by bicycle understand the support and incentive system available to them. In Western Pennsylvania, SPC has a ridesharing program, CommuterInfo (www.commuterinfo.org), for employers which offer rides home for cyclists under certain conditions.

Bike Friendly Employer

BikePGH's Bike Friendly Employer program works with local employers to help them assess how well they are meeting the needs of their bicycling employees. By demonstrating a supportive work culture and securing facilities that support the decision to bike to work, employers play a crucial role in advancing bicycling as a desirable mode of transportation.

To get started employers are provided with a questionnaire that informs them about the elements and resources that should be in place. The questionnaire is also used to evaluate how well the employer is doing. BikePGH follows each completed questionnaire up with a phone call to discuss whether or not the organization is meeting the basic bike friendly requirements. Following the conversation BikePGH provides the employer with an evaluation known as a Bicycle Action Plan. BikePGH then directly supports the organizations in meeting their goals.



BikePGH's Bike Friendly Employer Logo
Source: BikePGH

Employers are encouraged to apply to the League of American Bicyclist for recognition nationally as a *Bike Friendly Business*. By encouraging local employers to achieve this recognition, it sends a strong message about the priorities of the region. By empowering businesses to create a bike friendly culture at the workplace, the Bike Friendly Employer program supports organizations in their ability to have a positive impact on their employees and directly address the quality of life in the region.

2.5 EDUCATION, ENFORCEMENT & PUBLIC AWARENESS

Education of bicyclists and motorists, and enforcement of traffic laws and statues are important to supporting travel on a bicycle network. To properly plan for future growth of bicycle use, it is key to implement educational programs that encourage lawful and recommended practices among bicyclists and motorists. When educating a community, it is important to dispel myths, encourage behavior that follows the rules of the road, and enhance awareness. By utilizing the resources of the local police, schools, and libraries, education programs have the potential to reach a broader audience and cross-section of the community. In addition, Pennsylvania has the benefit of having an official Bicycle Driver's Manual.

The following four (4) primary groups should be educated about bicycle safety and awareness:

1. Young bicyclists
2. Parents of young bicyclists
3. Adult bicyclists
4. Motorists



Educational materials regarding recommended bicycle travel practices and behavior can be accessed at the following locations:

A Comprehensive Commuter Bicycle and Pedestrian Transportation Plan for Allegheny County

- BikePGH: <http://bike-pgh.org/>
- PennDOT, PACommutes: <http://www.pacommutes.com/biking/>
- USDOT, FHWA: http://safety.fhwa.dot.gov/ped_bike/ and <http://www.bicyclinginfo.org>
- Association of Pedestrian and Bicycle Professionals: <http://www.apbp.org>
- AAA: <http://www.aaacentral.com/community/safety/index.jsp>

The key to encouraging a safe and well-traveled transportation system is an enforcement program for traffic regulations applied to all roadway users: motorists, bicyclists, and pedestrians. Allegheny County and its municipalities can reduce poor travel behavior and encourage beneficial travel habits through enforcement. This process includes review of current local ordinances and traffic regulations to identify elements that may unnecessarily affect certain roadway users, such as bicyclists. As bicycle lanes and other designated bicycle facilities are installed, it is recommended that local ordinances and regulations be developed, or revised, to clarify items such as: application of vehicle laws to bicyclists, permitted movements on and across bicycle facilities (e.g. permitted motor vehicle movements across bicycle lanes), bicycling on sidewalks, and bicycle parking requirements.

Possible vehicle code references include the California Vehicle Code, Division 11, Chapter 1 (<http://www.dmv.ca.gov/pubs/vctop/vc/vctoc.htm>), the Pennsylvania Consolidated Statutes, Title 75, Chapter 35 (<http://www.dot.state.pa.us/bike/web/bikelaws.htm>) and the City of Cambridge, MA Traffic regulations Article XII (http://www.cambridgema.gov/cdd/et/bike/bike_reg.html). In addition, a review of enforcement regulations and practices may assist in identifying opportunities to partner with community, county, or state organizations to inform users about safe bicycle travel behavior, such as the use of helmets by bicyclists. Outreach and promotion through community channels and events is a critical piece in reminding all roadway users of existing laws and recommended travel practices.

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CHAPTER 3. *Walk and Roll Allegheny (ADA)*

ALLEGHENY PLACES identified key challenges to increasing pedestrian travel as a mode share. Some of those challenges included a “lack of continuous sidewalk network in new developments” and incorporation of pedestrian facilities into roadway projects. **ACTIVE ALLEGHENY** details the deficiencies and constraints for pedestrians in Allegheny County and offers solutions from engineering to education.

3.1 Walking & Rolling in Allegheny

Pedestrians, and wheelchair users, have identified desired access to destinations in the County. Although some sidewalks and pedestrian accommodations are available in high density retail areas and neighborhoods, in many cases there is a gap in the network. As a result, pedestrian trips are converted to vehicle trips to access destinations which may only be a half-mile long. PennDOT’s Design Manual^{xxxii} states “pedestrians are a part of every roadway environment and attention must be paid to their presence in urban as well as rural areas.”



Pedestrians, City of Pittsburgh

Desired Access

Through discussions with stakeholders and the public and analysis of the online survey results, pedestrian access and/or improvements are desired in nearly every municipality in Allegheny County. Destinations listed most frequently included access to schools, bus stops, hospitals, and commercial districts. Specific destinations identified for desired access are listed in **Table 3-1**.

Table 3-1. Desired Access for Pedestrians

To	From
Downtown	• Mt. Washington
Frank Curto Park (Bigelow Boulevard)	• Oakland
Chateau Trail (North Side)	• McKees Rocks
Hazelwood Business District	• Hazelwood Avenue • Eliza Furnace Trail
Bethel Park Giant Eagle	• Bethel Village T Station
Great Allegheny Passage	• East End Communities
West Busway Carnegie Station	• Panhandle Trail (Collier)
North Shore	• Mt. Washington
Millvale Riverfront Park	• Millvale
Duck Hollow Trail (Hazelwood)	• Frick Park • Eliza Furnace Trail
Schools in Mt. Lebanon	• Mt. Lebanon (specifically Washington Road)
South Park	• Library T Station
Pittsburgh Zoo & Aquarium	• Baker Street and vicinity (parallel to Route 8)
Route 48 (Monroeville)	• Penn Center Boulevard (Monroeville)
Oakland	• Polish Hill
Ben Avon & Avalon Parks	• Kilbuck

Identified Deficiencies

A review and analysis of existing conditions, survey results, pedestrian crashes and public feedback was performed to identify locations for potential pedestrian facility improvements. Although there are many locations throughout Allegheny County that could benefit from the installation of sidewalk, curb ramps, and crosswalks, 18 roadways and 19 intersections were identified in this plan. **Table 3-2** lists the corridors targeted for pedestrian enhancements and **Table 3-3** lists the intersections. Improvements at these locations can serve as a model for other locations throughout the County as they are identified in the future. The crash summary and pedestrian crash map are contained in **Appendix L**, which is included in a separate document.

Table 3-2. Pedestrian Corridors for Facility Improvements

Roadway	Municipality
Ardmore Boulevard	Forest Hills
Bates Street	Pittsburgh
Beadling Road	Mt. Lebanon
Beaver Street	Glen Osborne
Belmar Place	Swissvale
Bigelow Boulevard	Pittsburgh
Braddock Avenue	Braddock
Campbells Run Road	Robinson
Chartiers Avenue	McKees Rocks
Grove Road	Castle Shannon
Route 51/Island Avenue	McKees Rocks
Lincoln Way	White Oak
Mayview Road	Upper St. Clair
McLaughlin Run Road	Bridgeville and Upper St. Clair
Park Manor Boulevard	Robinson
River Road	Haysville
Robinson Town Center Boulevard	Robinson
Route 19 Truck/Washington Road	Mt. Lebanon
Route 19 Truck/West Liberty Avenue	Dormont
Business Route 22	Monroeville
Route 50	Heidelberg, Carnegie, Collier, and Bridgeville
Route 837	Clairton
Route 837	Duquesne
Steubenville Pike	Robinson

Table 3-3. Intersections for Facility Improvements

Intersection	Municipality
Thorn Run Road & Route 51	Coraopolis
Beadling Road & Washington Road	Mt. Lebanon
Negley Run Boulevard & Washington Boulevard	Pittsburgh (East Liberty)
Foster Street & 40 th Street	Pittsburgh (Lawrenceville)
Route 19 Truck & Brookline Boulevard	Dormont

Intersection	Municipality
Cochran Road & Washington Road/Route 19	Mt. Lebanon
Bethel Church Road & Broughton Road	Bethel Park
Smithfield Street & Carson Street	Pittsburgh (South Side)
Campbells Run Road & Steubenville Pike	Robinson
Route 837 & Amity Street	Homestead
Belmar Place & Woodstock Avenue	Rankin
Braddock Avenue & 4 th Street	Braddock
Braddock Avenue & 7 th Street	Braddock
Brinton Avenue & Ridge Avenue	East Pittsburgh
Boulevard of the Allies & Bates Street	Pittsburgh (Oakland)
Boulevard of the Allies & Halket Street	Pittsburgh (Oakland)
S. Bellefield Avenue & 5 th Avenue	Pittsburgh (Oakland)
Ardmore Boulevard & Yost Boulevard	Forest Hills
Castle Shannon Boulevard & Mt. Lebanon Boulevard	Castle Shannon
PJ McArdle Road & Liberty Bridge	Pittsburgh (Mt. Washington)
Route 28 & River Front Drive	Millvale
10 th Street & E. Carson Street (Crash Concentration)	Pittsburgh (South Side)
E. Ohio Street & Cedar Avenue (Crash Concentration)	Pittsburgh (North Side)
18 th Street & E. Carson Street (Crash Concentration)	Pittsburgh (South Side)
Baum Boulevard & Roup Avenue (Crash Concentration)	Pittsburgh (East Liberty)
Cherry Way & Boulevard of the Allies (Crash Concentration)	Pittsburgh (Downtown)
17 th Street & E. Carson Street (Crash Concentration)	Pittsburgh (South Side)
Main Street & 4 th Avenue (Crash Concentration)	Coraopolis
Butler Street & 45 th Street (Crash Concentration)	Pittsburgh (Lawrenceville)
S. Main Street & Wabash Street (Crash Concentration)	Pittsburgh (West End)
E. Ohio Street & Middle Street (Crash Concentration)	Pittsburgh (North Side)

Equal Opportunities

During the course of the ACTIVEALLEGHENY study, the Study Team worked closely with the City of Pittsburgh/Allegheny County Task Force on Disabilities and the Committee for Accessible Transportation (CAT) to incorporate deficiencies and opportunities for older adults and those with ambulatory, visual, hearing, or cognitive impairments. Several deficiencies and opportunities were raised by members of these groups and the general public, in both the online survey and at public meetings, including:

- ♿ ADA upgrades are vital for older neighborhoods and especially where there are concentrations of older adults.
- ♿ Connecting the Panhandle Trail from the trailhead in Collier to the Carnegie Busway Station on West Main Street would provide a safer route for those in electric wheelchairs currently forced to travel on Noblestown Road to reach bus service, due to recent service cuts.
- ♿ Bates Street in Oakland needs sidewalks and ADA compliant curb ramps.
- ♿ Ongoing improved access upgrades to Port Authority facilities (bus and light rail stops and busway, light rail, and incline stations) is a goal throughout the County. Access to Amtrak and Greyhound stations in Downtown Pittsburgh should also be improved to serve those with mobility impairments.
- ♿ Desire for “recharging stations” for electric wheelchairs in the City of Pittsburgh.
- ♿ Audible signals for signalized intersections are desired by those with visual impairments.

- ♿ Increasing the width of curb cuts and ramps to 4' (not including the flares) is crucial as narrow curb ramps present constraints for motorized wheelchairs.
- ♿ Perpendicular curb ramps are preferred by those with visual impairments.
- ♿ Increase availability, reliability, and timeliness of door-to-door ACCESS transit service or reduce the distance between bus stops in areas of concentrated disabled populations (recommendations in the PAAC Transit Development Plan may address these issues).
- ♿ Brownsville Road in Mt. Oliver needs ADA upgrades and improved maintenance.
- ♿ Business Route 22 and Old William Penn Highway in Monroeville need sidewalks and curb ramps to facilitate safe travel for those with visual impairments.
- ♿ Create a model ordinance for municipalities to maintain the sidewalk network, including curb ramps, and pedestrian signals.

Recommendations for ADA compliant improvements are contained in the Pedestrian Facilities Toolbox of this Plan. Additionally, the City of Pittsburgh ACCESS Transportation Board compiled a list of trip origins and destinations for pedestrians with visual impairments. This list, which is available in **Appendix M** (included in a separate document) identifies the need and location for accessible pedestrian signals and detectors in the City of Pittsburgh.

3.2 SYSTEM IMPROVEMENTS

It is recommended that responsible local municipalities and other agencies study the feasibility of pedestrian improvements listed in this plan, including ADA compliant upgrades to existing facilities. Existing roadways and intersections should be evaluated for design compliance with PennDOT and MUTCD guidelines.

Pedestrian Corridors

Pedestrian corridors for improvement should be studied relating to feasibility of sidewalk installation, curb ramp installation, and for crosswalks at any unsignalized intersection, where pedestrian activity indicates a need for their installation. Improvements should focus on sections of the corridors that facilitate pedestrian movements to transit, schools, commercial centers, and parks. Sidewalk and curb ramp installation are typically mid-term (3-5 years) to long-term (5+ years) improvements requiring design prior to construction. Potential constraints associated with sidewalk installation include utility relocation, environmentally sensitive areas requiring permits, stormwater management, and right-of-way acquisition. The corridors in this plan were investigated in the field and the results of those investigations are detailed in **Table 3-4**.

Table 3-4. Field Investigation Findings for Pedestrian Corridors

Roadway	Findings
Ardmore Boulevard (Forest Hills)	• Observed to be an active pedestrian corridor. • Transit stops and shelters exist. • Intermittent sidewalk. Desire lines are present where there is no sidewalk. • Connects southeast suburbs of Forest Hills and Chalfant with the City of Pittsburgh's Wilksburg neighborhood.
Bates Street (Pittsburgh – Oakland)	• Intermittent sidewalk exists. • Southbound sidewalk in disrepair (overgrowth, cracking, and obstacles) and then ends. • Northbound sidewalk passable, but width reduced to 3' due to utility poles.

Roadway	Findings
Beadling Road between Washington Road and Cedar Boulevard (Mt. Lebanon) 	• Sidewalk on one side of roadway. • Crosswalk at Markham Elementary School standard striped (potential improvement: restripe with high visibility crosswalk and overhead school warning signs with flashing beacons). • Safe Route to School investigation potential.
Beaver Street (Glen Osborne)	• Continuous sidewalks exist. • Yield to Pedestrian Channelizing Devices exist at Osborne Elementary School. • Transit route located along roadway. • Crosswalks at Osborne Elementary School are faded (potential improvement: restripe with high visibility crosswalk and overhead school warning signs with flashing beacons). • Safe Route to School investigation potential.
Belmar Place (Swissvale)	• Sidewalk in disrepair (cracking, overgrowth, and obstacles). • Connects to Woodstock Avenue for access to Swissvale East Busway Station and South Braddock Avenue (potential complete street).
Bigelow Boulevard (Pittsburgh – Hill District, Polish Hill, & Oakland)	• Posted speed limit of 35 mph, actual operating speed closer to 55 mph+. Speed control traffic calming measures should be investigated. • Intermittent sidewalks exist. • Sidewalks overgrown, resulting in reduced passable width, in some locations. • Provides a motor vehicle connection between Oakland to Downtown.
Braddock Avenue (Braddock)	• Continuous sidewalks exist in fair condition. • Transit route located along roadway. • Faded crosswalks at bus stops. • Connectivity to Swissvale via Belmar Place and South Braddock Avenue.
Campbells Run Road (Robinson)	• Connects hotels and businesses (Marquis Plaza and Bayer Corporation) to Steubenville Pike (Robinson Town Centre, the Pointe at North Fayette, and the Mall at Robinson). • No existing sidewalk.
Chartiers Avenue (McKees Rocks)	• Sidewalks continuous in fair condition. • Transit route located along roadway. • Connects Route 51 through business district. • Lane striping overlaps crosswalk striping at some intersections.
Grove Road (Castle Shannon)	• Provides connection between Baptist Road (Whitehall) and Route 88 (Castle Shannon). • Caste Village is located at Baptist Road, while a T Station is located at Route 88 (Memorial Hall). • Bicycle and pedestrian activity observed. • No shoulders or sidewalks.
Route 51/Island Avenue (McKees Rocks)	• Transit route located along roadway. • Cars observed parked on sidewalk. • Northbound sidewalk clear width reduced to 3' due to utility poles. • No buffer between sidewalk and roadway. • Faded crosswalks exist.

Roadway	Findings
Lincoln Way (White Oak) 	- No sidewalks or limited sidewalks. - Crosswalks faded. - Pedestrian activity observed. - Pedestrians with visual impairments (walking sticks) observed walking in street. - Children riding bicycles in street without helmets. - Yield to Pedestrian Channelizing Devices exist at locations without crosswalks.
Mayview Road (Upper St. Clair) 	- Provides connection between residential areas and Boyce Mayview Park. - Transit route located along roadway. - Bicycle and pedestrian activity observed. - No existing sidewalks.
McLaughlin Run Road (Bridgeville)	- Sidewalk in disrepair (cracking, overgrowth, and obstacles). - Cars parked on sidewalk. - Sidewalk replacement and maintenance needed.
McLaughlin Run Road (Upper St. Clair) 	5' sidewalk southbound along Recreation Center and park. - Sidewalk reduces to 4' southbound at municipal complex and ends at Route 19 interchange. - Sidewalk exists on north side of roadway through Route 19 Interchange. - Sidewalk provided to Route 19 Southbound; sidewalk not provided to Route 19 Northbound. - Limited number of crosswalks provided.
Park Manor Boulevard (Robinson)	- Connects retail to Montour Trail via Robinson Town Centre Boulevard. - Serves vehicular-oriented retail center. - Transit route located along roadway. - Unofficial park and ride facilities exist along corridor. - No existing sidewalk. - Pedestrian population consists of employees of local businesses and business patrons.
River Road (Haysville)	- One lane roadway with sidewalk northbound. - Cars drive on sidewalk and park on sidewalk. - Sidewalk limited on River Road to ¼ mile distance from Ohio River Boulevard.
Robinson Town Centre Boulevard (Robinson)	- Transit route located along roadway. - Connects to Robinson Town Centre, the Pointe at North Fayette, and the Mall at Robinson. - Serves vehicular-oriented retail center. - No sidewalk. - Pedestrians observed walking in median. - ACTA performed a pedestrian access improvement study for Robinson Town Centre, the Pointe at North Fayette and the Mall at Robinson.
Route 19 Truck/ Washington Road (Mt. Lebanon)	- Continuous sidewalk in good condition. - Standard striped crosswalks at intersections. - Curb ramps provided at most locations. - At unsignalized intersections upgrade crosswalks to high visibility or HAWK (high-intensity activated crosswalk) signals to accommodate level of pedestrian activity. - Washington Road is accessible from the Mt. Lebanon T Station.

Roadway	Findings
Route 19 Truck/ West Liberty Avenue (Dormont)	• Continuous sidewalk in varying condition. • Curb ramps at intermittent intersections. • Faded crosswalks at signalized intersections. • Strong pedestrian activity observed. • Transit route located along roadway.
Business Route 22 (Monroeville)	• Intermittent sidewalks exist. • Transit route located along roadway. • Observed pedestrian activity. • ADA accessibility concerns noted by the Committee for Accessible Transportation.
Route 50 (Heidelberg)	• Transit route located along roadway. • Yield to Pedestrian Channelizing Devices exist at unmarked midblock crossings. • Intermittent sidewalk. • Faded crosswalks at signalized intersections.
Route 837 (Clairton)	• Intermittent sidewalk in disrepair (cracking and overgrowth). • Cars observed parked on sidewalk. • Faded crosswalks at signalized intersections.
Route 837 (Duquesne)	• Sidewalk southbound, Steel Valley Trail northbound. • Transit route located along roadway. • Faded crosswalks at Grant Avenue.
Steubenville Pike / Route 60 (Robinson)	• Transit route located along roadway. • Intermittent sidewalk. • Connects to Robinson Town Centre, the Pointe at North Fayette, and the Mall at Robinson via Park Manor Boulevard

Pedestrian corridors are illustrated on the System Improvements Map – Pedestrian Corridors and Intersections contained in **Appendix N**, which is included in a separate document.

Intersections

Intersections recommended for improvement should be evaluated for upgrades to accommodate all types of pedestrians. Improvements should focus on striping crosswalks or restriping existing crosswalks; providing curb ramps with tactile surfaces and truncated domes; upgrading pedestrian signal heads to countdown signal heads with audible tones or vibrating surfaces; and evaluating signal timing directives and phasing to accommodate MUTCD guidelines. Intersection improvements are typically short-term (less than 3 years) to mid-term (3-5 years), with vehicle traffic analysis performed prior to design to determine impacts to motor vehicle level-of-service. The intersections in this plan, which were identified through the public involvement stakeholder process, were investigated in the field and the results of those investigations are detailed in **Table 3-5**. Intersections identified as having a concentration of crashes were not investigated in the field since they were automatically carried forward and recommended for further study to reduce the incidence of crashes.

Table 3-5. Field Investigation Findings for Intersections

Intersection	Findings
Thorn Run Road & Route 51 (Coraopolis)	• Signalized intersection. • Faded crosswalks. • Several “no pedestrian crossing” signs could cause confusion for pedestrians since there are striped crosswalks. • No curb ramps. • Sidewalk connection not provided to Coraopolis business district. • Sidewalk connection not provided to Sewickley Bridge.
Beadling Road & Washington Road/ Route 19 (Mt. Lebanon)	• Unsignalized intersection. • Recent pedestrian crash. • No crosswalks.
Negley Run Boulevard & Washington Boulevard/ Route 8 (Pittsburgh – East Liberty)	• Signalized intersection. • Proposed trail alignment through intersection desired. • Crosswalk faded. • Push button inaccessible (overgrowth) northbound. • No curb ramps or connecting sidewalk. • Pedestrian signals do not meet MUTCD 2009 guidelines.
Foster Street & 40th Street (Pittsburgh - Lawrenceville)	• Unsignalized intersection. • Wide intersection without refuge islands. • Transit route bus stops located near intersection. • Faded crosswalks. • Intermittent sidewalk does not provide complete connections between sidewalks and bus stops.
Route 19 Truck & Brookline Boulevard (Dormont)	• Signalized intersection. • Faded crosswalks. • Pedestrian activity observed. • Curb ramps are not ADA compliant. • Transit route bus stops located near intersection.
Cochran Road & Washington Road (Mt. Lebanon)	• Signalized intersection. • Faded crosswalks. • Pedestrian activity observed.
Bethel Church Road & Broughton Road (Bethel Park)	• Signalized intersection. • Faded crosswalks. • Curb ramps are not ADA compliant.
Smithfield Street & Carson Street (Pittsburgh – South Side)	• Signalized intersection. • Faded crosswalks. • Pedestrian activity observed. • PAAC identifies this location as a major transit hub in Pittsburgh with the Station Square T Station and lower station of the Monongahela Incline. All routes operating on the South Busway, as well as several local routes (e.g., #51 Carrick) have stops at this location.

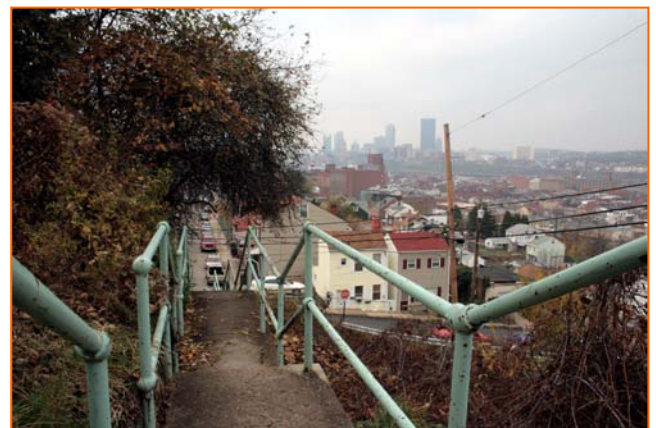
Intersection	Findings
Campbells Run Road & Steubenville Pike/ Route 60 (Robinson)	• Signalized intersection. • Accesses Robinson Town Centre, the Pointe at North Fayette, and the Mall at Robinson. • No crosswalks. • No curb ramps.
State Hwy 837 & Amity Street (Homestead)	• Signalized intersection. • Faded crosswalks. • Curb ramps not ADA compliant. • Pedestrian activity observed. • At-grade railroad crossings with heavy rail activity located near intersection.
Belmar Place & Woodstock Avenue (Rankin)	• Signalized intersection. • Faded crosswalks. • Curb ramps not ADA compliant.
Braddock Avenue & 4th Street (Braddock)	• Signalized intersection. • Faded crosswalks. • Curb ramps not ADA compliant. • Transit route bus stops located near intersection.
Braddock Avenue & 7th Street (Braddock)	• Unsignalized intersection. • Faded crosswalks on 7th Street. • No crosswalk on Braddock Avenue. • Transit route bus stops located near intersection serving Community College of Allegheny County facility.
Brinton Avenue & Ridge Avenue (East Pittsburgh)	• Unsignalized intersection. • Standard striped crosswalks with no connecting sidewalk. • No curb ramps. • Transit route bus stops located near intersection.
Boulevard of the Allies & Bates Street (Pittsburgh - Oakland)	• Signalized intersection. • Faded crosswalks. • Curb ramps are not ADA compliant. • Pedestrian signals do not meet MUTCD 2009 guidelines. • Transit route bus stops located near intersection.
Boulevard of the Allies & Halket Street (Pittsburgh - Oakland)	• Signalized intersection. • Faded crosswalks. • Curb ramps are not ADA compliant. • Wide intersection without refuge islands. • Transit route bus stops located near intersection serving Magee-Womens Hospital of UPMC.
S. Bellefield Avenue & 5th Avenue (Pittsburgh - Oakland)	• Signalized intersection. • Faded crosswalks. • Pedestrian signals do not meet MUTCD 2009 guidelines. • Pedestrian access to University of Pittsburgh.

Intersection	Findings
Ardmore Boulevard & Yost Boulevard (Forest Hills)	• Signalized intersection. • Faded crosswalks. • Intermittent curb ramps not ADA compliant. • Transit route bus stop located near intersection.
Castle Shannon Boulevard & Mt. Lebanon Boulevard (Castle Shannon & Mt. Lebanon)	• Signalized intersection. • Faded crosswalks. • No connecting sidewalk or curb ramp northbound. • Located near major light rail station, Park and Ride lot, and commercial center.
PJ McArdle Road & Liberty Bridge (Pittsburgh – Mt. Washington)	• Signalized intersection. • Signed “No Pedestrian Crossing.” • Connecting sidewalk on both sides of intersection. • No curb ramps. • Provides access to Downtown via Liberty Bridge sidewalk. • Access to Station Square area via “city steps” in very poor condition. • Guiderail is a barrier to pedestrian access.
E. Ohio Street & River Front Drive (Millvale)	• Unsignalized intersection. • Faded crosswalk. • No curb ramps. • Access to Millvale Waterfront Park and Three Rivers Heritage Trail. • No rail crossing. • Pedestrian Warning Signs exist. • Pedestrian access improvements currently under construction at this location as part of Route 28 corridor improvement project.
Steubenville Pike/ Route 60 & Park Manor Boulevard (Robinson)	• Gateway into vehicular-oriented retail center. • Pedestrian crossing of Steubenville Pike/Route 60 very difficult due to amount and speed of traffic. • Motorists often do not yield to pedestrians in crosswalk despite signage.

Recommended intersections for improvement are also illustrated on the System Improvements Map – Pedestrian Corridors and Intersections contained in **Appendix N**.

Public Steps

According to Bob Regan in his book “Steps of Pittsburgh,” the City of Pittsburgh has 912 sets of public stairs. This isn’t inclusive of Allegheny County, which has numerous sets of municipality-owned steps outside of the City of Pittsburgh. In the online survey for **ACTIVEALLEGHENY**, residents of Allegheny County were asked how frequently they utilize public steps (including steps at transit stops and stations). Twenty-five percent (25%) of respondents indicated they sometimes use them, while the remaining 75% indicated almost never or never using public steps/stairs. South Side Slopes were noted as the public steps used most frequently by respondents. The survey also requested the location of public steps/stairs that may need



Public Steps, South Side Slopes, Pittsburgh
Photo: Kevin Smay

maintenance. Public steps in the South Side Slopes, Troy Hill, Mt. Washington, and the Negley Avenue Bus Station were listed most frequently for needed maintenance.



*Public Steps and Pedestrian, South Side Slopes
Photo: Lynn Heckman*

To improve the public steps in Allegheny County, consideration must be given to accessibility. The Revised Draft Guidelines for Accessible Public Rights-of-Way (Draft PROWAG), released by the United States Access Board in November 2005, recommends that new construction or modifications to existing structures be met with equal opportunity with regards to access and mobility for those with disabilities. While these are currently guidelines, it is likely that they will eventually become mandates when the guidelines are formally adopted. Accessibility is a concern, however, many of the public step locations within Allegheny County have significant physical, site, and right-of-way constraints. If and when public steps are rehabilitated, an accessibility assessment should be conducted to determine if alternative accommodations are technically feasible. As with all accessibility design issues, appropriate documentation of the accessibility review is required, including the reasons for any determination that an ADA alternative is technically infeasible. In situations where a technically infeasible determination is made, the accessibility assessment may include a discussion of other ADA-compliant parallel or alternative routes, however, those routes are not necessarily justification for not providing ADA-compliant accommodations.

3.3 PEDESTRIAN FACILITIES TOOLBOX

Types of Pedestrians

The transportation network must accommodate a full spectrum of pedestrians. The American Association of State Highway and Transportation Officials *Guide for the Planning, Design and Operation of Pedestrian Facilities*^{xxxiii} identifies a variety of pedestrians including:

-  ↑ Children
- ↑ Adults (age 19-65)
- ↑ Older Adults (age 65+)
- ↑ Pedestrians with Ambulatory Impairments (wheelchairs, scooters, canes, prosthetics, etc.)
- ↑ Pedestrians with Hearing Impairments
- ↑ Pedestrians with Vision Impairments (white cane users, dog guide users, etc.)
- ↑ Pedestrians with Cognitive Impairments



Penn Avenue, City of Pittsburgh

It is important to recognize the different types of pedestrians when planning and designing pedestrian facilities since walking speed, spatial needs, and mobility issues are unique to each type of pedestrian. For example, pedestrians with ambulatory impairments require sufficient width to maneuver their devices, plus level and hard surfaces.

Sidewalks

PennDOT's Design Manual 2, Chapter 6, Section 6.6 establishes the criteria for sidewalks in PennDOT projects. The Department may participate when criteria are met and where communities support sidewalks and participate in cost sharing, ownership, and future maintenance. Municipal buy-in to add sidewalks is sometimes a challenge. Municipalities should be educated on Smart Transportation and the benefits of sidewalks.



Sidewalk activity in the Strip District

Locations identified for sidewalk construction or maintenance in this plan in Section 3-2 should follow guidelines set forth in PennDOT's Design Manual Part 2, Chapter 6; the MUTCD; and *AASHTO Guide for the Planning, Design and Operation of Pedestrian Facilities*. This includes constructing new sidewalk at a minimum of 5' to provide for a wheelchair to u-turn or pass another wheelchair. If 5' cannot be accommodated, then a minimum of 4' may be acceptable if passing spaces of 5' are provided every 200'.

In a central business district or Village Center, it is recommended that sidewalk widths be increased to 6' - 10' per AASHTO Design Guidelines and the Smart Transportation Guidebook. Robinson Town Centre is an example of a location where providing such

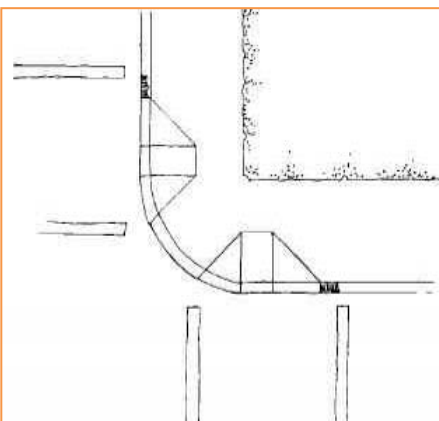
sidewalk widths would promote the use of the area as a village center.

PennDOT sidewalk design criteria further states that sidewalks must be physically separated from vehicles through the utilization of a buffer, curb, or other barrier depending on roadway conditions and pedestrian activity levels. Grade should not exceed 5% on a pedestrian access route and slope should not exceed 2% to accommodate different types of pedestrians. On bridges, sidewalk widths should be equivalent to the approaching sidewalk width. Eight feet (8') is desirable for the clear width of sidewalk on bridges per AASHTO guidelines, however 4' is the acceptable minimum clear width if 8' is not feasible. Most of the bridges within Allegheny County have narrow lane widths as compared to current standards. Motor vehicle travelers rarely have the choice to take alternate routes. For pedestrians and bicyclists, safe accommodations on bridges are critical and should be provided via wide sidewalks whenever possible.



University Boulevard, Moon Township

Curb Ramps



Two Perpendicular Curb Ramps

Source: FHWA

ADA Law, 28 CFR Part 35.151(e) maintains that new construction or alterations to streets, roads, or highways must have curb ramps at any intersection which has curbs or barriers to entry from a street level pedestrian walkway. PennDOT Design Manual, Part 2, Chapter 6, Sections 6.2, 6.3 and 6.9 further states that striping of a crosswalk (new installation) requires an upgrade and/or installation of curb ramps at the crossing. However, restriping does not constitute an alteration. A suggested revision of this policy will be evaluated in Section 3-4 of this plan. AASHTO guidelines specify that the grade of a curb ramp not exceed 8.33% and the cross slope be no greater than 2%. A curb ramp should be a minimum 4' wide, not including the flares. Detectable warning surfaces (e.g., truncated domes) must be provided per PennDOT and AASHTO Design Guidelines.

There are three (3) types of curb ramps: perpendicular, parallel, and diagonal.

However, diagonal ramps are not recommended since they may cause additional roadway exposure for pedestrians or aim a visually impaired pedestrian away from the crosswalk and into the intersection. PennDOT Design Manual, Part 2, Chapter 6, Section 6.9 provides additional guidance on adaptations with regards to curb ramp types which provides for various configurations in limited space locations.

Intersection Facilities

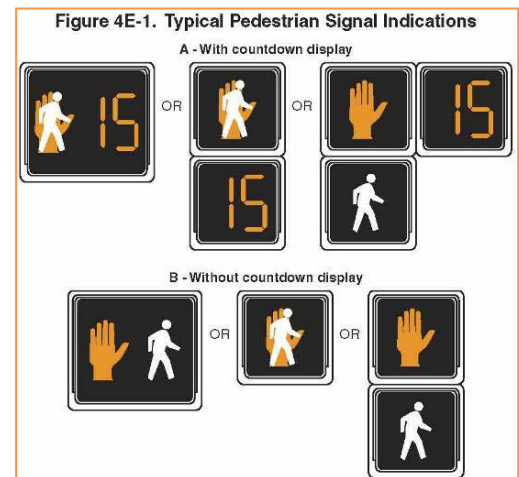
The Smart Transportation Guidebook, Chapter 7, Section 7.6, discusses the importance of designing intersections that balance the needs and features desired by different intersection users including motor vehicles, bicyclists, and pedestrians. Pedestrians desire short crossing distances, adequate time to cross the intersection through proper signal timing, and marked crosswalks. Additionally, movements at the intersection are facilitated by providing sidewalks approaching the intersection and curb ramps.

Traffic Control Signals

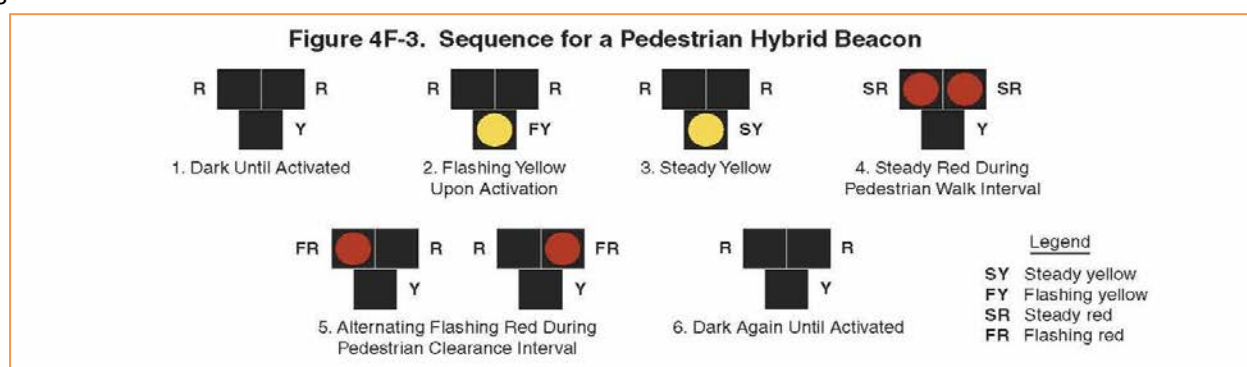
MUTCD, Chapter 4, Section 4D.03 provides standards and guidance for the design and operation of traffic controls for pedestrians. It states specifically, “where pedestrian movements regularly occur, pedestrians should be provided with sufficient time to cross the roadway by adjusting the traffic control signal operation and timing to provide sufficient crossing time every cycle or by providing pedestrian detectors.” MUTCD guidelines mandate a walking speed of 4’ per second be used to determine pedestrian clearance times. Where slower pedestrian traffic is anticipated, as is the case with older adults and those with ambulatory impairments, a walking speed of 3.5’ per second should be considered. PennDOT Publication 149, *Traffic Signal Design Handbook* states that a walking speed of 3.5’ per second should typically be used.

Revisions to the 2009 Edition of the MUTCD include the use of pedestrian change interval countdown displays on pedestrian signal heads used at crosswalks where the pedestrian change interval is more than 7 seconds. If less than 7 seconds, the countdown display should still be considered to inform pedestrians of remaining time in the change interval.

Accessible pedestrian signals and detectors providing audible tones, vibration, or speech should be considered for intersections where pedestrians with visual disabilities need to cross. MUTCD guidelines state that an engineering study should be performed in these cases to determine the needs of pedestrians to be accommodated at the intersection, and the need for accessible pedestrian signals. Several factors should be considered prior to the installation of accessible signals including pedestrian volumes, traffic volumes, the complexity of the intersection geometry, and the complexity of traffic signal phasing. MUTCD cites the importance of involving local organizations who provide support services for pedestrians with hearing and/or vision impairments. At unsignalized crosswalks, Pedestrian Hybrid Beacons are an option for interrupting traffic flow to provide pedestrians with priority movement. MUTCD 2009 Edition provides warrants for their installation. The following graphic illustrates how the hybrid sequence functions at a midblock crossing.



Pedestrian Signal Indications
Source: MUTCD 2009 Edition



Pedestrian Hybrid Beacon Sequence
Source: MUTCD 2009 Edition

Crosswalks

The Smart Transportation Guidebook, Chapter 8, Section 8.1.4 states that “crosswalks should be present on all legs of a signalized intersection, unless hazardous conditions make one or two legs unsuitable for installation.” MUTCD and AASHTO agree that crosswalks provide a pedestrian guidance and right-of-way when crossing roadways. Crosswalk design and striping can range from transverse (standard) lines to colored or textured treatments. This toolbox recommends crosswalks with longitudinal (ladder) striping for the following reasons:

- 1) Due to the parallel line positioning, fading from vehicle wheel paths is not as prevalent.
- 2) Motorists can see a longitudinal striped crosswalk from a further distance than a diagonal or standard striped crosswalk.

Crosswalks, regardless of type, should include curb ramps to facilitate pedestrian movements to and from the push button assemblies and to the connecting sidewalk. Crosswalks should also be high visibility and striping should be thermoplastic for longevity.



Rendering of a Midblock Crosswalk

Midblock crosswalks should be considered for unsignalized intersections if pedestrian demand and activity supports their installation. The placement of a crosswalk is dependent upon the context of the area, however, a midblock crosswalk should have adequate sight distance and it should not be located less than 200' from a signalized intersection. Pedestrians crossing less than 200' from a signalized intersection should be encouraged or channeled to use the signal crossing, if needed. Pedestrian warning signs and longitudinal striped crosswalk with ADA compliant curb ramps and truncated domes should be installed at midblock crosswalks at uncontrolled locations.

Signs, Direction and Reflection

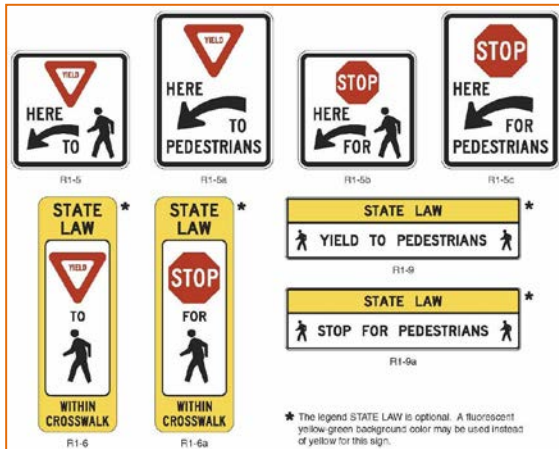
Signs not only provide direction to pedestrians, but also serve to alert motorists to the presence of pedestrians. At signalized intersections, pedestrian signs provide direction on push button operation, preferred crossing location, and pedestrian signal head use. Regulatory signs, such as “No Turn on Red” (MUTCD R10-11a) and “Turning Vehicles Yield to Pedestrians” (R10-15) at intersections alert motorists to the presence of pedestrians and provide direction on priority movements at the intersection.



* A fluorescent yellow-green background color may be used instead of yellow for this sign.

From Left to Right: Pedestrian Warning Sign, Handicap Warning Sign, Turning Vehicles Yield to Pedestrians for use at Signalized Intersections

Source: MUTCD 2009 Edition



Pedestrian Channelizing Devices for MidBlock Crosswalks
Source: MUTCD 2009 Edition

At unsignalized intersections, pedestrian warning signs and/or bollards alert motorists to yield or stop for pedestrians in the crosswalk. The Yield signs are used in the state of Pennsylvania, rather than the Stop signs. Yield to Pedestrian Channelizing Devices should be placed on the approach to a midblock crosswalk to increase motorist awareness of the crossing and to slow vehicular traffic (traffic calming effect).

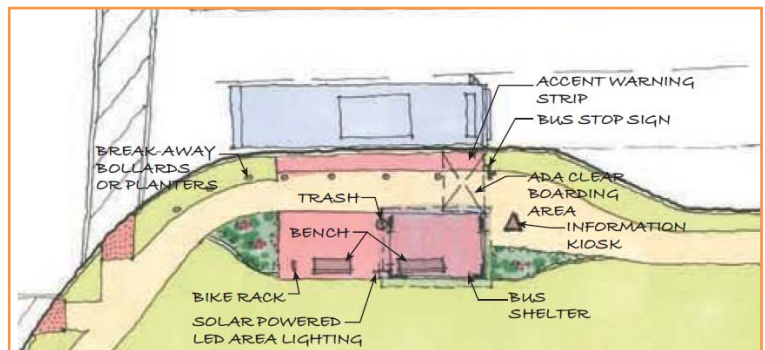
These pedestrian bollards were observed throughout Allegheny County, however they had been moved to the sidewalk or curb side at many of those locations. This observed displacement of the bollards should be monitored by municipalities who utilize the channelizing devices at unsignalized crossing locations.

Traffic Calming and Streetscapes

Speed control traffic calming measures and streetscapes provide for an improved pedestrian environment by reducing vehicular speeds, increasing pedestrian activity and providing an economic stimulus to the corridor through the use of decorative materials and pedestrian amenities (e.g., street furniture, landscaping, etc.).

Transit Stops and Stations

Transit facilities are an important component of active transportation and transit stops and stations facilitate connectivity and encourage multimodal trips (walking/bus or biking/bus). According to the *Smart Transportation Guidebook*, bus stops should be well-lit, provide safe access, and facilitate mobility for a variety of pedestrians in order to encourage trips by bus. Properly designed passenger waiting shelters provide pedestrians rest and cover from the weather, hence a more desirable and comfortable trip. Transit-Oriented Development (TOD) strategies further enhance the accessibility of transit stops and stations. The ACTA (www.acta-pgh.org/) has developed a prototype bus stop design in an attempt to improve the user experience. Additionally, the *20/20 Transit Vision Study* (developed by the SPC, Port Authority of Allegheny County, and the region's other transit providers) includes a TOD Toolbox with a section on typology for passenger facilities (http://www.portauthority.org/PAAC/Portals/Capital/VisionStudy/images/Toolbox_C.pdf).



Suburban Bus Stop Concept
Source: ACTA

Order of Magnitude Costs

Costs associated with implementing pedestrian facility improvements will vary. Interim improvements (e.g., crosswalk striping and signage) will have less design requirements and will therefore be less expensive than an improvement that would need to go through feasibility assessment and design before obtaining funding for construction (e.g., sidewalk construction). Typical costs are contained in **Table 3-6**.

Table 3-6. Typical Costs for Pedestrian Facilities

Item	Cost Per Unit	Quantity	Cost	Source
Cost Per Lane Mile				
5' Concrete Sidewalk	\$5.19 / SF	26,400 SF	\$137,016.00	2010 Means Sitework and Landscape Cost Data: G 2030 120 1580 Concrete Sidewalk 4" on 4" converted to SF
Crosswalk Striping	\$3.00 / LF	240 LF	\$720.00	12" White Hot Thermoplastic Pavement Markings (ECMS)
Cost Per Item				
Curb Ramp	\$1,500 EA	1 Ramp	\$1,500.00	Walkinginfo.org
Pedestrian Warning Sign	\$25.00 / SF	6.25 SF	\$156.25	30" X 30" Pedestrian Warning Sign, Type B Post Mounted (W11-2) (ECMS)
Pedestrian Signal Head	\$650.00 EA	1 Signal Head	\$650.00	Pedestrian Crosswalk Signal Head (LED), Type B, Countdown (ECMS)
Audible & Vibrating Push Button	\$1,400.00 EA	1 Pushbutton	\$1,400.00	Pedestrian APS Push Button (ECMS)
In-Street Pedestrian Channelizing Device	\$300.00 EA	1 Device	\$300.00	12" X 36" In-street Crossing Sign per MUTCD 2009 (private supplier)
Bench	\$810.00 EA	1 Bench	\$810.00	2010 Means Sitework and Landscape Cost Data: 12 93 43.13 0610 Site Seating
Pipe Bollard	\$525.00 EA	1 Bollard	\$525.00	2010 Means Sitework and Landscape Cost Data: 32 17 13.13 1300 Pipe Bollards

Source: PennDOT, <http://www.dot14.state.pa.us/ECMS/> and 2010 RS Means Site Work and Landscape Cost Data, 29th Edition

Pedestrian Innovation

Both ITE and FHWA offer alternative treatments to existing pedestrian facility improvements. ITE's *Innovative Treatments at Unsignalized Pedestrian Crossing Locations* offers treatments specifically for unsignalized locations. FHWA's *Study Tour for Pedestrian and Bicyclist Safety in England, Germany and the Netherlands* showcases innovative European approaches to pedestrian facility design. Some innovative pedestrian facility designs and/or treatments are detailed in **Table 3-7**.

Table 3-7. Innovative Pedestrian Facilities and Treatments

Pedestrian Zones	A Pedestrian Zone or Mall blocks access to an area or roadway for vehicles, and in some cases bicycles. Pedestrian zones create a sense of place where pedestrians are not in conflict with motor vehicles, and an otherwise vehicle congested area is transformed into an area with exclusive pedestrian access. Times Square in New York City has embraced the vehicle free pedestrian zone, and locally the City of Pittsburgh recently redeveloped Market Square to be more pedestrian-oriented. The square has already generated increased pedestrian activity with its picnic tables. Market Square, when complete, will boast a children's play area, garden, and public art. xxxiv	 Market Square, Pittsburgh
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Zebra, Pelican, Toucan and Puffin Crossings	Zebra, Pelican, Toucan and Puffin Crossings are different types of nonstandard crosswalk striping. According to the FHWA Study Tour, Zebra crossings are specifically used in uncontrolled midblock locations and are a zig-zag pattern at vehicle approaches indicating that pedestrians have the right-of-way. Pelican crossings are dotted lines marking the crosswalk and controlled by traffic signals and push buttons. Puffin (Pedestrian User-Friendly Intersection) crossings are a combination of traffic and pedestrian signals with infrared detectors and barriers to channel pedestrians into the crosswalk. Finally, Toucan (cyclists “too can” cross together) crossings are a shared pedestrian and bicycle treatment with tactile surfaces, audible signals, push buttons, monitors, and detection.	 <i>Zebra Crossing, Abbey Road, London, UK</i> Photo: www.Wired.com  <i>Toucan Crossing, Longden Road, Shrewsbury, UK</i> Photo: www.Shropshire.gov.uk
Pedestrian Pavement Messages	Utilized most frequently in Europe, these pavement messages alert the pedestrian to “Look Right” and/or “Look Left” before proceeding through the crosswalk.	 <i>Pavement Messages in London, UK</i> Photo: FHWA

3.4 POLICY & PROGRAMMATIC CONSIDERATIONS

This plan was developed to serve as a guide for Allegheny County and its local municipalities to assist in improving active transportation facilities and encouraging active transportation. Based on the public outreach activities and document comments, there is growing public support for improved pedestrian facilities for access and mobility. Policies and programs that support walking improve the built environment for all transportation modes. Everyone starts their trip, regardless of trip purpose, as a pedestrian (even if you are getting in your car, you walk and/or roll there).

Policies

PennDOT's Design Manual

Sidewalks – PennDOT's Design Manual 2, Chapter 6, Section 6.6 establishes the criteria for sidewalks in PennDOT projects. The Department may participate when criteria are met and where communities support sidewalks and participate in cost sharing, ownership and future maintenance. Municipal buy-in to add sidewalks is sometimes a challenge. Municipalities should be educated on Smart Transportation and the benefits of sidewalks:

- To emphasize the importance of pedestrian safety,
- To advance the implementation of smart and healthy transportation principals,
- To assist with placemaking and to discourage sprawl, and
- To reduce fossil fuel consumption and to reduce costs in the long term.

Municipalities should be further encouraged to support the local match with applicable local funding sources or creative funding programs, such as grants, donation credits, and/or private and business participation.

Curb Ramps – PennDOT Design Manual, Part 2, Chapter 6, Sections 6.2, 6.3 and 6.9 states that striping of a crosswalk (new installation) requires an upgrade and/or installation of curb ramps at the crossing. Within the next ten years, PennDOT will be upgrading curb ramps for ADA compliance per federal regulations. However, if specific locations are identified as a concern by a user or an evaluation is requested, PennDOT or the local municipality will address a potential location upgrade expeditiously.

Pennsylvania Driver's Manual

Chapter 2: PA's signal, sign, and pavement marking requirements state "always yield to pedestrians" as they may not know their responsibility at an intersection with regards to pedestrian signals. Pedestrian crossing and school crossing signs are also addressed. It is recommended that when the Driver's Manual is revised, it incorporate language advising drivers not to stop in a crosswalk as it hinders pedestrian mobility at an intersection.

Hazardous Walking Routes



Pennsylvania code sections 506 and 2001 of the Administrative Code of 1929 and sections 1362 and 2541 of the Public School code of 1949 empowers schools to designate a school student walking route along a public highway as hazardous provided they meet the outlined criteria. The hazardous walking routes designation discourages students from walking to school on roadways without sidewalks, shoulder provisions, or designated crossings. While the code has the best interest of students in mind, schools should be required to evaluate potential safe routes to school prior to designation of hazardous walking routes. If funding is available for improvements, it is recommended that the potential hazardous walking routes be improved. If hazardous route designation is unavoidable, it is recommended that parallel routes to school be evaluated and designated by the school. It is further recommended that hazardous designations be reviewed periodically, and if necessary updated, to reflect infrastructure improvements.

State Bicycle and Pedestrian Coordinator

The Federal Highway Administration (FHWA) indicates that each state should have a Bicycle and Pedestrian Coordinator "to promote and facilitate the increased use of non-motorized transportation, including developing facilities for the use of pedestrians and bicyclists and public educational, promotional, and safety programs for using such facilities." For Pennsylvania, that person is based out of PennDOT's Bureau of Design, Highway Quality Assurance Division in Harrisburg. The Bicycle and Pedestrian Coordinator position facilitates the following:

- ✓ "Increased use of nonmotorized transportation, including developing facilities for the use of pedestrians and bicyclists and public education, promotional and safety program for using such facilities." ^{xxxv}
- ✓ Implementation of the goals and objectives set forth in the 2007 PennDOT Bicycle and Pedestrian Plan
- ✓ Information exchange among public agencies with regards to bicycle and pedestrian travel.

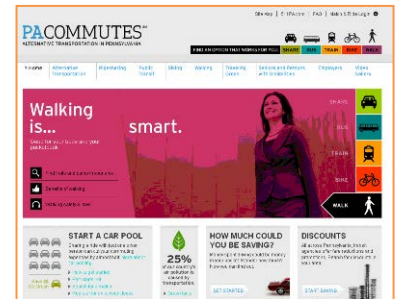
Given this ambitious program, the implementation of this plan may require additional staff resources to achieve the program goals.

Programs

United States

Many states and their communities have developed programs in the past decade to encourage walking for the numerous benefits it provides to an individual and the environment. Walkinginfo.org^{xxxvi} offers strategies to promote walking including special events, clubs, incentives, programs, and maps. Some examples of successful programs across the country are detailed in this section.

Pennsylvania – “PACommutes” caters to alternative transportation in Pennsylvania, including ridesharing, transit, bicycling, and walking. Resources for pedestrians are provided on www.PACommutes.com including the benefits of walking, safety and laws in the state, information and links to trails and paths, and “walkscore.” “Walkscore” provides pedestrians with a checklist to evaluate their community for walkability and suggests ways to take action if their community scores low. A tool is also available on the PACommutes website to calculate how much money an individual could save through switching from single occupancy vehicle commutes to alternative modes of transportation.



PA Commutes Website

Source: www.pacommutes.com

Allegheny County, Pennsylvania – The TMAs in Allegheny County (Oakland Transportation Management Association, the Pittsburgh Downtown Partnership and Airport Corridor Transportation Association) have been developing programs to raise awareness of walking as a viable mobility alternative. The following programs were provided by ACTA for the Plan:

- † Walk Challenge (www.walk-challenge.org) – An eight-week Summer Walk Challenge “Ready, Set, Walk” and a Winter Walk Challenge “See a Smaller Shadow” are held jointly by the TMAs to promote walking as a means of commuting to destinations. People can track their steps via provided pedometers and record their numbers on the Walk Challenge website for weekly prizes and a grand prize.

- † Walk Pittsburgh – CMAQ funded project under development via website to encourage walking. Walk Pittsburgh will provide walking maps of the Greater Metropolitan Area, as well as other resources once publicly available.

- † Walkability Audits – Utilizing grant monies recently applied for by the TMAs, there are plans to assist Allegheny County municipalities in performing walkability audits in their communities. Walkability Audits focus on improving and enhancing pedestrian conditions along corridors and at intersections to improve access, mobility and safety.



Walk Challenge Poster

Source: walk-challenge.org/sass2011/default.asp

Cambridge, Massachusetts – “Cambridge Walks and the Hunt for the Golden Shoes” is an incentive program by the Cambridge Public Health Department to increase walking as a mode share through rewards. It enlightens people on the benefits of walking, while providing an interactive experience of hiding gold shoes around the city where people may walk. Shoes can then be turned in by the finder for prizes donated by local merchants. For more information, visit www.cambridgema.gov/cdd/et/ped/index.html.

Yolo TMA, California – Yolo Transportation Management Association has a pedestrian incentive program that provides financial compensation to single occupancy vehicle commuters who switch to walking as a mode of commuting. There are guidelines for participation including an application process. Compensation includes \$10 per month for three months and bonuses for continued participation of ten days of commuting at least one mile each way for a period of one year.

Wilsonville, Oregon – Wilsonville implemented the “WalkSmart” program, which provides residents and employees with a pedometer and log book to encourage walking within the community. Participants in the program who submit their logs are eligible to receive prizes in a monthly drawing as well.

Canada

With high concentrations of pedestrian activity in their cities, Canada has several programs for its residents who choose to walk. The “Commuter Challenge” is a national one week event urging commuters to leave their vehicles parked and travel to work by other means. The goal is to educate commuters on the numerous benefits of commuting by alternative transportation and encourage trips by walking, biking, or transit.

Toronto and Hamilton Metropolitan Area – “Smart Commute” assists employers and commuters through emergency ride home services, incentives, and promotions for those who are interested in commuting through alternative transportation.

Vancouver – The Greater Vancouver Transportation Authority piloted a program “TravelSmart” to promote sustainable travel through education and incentives. “TravelSmart” participants who requested additional information about walking as a travel mode choice were rewarded with prizes.

TravelSmart

TravelSmart helps you achieve better, healthier, more sustainable transportation. There are lots of great options to driving alone, and we can help!

You made great travel choices during the 2010 Winter Games and saw first-hand the role travel choices play in helping our region remain one of the most beautiful and livable places on earth.

Choosing to TravelSmart not only reduces your carbon footprint, it can save you time, money and stress. With so many smarter travel options available, we're positive there's one out there that suits your lifestyle.



TravelSmart Summary
Source: travelsmart.ca/

Alberta – Alberta Winter Walk Day is an event held in February to promote active transportation, including school trips. Schools can become a member of “Safe Healthy Active People Everywhere (SHAPE)” and information for students will be sent to the school. Total number of minutes walked to school is logged into the membership profile of SHAPE, who promotes active and safe routes to schools in Alberta.



3.5 EDUCATION, ENFORCEMENT & PUBLIC AWARENESS

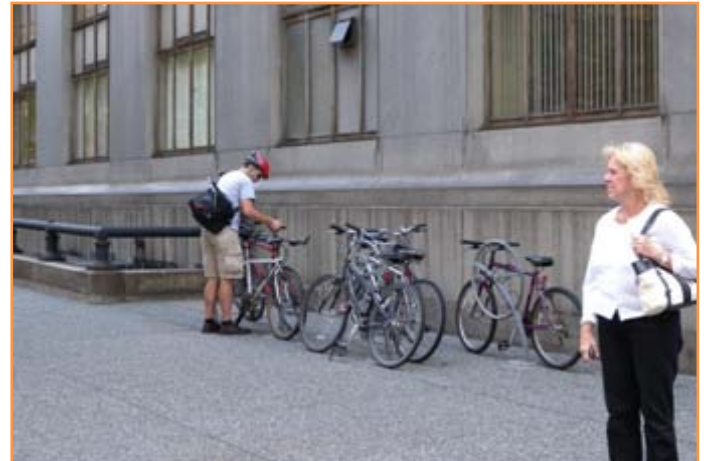
To properly plan for future growth of pedestrian facilities usage in the county, it is important to implement educational programs that encourage proper safety techniques among pedestrians and motorists statewide, countywide, and on a municipal level. When educating the targeted users of the transportation network, it is important to dispel myths, encourage courteous, respectful and lawful behavior, and enhance awareness and acceptance of alternative modes. By utilizing the resources of the police, schools, municipal administration, and libraries, educational programs have the potential of reaching a broader audience and cross section.

It is important to educate each group on the most frequent causes of crashes and injuries. Pedestrian safety programs and educational materials should be made available to everyone. The U.S. Department of Transportation (USDOT) – Federal Highway Administration (FHWA) provides materials and training to assist states, counties, and local jurisdictions in enhancing pedestrian access, mobility, and safety. These materials and/or training are detailed on FHWA’s website and include:

- † **FHWA University Course on Bicycle and Pedestrian Transportation** – Provides current information on pedestrian planning and design techniques, as well as practical lessons on how to increase walking facilitated by land use and engineering. (www.fhwa.dot.gov/publications/research/safety/pedbike/05085/)
- † **National Highway Institute (NHI) Pedestrian Facility Design Course** – A one and a half day session that provides information and application opportunities for those involved in the design of pedestrian facilities. (www.nhi.fhwa.dot.gov/training/course_detail.aspx?num=FHWA-NHI-142045&num=)
- † **Safer Journey CD-ROM** – An interactive CD-ROM that takes the user through various pedestrian safety scenarios encountered every day by pedestrians. (http://safety.fhwa.dot.gov/ped_bike/ped_bike_order.cfm)
- † **PEDSAFE** – The Pedestrian Safety Guide and Countermeasure Selection System (FHWA-SA-04-003) is intended to provide practitioners with the latest information available for improving the safety and mobility of those who walk. (www.walkinginfo.org/pedsafe/about.cfm)
- † **Pedestrian Road Safety Audit Guidelines and Prompt Lists** – This publication (FHWA-SA-07-007) provides transportation agencies and teams conducting road safety audits with a better understanding of the needs of pedestrians and their abilities. (<http://drusilla.hsrrc.unc.edu/cms/downloads/PedRSA.reduced.pdf>)
- † **A Review of Pedestrian Safety Research in the United States and Abroad** – This report (FHWA-RD-03-042) examines pedestrian safety across the 50 states and in countries abroad to draw comparisons and conclusions. (www.fhwa.dot.gov/publications/research/safety/pedbike/03042/)

Several other organizations and agencies provide information regarding walking including the National Center for Bicycling and Walking (www.bikewalk.org), the Pedestrian and Bicycle Information Center (www.walkinginfo.org), America WALKS (www.americawalks.org), Perils for Pedestrians (www.pedestrian.org), and Walkable Communities (www.walkable.org).

In Allegheny County, the SPC (www.spcregion.org), BikePGH (www.bike-pgh.org) and Friends of the Riverfront (www.friendsoftheriverfront.org) provide pedestrian resources as well as bicycle resources detailed in Chapter 2.



Pedestrian and Bicyclist, City of Pittsburgh

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CHAPTER 4. Other Active Transportation Opportunities



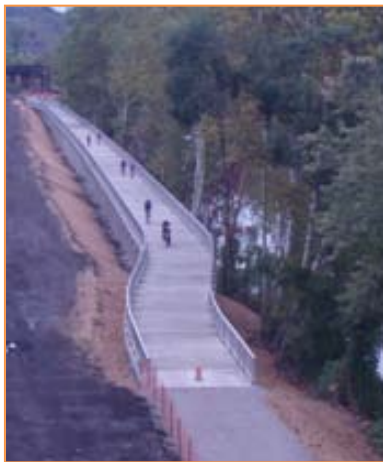
Although bicycling and walking are the most prevalent modes of active transportation in Allegheny County, other modes are emerging as not only a recreational preference, but as commuter options. A system of water trails is available to kayakers and non-motorized watercraft. While on land in-line skating has become increasingly popular partly due to the network of paved paths.

4.1 THREE RIVERS WATER TRAIL

The Three Rivers Water Trail is a system of access points and accommodations for water sport access for the Allegheny, Monongahela, and Ohio Rivers in Allegheny County. The water trail was developed and is maintained by Friends of the Riverfront. According to their website (www.friendsoftheriverfront.org/), “kayaking and canoeing were among the top five sports that had the most first-time participants in 2001.” The Three Rivers Water Trail serves these emerging non-motorized water users with signed routes and amenities.

Connectivity

The Three Rivers Water Trail system provides connectivity to the Three Rivers Heritage Trail and Steel Valley Trail (part of the Great Allegheny Passage) (www.gaptrail.org/), and offers access to bike rentals and programs in South Side and Millvale. As part of ACTIVEALLEGHENY, countywide bicycle routes were drafted to connect active



Route 28 Trail “Missing Link”

transportation opportunities (Chapter 2-2), including access to the Three Rivers Heritage Trail and Steel Valley Trail, which both offer access to the Three Rivers Water Trail system. ACTIVEALLEGHENY also recommended locations for complete street improvements, and pedestrian access and mobility at intersections and along corridors in Allegheny County (Chapter 3-2). A few of these locations serve as potential access to the rivers and more specifically the Three Rivers Water Trail. For example, the missing link along the Route 28 Trail has recently been constructed and additional improvements are currently under construction to enhance access at River Front Drive near Route 28. These improvements have and will continue to facilitate mobility and access to the Millvale Riverfront Park, the Three Rivers Water Trail and the Three Rivers Heritage Trail.



Water Trail Route Sign

Accessibility

ACTIVEALLEGHENY online survey respondents told a story of active and enthusiastic kayakers. A desire was expressed by kayakers in the survey for additional access points and convenient parking near the rivers. Issues with kayaking mentioned by survey respondents included:

- Metered parking on Federal Street is restricted to 2 hours, limiting time on the Three Rivers Water Trail (Kayak Pittsburgh is located underneath the Robert Roberto Clemente Bridge on the north side of the Allegheny River).
- Secure kayak parking is desired in Verona and Oakmont to enjoy nearby restaurants and retail.
- More kayak rental locations are desired in addition to the rentals offered by Kayak Pittsburgh (<http://www.kayakpittsburgh.org/>).
- There needs to be more publicity and public awareness for docking and launching locations.
- Add an access point at Chapel Harbor, where available vacant land use could be utilized.

Planning and Development

The Three Rivers Water Trail is expanding in Allegheny County and new access sites and amenities are planned. According to the Friends of the Riverfront website, the following access sites are under development:

Monongahela River

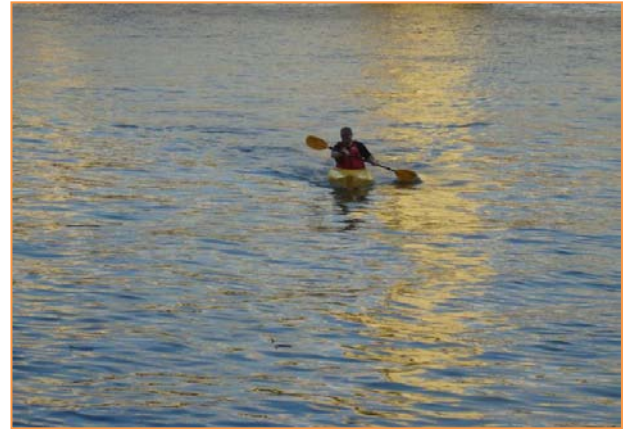
Braddock
Elizabeth
Mon Wharf

Allegheny River

Sharpsburg
O'Hara
Sycamore Island
Oakmont

Ohio River

Sewickley

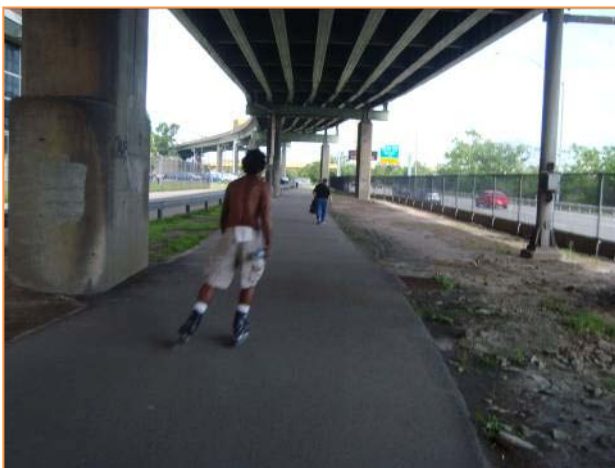


Kayaker in the Allegheny River

Updates and progress with regards to expansion of the Three Rivers Water Trail are available on the Friends of the Riverfront website (www.friendsoftheriverfront.org/) under "Trails" and then "Water Status."

4.2 IN-LINE SKATING & SKATEBOARDING

In-line skating has emerged as an active transportation option in Allegheny County due in part to the network of paved paths available as part of the trail system. In-line skaters in the region are supported through the Three Rivers Inline Club, which holds activities and events to promote and teach skating to cater to a variety of skill levels. Both in-line skating and skateboarding were mentioned by survey respondents as an activity they enjoy and would like better accommodated by additional paved paths. Two paths where in-line skaters were observed in the City of Pittsburgh are shown below.



In-line Skater on Eliza Furnace Trail



Group of In-line Skaters, Station Square Trail

CHAPTER 5. Complete the Street



5.1 WHAT IS A COMPLETE STREET?

The term “complete streets” is relatively new; it was coined in 2003 by the advocacy group “America Bikes” as it worked to include pedestrians, bicyclists, and transit users in SAFETEA-LU, the federal transportation funding bill. The term was defined as follows: “A complete streets policy ensures that the entire right-of-way is routinely designed and operated to enable safe access for all users. Pedestrians, bicyclists, motorists, and transit riders of all ages and abilities must be able to safely move along and across a complete street.”²

This policy was not included in SAFETEA-LU when the bill passed in 2005, but the effort sparked the formation of the National Complete Streets Coalition (www.completestreets.org/) and a nationwide movement to enact complete streets policy at the municipal and state level. Early members of the Complete Streets Coalition included the Institute of Transportation Engineers, AARP (formerly known as the American Association of Retired Persons), the American Society of Landscape Architects, the American Planning Association, Smart Growth America, Paralyzed Veterans of America, and the American Public Transportation Association.



Sidewalk with adequate clear width and buffer

The impetus for complete streets grew partly out of the recognition that the previous approach for accommodating pedestrians and bicyclists on federally funded studies – arguing for the inclusion of pedestrian and bicycle facilities on a project-by-project basis – had limited potential for changing infrastructure. Under the complete streets approach, all projects begin with the assumption that pedestrians, bicyclists, and transit users of all ages and abilities should be accommodated.

In this chapter, complete streets practice is defined, along with a discussion of complete streets policy elements. This is followed by a discussion of the potential for re-visioning three example projects that are serving as prototypes for ACTIVEALLEGHENY using complete streets principles.

There is no standard design or template for a complete street, and the National Complete Streets Coalition has actively discouraged attempts to prescribe specific roadway components. The emphasis of the complete streets movement has been primarily on policy, and less on design practices. As noted in *Complete Streets: Best Policy and Implementation Practices* (APA 2010), “changing the overall approach to design is of greatest importance.” Practitioners thus have great flexibility in working toward the goal of creating roadways that provide safe mobility for all modes. The Coalition has indicated that examples of features that might be included in a Complete Street are sidewalks, bike lanes (or wide paved shoulders), special bus



Separated bike lane in NYC

² Information on the history of complete streets, and complete streets practices discussed in this chapter, is found in the text *Complete Streets: Best Policy and Implementation Practices*, Planning Advisory Service Report 559, American Planning Association, March 2010. <http://www.planning.org/apastore/search/Default.aspx?p=4060>.

lanes, comfortable, and accessible transit stops, frequent crossing opportunities, median islands, accessible pedestrian signals, and curb extensions.

In short, the tools to be used in designing complete streets are not unique to roadways designated as complete streets. They include planning and design techniques that are regularly used to develop pedestrian and bicycle facilities; those discussed elsewhere in the ACTIVEALLEGHENY Plan. In complete streets, there is also a strong emphasis on accommodating the more vulnerable groups in our population: children and older adults and persons with disabilities.

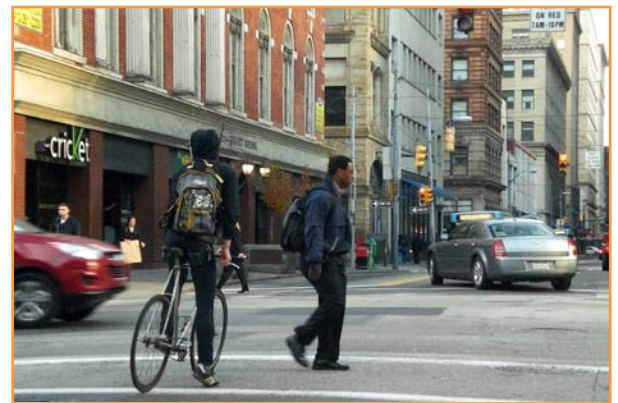
Along with facilities regularly incorporated into roadway projects (curb ramps), consideration should be given to facilities such as audible traffic signals, preferably with vibrating arrows, as intersection improvements. A sidewalk may exist on a corridor, but if the clear width is impeded by utilities and street furniture or if sidewalk panels are uneven, travel by persons in wheelchairs is difficult. As part of the public involvement process on projects, input should be gathered on the presence of disabled persons in the vicinity and facility needs.

5.2 THE SMART TRANSPORTATION CONNECTION

A complete streets approach is consistent with PennDOT's Smart Transportation Initiative. That initiative is built around 10 Smart Transportation themes, including the theme "accommodate all modes." The *Smart Transportation Guidebook* (<ftp.dot.state.pa.us/public/Bureaus/design/SMART%20TRANSPORTATION%20.pdf>) was jointly developed by PennDOT and NJDOT to guide the planning and design of all land service roadways. The Guidebook is essentially a complete streets practice in its emphasis on flexibility in creating transportation facilities that work well for all users, and in balancing trade-offs between vehicular, pedestrian, bicycle, and transit mobility.

For example, the Guidebook does not specify the type of bike facility that should be provided on roadways to accommodate bicyclists; rather, the planner or designer must evaluate all pertinent factors in selecting an outside travel lane width, bike lane width, or shoulder width that would be compatible with bicycle travel.

Similar flexibility is offered in the Guidebook for pedestrian facilities. Sidewalks are the cornerstone of any pedestrian network, but their width, and their setback from the roadway, will vary depending upon roadway type and land use context). Although critical on urban and most suburban roadways, sidewalks are not vital to many roadways in rural areas. Further, there will be choices for how pedestrian travel throughout the community: signalized or unsignalized crossings, the frequency of designated pedestrian crossings, the provision of medians for multilane roadways, accessible pedestrian signals, and curb extensions are examples of how pedestrian movement can be accommodated.



Bicyclist, pedestrian, and motor vehicles in Pittsburgh
Photo: Lynn Heckman

Accommodation of transit service is another component of a complete street. This can be viewed from two perspectives:

- Accommodation of transit riders traveling to and from bus stops
- Accommodation of the transit vehicle

The ability of transit riders to safely access bus stops and rail stations depends in large measure on the provision of adequate pedestrian and bicycle facilities proximate to the stops and stations. As discussed in the *Smart Transportation Guidebook*, there are unique issues associated with bus boarding that must be taken into account. Bus stops are

typically better positioned at intersections than mid-block locations, since they offer the best pedestrian access from both sides of the street as well as cross-street locations, and will reduce the tendency for jaywalking. However, at some locations, major land use generators will suggest the need for midblock bus stops; in these cases, the accommodation of safe midblock crossings must be evaluated.

The need for balancing modes in complete streets is illustrated by the issue of transit. Pedestrians find it easier to cross roadways with narrow travel lanes, and modest curb radii at intersections (smaller curb radii result in reduced pedestrian crossing length, and also slow vehicular speeds through intersections). However, buses have different needs than passenger vehicles. The roadway width needed to accommodate buses depends in part on the frequency of service. As discussed in the Guidebook, on arterial roadways with regular bus service, a travel lane width of 12' is recommended. On collector roadways, travel lane width of 11' is recommended. At intersections where buses make regular turning movements, a curb radius of at least 25' may be needed.



Suburban roadway with high visibility crosswalks, median and bike lanes

5.3 POLICY

Complete Streets Policy

The most fundamental step that Allegheny County and its constituent local municipalities can take to advance complete streets practice is to adopt and implement a complete streets policy. **ALLEGHENY PLACES**, the County's comprehensive plan, advocates for complete streets. The National Complete Streets Coalition recommends consideration of 10 elements in a comprehensive complete streets policy document. Following is a summary of these 10 elements, along with a discussion of each element.

- *A vision for how and why the community wants to complete its streets.* The primary purpose of the complete streets policy should be identified.
- *Specifies that "all users" includes pedestrians, bicyclists, and transit passengers of all ages and abilities, as well as automobile drivers and transit vehicle operators.* This provision is particularly important given the significant concentrations of disabled persons living in some Allegheny County municipalities. As noted on the Complete Streets website (www.completestreets.org), narrow sidewalks or sidewalks with obstacles and uneven surfaces can discourage wheelchair travel, and signalized intersections without audible signals can increase the danger for blind pedestrians.
- *Encourages street connectivity and aims to create a comprehensive, integrated connected network for all modes.* A well-connected network greatly enhances the ability of pedestrians and bicyclists to move around a community, and is a vital complement to complete streets.
- *Is adoptable by relevant agencies to cover all roads.* Local officials should work with state officials to ensure that complete streets principles are applied to state roadways within their jurisdiction.
- *Applies to both new and retrofit projects, including design, planning, maintenance, and operations, for the entire right-of-way.* Even a relatively simple resurfacing project should involve an evaluation as to whether the roadway can better accommodate bicyclists; for example, can travel lanes be narrowed to provide wider shoulders or bike lanes?
- *Makes any exceptions specific and sets a clear procedure that requires high-level approval of exceptions.* Many jurisdictions with complete streets policies recognize the need for exceptions on some projects; this policy should be clearly stated in advance to avoid confusion.

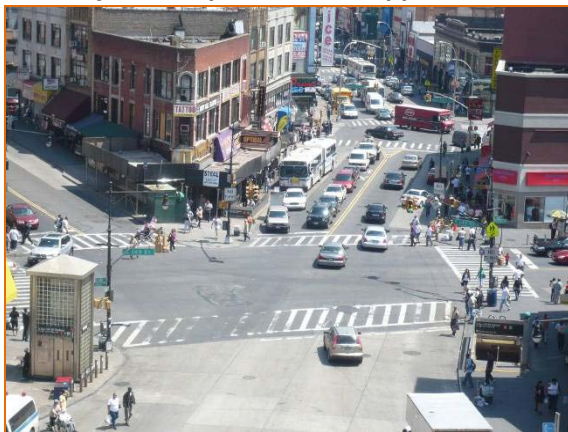


*Intersections developed under a Complete Streets Policy accommodate all users
Photo: FHWA*

- *Directs the use of the latest and best design standards while recognizing the need for flexibility in balancing user needs.* Detailed design standards should not be included in an ordinance or resolution, but there should be a commitment to revisit existing standards in municipal subdivision and land development ordinances or design manuals.
- *Directs that complete streets solutions will complement the context of the community.* Roadway design that is tailored to fit the context of the surrounding community is at the heart of Pennsylvania's Smart Transportation initiative, and the key guiding principle to the *Smart Transportation Guidebook*.
- *Establishes performance standards with measurable outcomes.* Goals should be set on improving pedestrian and bicycle facilities, and existing standards for vehicular service may need to be revised. For example, vehicular levels of service may need to be lowered if these promote major roadway improvements at the expense of pedestrian or bicycle facilities.
- *Includes specific next steps for implementing the policy.* The policy should state other steps, such as revision of manuals or procedures that will be necessary to better implement policy.

There are a variety of measures by which local municipalities can adopt a complete streets policy, and not all of the above 10 elements are vital to every measure. For example, the establishment of performance measures would be appropriate for a complete streets plan, but detailed performance measures should be left out of an ordinance.

Before Complete Streets Application



After Complete Streets Application



Photos: CompleteStreets.org

Complete streets practice can be adopted in a wide variety of ways:

- Ordinance or Legislation
- Resolution
- Executive Order
- Internal Policy
- Plan

Ordinances and resolutions are the preferred means for adopting complete streets policies, since they provide a concise, direct declaration of municipal intent by the municipality's governing body. Resolutions have been chosen by the largest plurality of municipalities, representing 47% of municipalities with complete streets policies. An ordinance is second in popularity, being adopted by 22% of municipalities.³ Plans and internal policies can be useful in providing guidelines for implementing ordinances, resolutions, or executive orders.

³ Percentages were calculated based on summary of adopted policies, www.completestreets.org.

Below is a model ordinance recommended for adoption by Allegheny County's local municipalities. The text is based on model policy language recommended by the National Policy and Legal Analysis Network to Prevent Childhood Obesity (NPLAN), and also incorporates language from adopted policies for Rochester, Minnesota and Seattle, Washington. The model ordinance is concise by intent, focusing on the simple principle that roadway projects should accommodate all users. The language can be modified for use on resolutions and executive orders.

Complete Streets Model Ordinance

AN ORDINANCE relating to complete streets policy for the ____ of ____, stating guiding principles and practices so that transportation improvements are planned, designed and constructed to encourage walking, bicycling and transit use while promoting safe operations for all users.

WHEREAS, implementing transportation improvements that are planned, designed and constructed to safely accommodate walking, bicycling, and transit use increase the general safety, health and overall welfare of the citizens of and visitors to the ____ of ____; and,

WHEREAS, the ____ of ____ will seek to enhance the safety, access, convenience and comfort of all users, including pedestrians, bicyclists, transit users and drivers, motorists and freight drivers, and people of all ages and abilities, including children, older adults, and persons with disabilities, through the design, operation and maintenance of the transportation network so as to create a connected network of facilities accommodating each mode of travel; and,

WHEREAS, transportation improvements are to be planned and designed in a manner consistent with, and supportive of, the surrounding community, recognizing that all streets are different and that the needs of various users will need to be balanced in a flexible manner;

NOW, THEREFORE, BE IT ORDAINED BY THE ____ OF ____ AS FOLLOWS:

Section 1. All roadway projects, including construction, re-construction, re-paving and rehabilitation, will provide appropriate accommodation for pedestrians, bicyclists, transit riders and drivers, motorists and freight drivers, and people of all ages and abilities, including children, older adults and persons with disabilities, except under one or more of the following conditions:

- The roadway project is comprised of ordinary maintenance activities designed to keep assets in serviceable condition (e.g., mowing, cleaning, sweeping, spot repair and surface treatments such as chip seal);
- Where use by nonmotorized users is prohibited by law;
- The cost would be excessively disproportionate to the need or probable future use over the long term;
- There is an absence of current and future need.

Section 2. Appropriate accommodations include facilities and amenities that are recognized as contributing to complete streets, which may include sidewalks and pedestrian safety improvements such as median refuges, pedestrian signals, bulbouts and crosswalks; street and sidewalk lighting; improvements that provide ADA (Americans with Disabilities Act) compliant accessibility; transit accommodations including improved pedestrian access to transit stops and bus shelters; bicycle accommodations including shared-use lanes, wide travel lanes or bike lanes as appropriate; paved shoulders; bicycle parking; street trees, landscaping, street furniture and adequate drainage facilities; and other facilities.

Section 3. Complete streets principles will be incorporated into the comprehensive plan, subdivision and land development ordinance, and other plans, manuals, regulations and programs as appropriate.

The first paragraph of the model ordinance summarizes the ordinance, and indicates the purpose. The preamble ("Whereas" clauses) indicate the reasons why the municipality is adopting a complete streets ordinance; it is recommended that officials of local municipalities in Allegheny County add reasons specific to their community, if possible.

Section 1 indicates that complete streets policies will be followed on roadway projects in the municipality, with the exception of simple maintenance projects, and projects where there is no need or where implementation of the policy will result in disproportionate costs. These exceptions are common provisions in adopted complete streets policies nationwide. They help address concerns on the part of some that implementation of a complete streets policy will significantly increase costs.

Section 2 provides examples of complete streets facilities. The examples are similar to those found in the NPLAN model ordinance, and in adopted policies. However, not every local municipality has listed typical examples of complete streets facilities in their adopted policies.

Section 3 indicates that the local municipality will incorporate complete streets principles into other municipal ordinances, plans, and standards as appropriate. Although, as discussed earlier, there is not a prescribed complete streets treatment, the municipal standards should be reviewed to determine whether there are basic standards for sidewalks and bike facilities. Further, there should not be one set standard for travel lanes; flexibility for this feature is desirable.

In addition to local municipalities, developers should consider Complete Streets approaches when planning, designing and building large scale projects (e.g., The Waterfront at Homestead and South Side Works).

Where Have Complete Streets Policies Been Adopted?

According to the National Complete Streets Coalition, over 150 jurisdictions across the country – state, county, and local governments – have adopted complete streets policies, or committed to doing so. PennDOT does not have a formal complete streets policy, but it has increasingly followed a complete streets approach to roadway projects in policy and implementation. This approach is exemplified in its Bicycle and Pedestrian Checklist. When issuing the Checklist, PennDOT stated:

“Department policy requires the evaluation of the access and mobility needs of pedestrians and bicycle users in highway and bridge transportation corridors. This revised policy mandates that highway and bridge projects must evaluate the existing, latent, and projected needs of pedestrians and bicycle users. It requires the integration of the identified needs into project planning and design processes.”



Google Map showing where Complete Streets Policies have been enacted

Source: CompleteStreets.org

At the local level in Pennsylvania, interest in complete streets is also growing. The City of Philadelphia was the first municipality in the state to adopt a complete streets policy, issuing an Executive Order in 2009. The City directed all city departments and agencies to:

- Give full consideration to the safety and convenience of all users of the transportation system, be they pedestrians, bicyclists, public transit users, or motor vehicle drivers;
- Place a high priority on the safety of those traveling in the public right-of-way, and in particular the safety of children, the elderly, and persons with disabilities.



Philadelphia is also currently preparing a complete streets handbook. The City of Franklin in Venango County recently became the second municipality in Pennsylvania to adopt a complete streets policy, in its case doing so through a resolution (<http://www.completestreets.org/webdocs/policy/cs-pa-franklin-resolution.pdf>). The City of Allentown has been preparing a complete streets policy to incorporate into *Connecting our Community*, which is the city's bicycle and pedestrian trail plan.

5.4 IMPLEMENTATION

Following approval of a complete streets policy, a local municipality can move immediately to implementation. Depending upon the level of detail of the adoption instrument, the municipality may wish to consider preparation of a plan or policy at this point. As discussed above, a plan or policy is not recommended as the primary adoption instrument.

Transportation Plan

A municipality's transportation plan or transportation element of a comprehensive and/or master plan should express support for complete streets goals and establish a framework for improving pedestrian, bicycle, and transit facilities. It should also identify areas of the municipality where transit service is needed, or where greater frequency/extended service periods are needed. Safe and accessible routes to transit stops or stations are needed, with consideration of relocating transit stops, if safe routes to transit stops are not feasible at the current location. Although the transportation plan is the most common plan used to express municipal goals for roadways, some municipalities have adopted a complete streets implementation plan.



*Transportation elements play a vital role in Complete Streets Policy making.
Photo: CompleteStreets.org*

Programs complementary to pedestrian and bicycle facility improvements, such as travel demand management and parking management programs, should be included in a complete streets plan where appropriate.

Performance measures may also be included. For example, the plan could state that by 2020, 20% of trips will occur by bicycling or walking; the number of injuries to pedestrians and bicyclists will be reduced by 20%, and the number of miles of roadways with 5' sidewalk will increase by 20%.

SALDO and Design Standards

An important step will be to revise every local municipality's Subdivision and Land Development Ordinance (SALDO) or their public and private improvements codes to provide standards for pedestrian and bicycle facilities. Equally important, the standards should be revised to approve roadway design supportive of pedestrian and bicycle movement, ranging from a well-connected street network, to discouraging excessively wide roadways, large curb radii, and other features that promote vehicular speeding. One of the most important components of a complete streets approach is

moving away from a “one size fits all” idea that permeates the SALDO standards in many municipalities, in which existing standards for roadway design are narrowly prescriptive. For example, it is not unusual to encounter travel lane widths of 12’ as a minimum standard for arterial roadways, regardless of whether the roadway is along a fast-moving commercial corridor or in a busy downtown, and whether the travel lanes are accompanied by bike lanes. In complete streets, the entire context of the roadway is taken into consideration, and the travel lane width should change depending upon the need to complement the surrounding land use context, manage vehicular speeds, provide room for bike lanes, and other factors.

An important resource for local municipalities that wish to revise their roadway design standards is the *Smart Transportation Guidebook* (<ftp.dot.state.pa.us/public/Bureaus/design/SMART%20TRANSPORTATION%20.pdf>). Table 5-1 shows suggested design values for arterial and collector roadways, based on the Guidebook:

Table 5-1. Smart Transportation Guidebook Design Values

Feature	Land Use Context		
	Urban	Suburban	Rural
Travel Lanes	Evaluate 10’ to 11’ lanes for roadways of 35 mph or below; 12’ for roadways of 35 mph or above and high traffic volumes and heavy vehicles. 14’ shared lane recommended for bike compatible treatment on roads without shoulders.	11’ to 12’, with 14’ shared lane recommended for bike compatible treatment, on roads without shoulders.	Evaluate 10’ for lightly trafficked roadways; 11’ to 12’ for roadways with regularly trafficked roadways, or with speeds above 35 mph.
Shoulders	4’ to 6’ on roadways where sidewalks are not provided.	8’ to 10’ for suburban corridors; 4’ to 8’ for suburban neighborhoods.	8’ to 10’ for arterials; 4’ to 8’ for collector roadways.
Medians	Provide depending upon access control, left turn and “pedestrian refuge” needs. Left turn medians are 12’ to 18’; pedestrian refuges 4’ to 8’.	Provide depending upon access control, left turn and “pedestrian refuge” needs. Left turn medians are 12’ to 18’; pedestrian refuges 4’ to 8’.	Design depending upon access control, left turn needs.
On-Street Parking	7’ to 8’ parallel parking.	Provide on-street parking as needed.	NA
Grass buffer	4’ to 6’ along neighborhood streets; typically absent in urban/town centers.	4’ to 8’.	NA
Clear Sidewalk Width	6’ to 14’ in urban/town centers, 5’ to 8’ in urban/town neighborhoods.	Min. 5’.	NA

Checklist

As part of the normal review process of projects advanced by private developers, a checklist should be used to ensure that developers have considered the needs of pedestrians, bicyclists, and transit users. Further, the municipal capital improvements program should be reviewed to determine whether all projects are consistent with complete streets principles. The implementation of complete streets goals should be coordinated with the planned resurfacing or reconstruction of streets, utility projects, or other public improvement projects.

Training

Some municipalities that have adopted complete streets policies have conducted training sessions on the design of pedestrian and bicycle facilities for planners and engineers. Other municipalities have emphasized procedural training, provided to all members of staff that might have a hand in implementation, such as zoning officers that do site plan review. Training has also been provided to contractors and consultants who regularly provide municipal services.

5.5 PROTOTYPES IN ALLEGHENY COUNTY

Criteria

In this section, recommendations are provided for three (3) roadways using complete streets principles. The three prototypical projects are representative of different cross sections in Allegheny County and represent what could be implemented for a complete street candidate. The complete streets candidates (**Appendix O**, which is included in a separate document) were selected after input from the ACTIVEALLEGHENY Core and Study Advisory Committees and an extensive review process. The following are the primary criteria used to evaluate candidate projects:

- Arterial or collector roadway
- Average daily traffic greater than 5,000
- Volume demand does not exceed capacity
- At least minimal pedestrian activity
- Proposed bike route and/or existing transit route
- Existing pedestrian and bicycle facilities are not ideal
- Economic growth area (identified as “places” in ALLEGHENYPLACES), or otherwise identified as proposed and targeted development areas

There is a preference for applying complete streets treatments to arterial and collector roadways, and roadways with daily volumes greater than 5,000, since the potential exists for a higher number of conflicts between motorists and non-motorized travelers on these roadways. However, it is recommended that when beginning a complete streets program, municipalities concentrate on roadways where traffic volumes do not significantly exceed capacity. It will be more difficult to reconfigure roadways where existing traffic congestion is seen as a serious issue, potentially shifting right-of-way to non-motorized modes or slowing vehicular traffic.

Complete streets implementation is recommended for roadways with at least minimal pedestrian activity, to address situations with demonstrated existing need. For the same reason, municipalities should focus on roadways where existing accommodations for pedestrians or bicyclists are not already ideal. Roadways with transit routes are preferred, to help foster multi-modal connections between transit and non-motorized modes. Finally, treatments should be prioritized for economic growth areas, since infrastructure improvements have the ability to help spur reinvestment.

Similar criteria can be used by study area municipalities when evaluating complete streets projects in the future. As these projects are advanced, it is recommended that local municipalities focus on those areas that meet vital planning goals, and thus consider the additional contextual criteria below:

- *Alignment of other public investments:* Is the county, local municipality or another public entity planning to make investments in the areas under study?
- *Land use impacts:* Will the complete street designation impact land use decisions in the areas adjacent to the designated streets?
- *System preservation and enhancement:* Are other transportation investment scheduled for the areas being proposed, and if so, can a complete streets approach be incorporated into the planned improvement?
- *Density and capture area:* Is the roadway in proximity to a good population base of pedestrians and bicyclists?

Complete Streets Prototypes

The complete streets prototypes for **ACTIVEALLEGHENY** are:

- Freeport Road, Blawnox Borough and O'Hara Township
- South Braddock Avenue, Swissvale Borough
- Broadway Avenue, Beechview, City of Pittsburgh

The three prototypes, selected from the candidate list as representative projects, were chosen to serve as examples for the other candidates based on diversity in location and roadway type. Freeport Road dramatically changes character as it transitions from the traditional downtown center of Blawnox Borough to the strip commercial corridor of O'Hara Township (just west of the Waterworks Mall); solutions discussed for this corridor can be applied to many other locations in the region where urban areas adjoin busy suburban retail corridors. South Braddock Avenue is a constrained roadway in an older urban neighborhood, with a poorly maintained pedestrian infrastructure in Swissvale. Options for major roadway reconfiguration are limited, but the pedestrian infrastructure can be upgraded, making a clear difference in the lives of the many residents that depend upon walking (including walking to transit) as their primary mode of transportation. Broadway Avenue has more roadway capacity than needed to accommodate vehicular traffic, and has potential to upgrade bicycle accommodations, in particular.

1) Freeport Road, O'Hara Township, Blawnox Borough

Freeport Road is an ideal example of why a complete streets approach is needed in transportation planning. Sidewalks or other pedestrian infrastructure are missing on significant sections of the roadway in O'Hara Township, although they could easily have been installed as part of the original land developments along the corridor. In this pilot project, the condition of Freeport Road, a state owned road, is reviewed in both O'Hara Township and Blawnox Borough, as the roadway changes quite markedly between the two municipalities. It also illustrates a common problem: it is difficult for pedestrians and bicyclists living in urban areas to access the major retail areas that have migrated to the outside of boroughs and cities.



Land uses along Freeport Road in Blawnox Borough are mixed: residential, retail, commercial, and institutional. Freeport Road serves as the “main street” of the community. Different land uses are also found along Freeport Road in O'Hara Township, however, they are not as mixed as they are in Blawnox Borough. The Waterworks Mall, a large retail center located west of Fox Chapel Road and on the north side of Freeport Road, is near a compact residential subdivision to the east, and other scattered retail uses. Closer to Blawnox, a number of office and industrial uses are present.

Sidewalk clear width in Blawnox is reduced by utilities and parking meters. Maintenance is needed for some sections.



Freeport Road in Blawnox

Freeport Road in Blawnox Borough is typically 40' wide, comprised of two travel lanes of about 12' in width and two parking lanes of about 8' in width. The roadway is posted at 25 mph. In O'Hara Township, the total pavement width of Freeport Road west of the Route 28 ramps is typically 40' to 42', with 13' to 15' travel lanes and 5' to 6' shoulders. However, the shoulder width is not consistent on this section, and shoulders are narrow or non-existent closer to Fox Chapel Road. Our survey indicates that Fox Chapel Road is a very popular biking destination for riders from the East End of Pittsburgh.

To the east of the Route 28 ramps, the total roadway width is 44', with four 11' travel lanes, and no shoulders. The AADT (Annual Average Daily Traffic) along the corridor is 32,000 west of the Route 28 interchange, and 16,000 east of the Route 28 interchange. The corridor is very heavily trafficked by motor vehicles. Access management along Freeport Road is poor, and there are many curb cuts/driveways intersecting with Freeport Road along the corridor.

Pedestrian Conditions – Sidewalks are present along Freeport Road in Blawnox Borough. The typical width is 4' to 5', with the presence of utility poles, street lights, meters and other street furniture reducing the clear width to 2.5' to 3'. Sections of the sidewalk are deteriorated. Curb ramps are typically present, with some in poor condition. Crosswalks are provided across Freeport Road at regular intervals at both unsignalized intersections and at mid-block locations. Standard crosswalks with parallel stripes are provided, as well as a unique crosswalk design with diagonal crossing lines and foot patterns.

In O'Hara Township, sidewalks are missing from the vast majority of the corridor. They are present on the north side of Freeport Road, just east of the Route 28 ramps. However, based on the presence of highly worn foot paths, many pedestrians prefer to walk on the south side of Freeport Road, in part to avoid having to cross the ramps to Route 28. During field investigation, pedestrians were observed walking in the shoulder of several sections of the roadway. Curb ramps are missing from the corridor, largely due to the lack of sidewalks.



Crosswalk with foot pattern in Blawnox



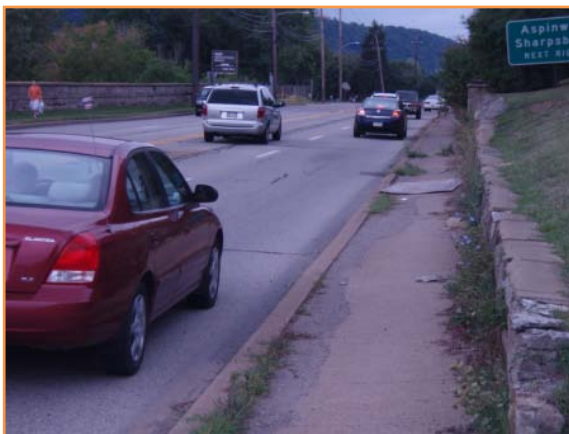
Freeport Road in O'Hara Township.

Bicycle Conditions – There are no designated bicycle facilities along the corridor. Based on PennDOT's *Bicycle Guidelines*, the section of Freeport Road west of Route 28 is typically bicycle compatible, due to the presence of shoulders at least 5' to 6' in width. However, shoulders are not consistent on this section. East of Route 28, the roadway is not bicycle compatible, since shoulders are absent. Freeport Road is not bicycle compatible within Blawnox Borough; for that context, 14' shared lanes are recommended in the *Bicycle Guidelines*.

Transit Conditions – The Route 1 bus between Tarentum and Downtown serves this corridor. Bus stop shelters are located on the corridor adjacent to the edge of cartway in O’Hara Township.

Complete Streets Analysis – Freeport Road presents very different issues for non-motorized travelers within the two municipalities. Most pedestrians would prefer walking on Freeport Road through Blawnox Borough to walking in O’Hara Township. Blawnox Borough has a well-connected sidewalk system with and well-defined midblock crossings, and the presence of parked cars shields pedestrians on the sidewalk from passing vehicles. The sidewalk clear width is not ideal; at 2.5’ to 3’, it is less than the recommended minimum of 5’, and would discourage two pedestrians from walking side-by-side. Sidewalks should be widened where readily permitted, but will be difficult to widen in most places. The presence of slopes and a retaining wall on the north side of Freeport Road through much of the Borough will make widening on this side challenging. Shifting the curb line and extending the sidewalk into the existing roadway would reduce the width of Freeport Road below 40’, considered the minimum ideal width for a “main street” with on-street parking. It would also further reduce the compatibility of the roadway for bicyclists and transit vehicles. Deteriorated sidewalk should be replaced to improve travel, particularly for persons with disabilities.

Increasing the compatibility of the roadway for bicyclists within Blawnox would also prove difficult. Shifting on-street parking to only one side of the roadway would free-up roadway space to create bike lanes, but few urban commercial districts have shown interest in this strategy. A more feasible strategy would involve re-striping parking stalls from their existing width of 8’ to 7’. Studies show that motorists park slightly closer to the curb with narrower parking stalls, thus reducing the potential for a car door to be opened into the path of a passing bicyclist.⁴ This treatment should be combined with the use of shared lane (sharrow) markings (centered 11’ from the curb), to encourage bicyclists to appropriately position themselves in the roadway to avoid being “doored.” Since only limited improvements are possible in Blawnox, consideration should be given to traffic calming treatments to reduce the speed of vehicles moving through the Borough. Landscaped curb extensions and other measures could be employed downtown. Based on recent field views, parking space occupancy is low for several blocks downtown; well-used on-street parking spaces are an effective traffic calming measure.



A sidewalk is provided on the north side of Freeport Road east of Route 28, however sidewalk does not exist on the south sides where many pedestrians prefer to walk.

In O’Hara Township, the most pressing need is to begin development of a sidewalk network. Sidewalks are missing along the entire corridor, so it will be necessary to prioritize sections, if public funding is desired. Local municipal officials should ask landowners along the corridor to install sidewalks as part of all future developments and redevelopments. An example of a high priority for a new sidewalk is in front of the Community Supermarket shopping center at the northeast corner of Fox Chapel Road and Freeport Road. A new sidewalk could be readily installed here, with space created by restriping the parking lot. Unlike Blawnox Borough, where widening a sidewalk would present significant challenges, sidewalks could readily be installed along much of the corridor in O’Hara Township, with minimal impact to existing land uses.

As noted, most of Freeport Road west of Route 28 is bike compatible due to the presence of shoulders. However, to provide a greater setback for bicyclists from the high traffic volumes – especially since

⁴ P. Furth, D. Dulaski, M. Buessing, and P. Tavakolian, *Parking Lane Width and Bicycle Operating Space*, Transportation Research Board 2010 Annual Meeting.

pedestrians are equally dependent upon the shoulders for travel, given the absence of sidewalks – shoulders should be expanded to 8’ as part of future roadway improvements. Shoulders of this width would also enable buses to pull completely off the roadway at transit stops along the corridor. Shoulders are needed on Freeport Road east of Route 28; the lack of shoulders on this section, combined with vehicles moving onto the Route 28 ramps, could discourage bicycle traffic.

Some shoulder improvements could take part in the absence of roadway widening projects. As noted, travel lanes are about 13’ to 15’ in width in sections of O’Hara Township close to Fox Chapel Road. While a travel lane of this width permits safer side-by-side travel with bicycles than the standard lane width of 11’ to 12’, a shoulder would be considered more desirable for bicyclists on a high volume corridor. The travel lanes should be reduced in size, and the width added to shoulders.



Bicycles are prohibited at Waterworks Mall

As part of future redevelopment, retail uses along the corridor should also be required to install bike racks, with the number of bike rack spaces proportionate to the number of vehicle parking spaces. At a minimum, all land uses should be prevented from prohibiting bicycle access. Such a prohibition is in place at the Waterworks Mall.

Transit shelters are present along Freeport Road in O’Hara Township. More shelters are needed in Blawnox Borough. There is no room to install shelters, so officials might wish to identify landowners willing to host shelters.

An Access Management Plan should be instituted for the Freeport Road corridor, encompassing O’Hara Township and other sections along the roadway, including immediately to the west – in Pittsburgh, adjacent to the Waterworks Mall – and to the east, such as the retail strip in Harmar Township. By better controlling the frequency, location, and design of driveways along the corridor, an Access Management Plan can reduce the number of conflicts between turning vehicles, bicycles and pedestrians walking along the corridor. An Access Management Plan could also improve mobility for motorists, and has the potential to lower crash rates.

Conclusion – In O’Hara Township, Freeport Road is a typical suburban strip corridor, built primarily to accommodate motorists. Complete streets improvements are needed, to enable pedestrians and bicyclists to travel between land uses along the strip corridor, and to enable persons to travel here from adjacent urban communities, such as Blawnox. The most basic improvements are sidewalks for pedestrians and shoulders of an adequate width for bicyclists. Further, the proposed improvements would enable pedestrians and bicyclists to better access transit service along the corridor. Although facilities have not been installed to accommodate pedestrians and bicyclists, both groups travel the corridor today. Their numbers could increase with these improvements. In Blawnox, options for improvements are more limited, but upgrades of pedestrian facilities are desirable. Installation of such measures as shared lane markings (sharrows) would increase motorists’ awareness of bicyclists.

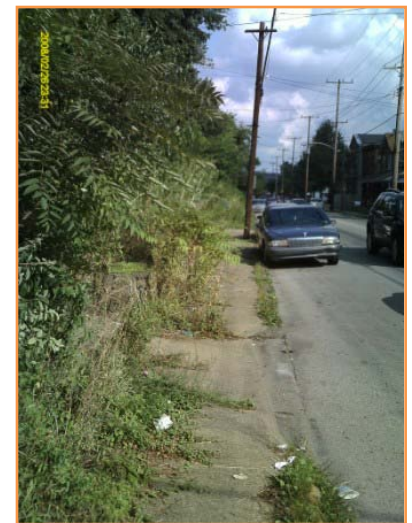
2) South Braddock Avenue/Belmar Place, Swissvale

The study area on South Braddock Avenue, a locally owned road, extends from immediately east of the interchange with I-376 to Woodstock Avenue, for a distance of about one mile. A small section of Belmar Place (at the eastern end of South Braddock Avenue) is also included in this analysis. That extends the corridor to just past the driveway for the MLK East Busway Swissvale Station. East of Edgewood Avenue, South Braddock Avenue lies wholly within Swissvale Borough. West of Edgewood Avenue, South Braddock Avenue lies between Swissvale Borough and Edgewood Borough. Land uses

are mixed along the corridor, although residential uses are predominant east of Roslyn Street. SPC demographic data indicates that Swissvale has a high concentration of low-income, disabled, and minority populations. Based on field observation, there is a presence of older adults and minority persons in the study area. Swissvale is adjacent to the Edgewood Towne Center retail complex, and as such, it would be desirable to facilitate the ability of Swissvale residents to walk or bicycle to these retail services.

There are two typical roadway sections in the study area. On the western half of the corridor, the roadway width is 30', consisting of two 15' travel lanes, without parking. However, the width increases proximate to I-376 and Edgewood Avenue, due to the presence of auxiliary lanes. On the eastern half of the corridor, the roadway width is also 30', with the cross-section comprised of an 11' eastbound travel lane and a 19' westbound travel lane, with on-street parking permitted but not striped. The roadway is posted at 25 mph east of Edgewood Avenue.

Pedestrian Conditions – Much of the pedestrian infrastructure in the study area east of Church Street is in poor condition. Many of the crosswalks, which consist almost exclusively of standard parallel stripes as opposed to high-visibility treatments, are badly faded. Crosswalk markings are missing on the approaches at some signalized intersections, such as Center Drive. Curb ramps are missing at many intersections, and of the existing curb ramps, few have detectable warning pads. Sidewalk width varies from 6' to as little as 3'. Even on sections of 5' to 6' width, the presence of utilities and other obstructions often effectively narrow the sidewalk clear width to 3'. Weeds and other growth have encroached on the sidewalks at a number of locations. Sidewalks are absent on the north side of South Braddock Avenue in several sections, such as adjacent to wooded lots at Ormond Street and Melrose Street. Portions of the sidewalk are badly overgrown, with uneven panels.



Many sidewalks along S. Braddock Avenue are in poor condition

West of Church Street, the pedestrian infrastructure is in much better condition. The sidewalk is relatively new, and “Yield to Pedestrians” channelizing devices alert motorists to the potential presence of pedestrians.

Bicycle Conditions – There are no designated bicycle facilities on the corridor. The section of South Braddock Avenue comprised of two 15' lanes is bicycle compatible, based on PennDOT *Bicycle Guidelines*. The section of South Braddock with parking and two 11' lanes is not bicycle compatible. On field views, children were seen riding their bicycles on the sidewalk.

Transit Conditions – The corridor enjoys access to good transit service. The Swissvale Station on the East Busway is one block removed from the corridor.

Complete Streets Analysis – The study area is characteristic of many roadways in older urban areas: a constrained roadway with existing land uses in close proximity, limiting options for improvements. The most significant complete streets improvements possible on this corridor would involve addressing the pedestrian infrastructure. The installation of sidewalks is needed on the north side of South Braddock Avenue where missing. Sidewalk should also be replaced where deteriorated. Sidewalk clear width is preferably 5', and should be 4' at a minimum. Curb ramps need to be installed at intersections where missing. High-visibility crosswalks are recommended for signalized intersections with higher pedestrian volumes. Crosswalks are also recommended for uncontrolled intersections with higher pedestrian volumes, accompanied by “Yield to Pedestrian” channelizing devices. At signalized intersections, auditory pedestrian signals are recommended.

Given the high concentrations of elderly and disabled persons in Swissvale, the above steps for improving the pedestrian infrastructure are key recommendations. Although curb ramps are perhaps the most visible example of facilities for disabled persons, sidewalks of an even grade, free of debris and obstacles, are also important in facilitating travel by this population.

Pedestrians can cross South Braddock Avenue at a number of stop-controlled intersections between Roslyn Street and Woodstock Avenue. These intersections are Vernon Avenue, Melrose Street, and Cheyenne Street, all of which are T-intersections with South Braddock Avenue. It should also be noted that no stop bars or crosswalks are located on South Braddock Avenue at these intersections. Based on field views, motorists on the corridor regularly roll through these stop signs without coming to a complete stop. This activity should be evaluated, since it may increase the potential for conflicts, with both cars and pedestrians. If volumes on South Braddock Avenue are significantly higher than side street volumes, consideration should be given to shifting the stop control to the side street. However, a crosswalk with “Yield to Pedestrians in Crosswalk” sign could then be installed for South Braddock traffic to facilitate safer pedestrian crossings.

Although, as noted earlier, sidewalk condition is typically in good condition west of Church Street, steps can be taken here to improve pedestrian infrastructure as well, to facilitate travel between the Edgewood Towne Center and the dense residential uses in Swissvale. This area is marked by roadway design typical of newer suburban areas, including large curb radii and channelized islands. For example, the Towne Centre driveway intersecting with South Braddock Avenue between Schoyer Avenue and McClure Avenue is channelized (using striping only), with no crosswalk along South Braddock. A stop sign does control vehicular egress here, but it is placed close to the beginning of radius curvature, and the natural motorist movement would be to roll through this stop sign (no stop bar) and be in motion at the point of pedestrian crossing. It is recommended that a physical island and crosswalks be installed here to improve pedestrian crossings. Similar reviews should occur for the newer, suburban-type infrastructure west of Church Street.



Towne Center Driveway Stop Sign Location

Options for specifically accommodating bicyclists on the corridor are limited. Where two 15' travel lanes are present, it would be possible to stripe 5' shoulders and 10' travel lanes. These shoulders could essentially serve as bike lanes. However, the extent of this treatment would be limited; only from Roslyn Street to Ormond Street on the west, and east of Vernon Avenue. This treatment is not possible for the entire length of the corridor due to the presence of parked vehicles in front of two section of row houses on the north side of South Braddock Avenue, where the cross-section consists of an 11' eastbound travel lane and a 19' westbound travel lane. This cross-section represents an appropriate response to the need to accommodate parked vehicles on a section with no off-street parking, and where the roadway width is constrained.

Although buses do not travel the corridor between Roslyn Street and the Swissvale MLK East Busway Station driveway, residents along the corridor have access to excellent transit service in close proximity.

Conclusion – Major changes in roadway configuration are not envisioned for South Braddock Avenue. This will be characteristic of many complete streets projects in the region; major upgrades will be possible for one non-motorized mode, but not for all. In the case of South Braddock Avenue, there is potential to provide significant improvements for pedestrians, particularly those with disabilities.

3) Broadway Avenue, Beechview, City of Pittsburgh and the Borough of Dormont

Broadway Avenue, is located in the City of Pittsburgh (Beechview neighborhood) and Borough of Dormont, extending 1.2 miles in length from Fallowfield Avenue in the north to Potomac Avenue on the south. North of Neeld Avenue, Broadway is 52' in width, with two travel lanes. In this section, the northbound and southbound lanes share the same roadway space as the T Red Line northbound and southbound tracks. Between Potomac Avenue and Neeld Avenue, the rail lines occupy dedicated right-of-way, with northbound and southbound travel lanes of 20' in width on either side.

North of Neeld Avenue, Red Line passenger platforms are located at the intersections of a number of roadways with Broadway. The platforms are 6' in width, and placed 9' from the curbline. On-street parking is permitted along most of Broadway Avenue. Parking spaces are striped south of Neeld Avenue; unstriped between Neeld Avenue and Coast Avenue; and striped north of Coast Avenue. Parking is typically prohibited proximate to intersections with streetcar platforms. The roadway between streetcar platforms and curbing is used by many motorists as an auxiliary lane, although it is not striped as such. Some motorists use these areas to turn right onto side streets. Other motorists essentially use these areas as a passing area. For example, if the streetcar stops at the platform, some motorists circumvent the slower-moving streetcar by driving to the right of the platform. Based on field views, other motorists use these areas to bypass slower motorists.



A vehicle drives to the right of the T passenger platform before turning right

The large majority of Broadway Avenue is posted at 25 mph, but a small northbound section of Broadway, south of Neeld Avenue, is posted at 20 mph. North of Coast Avenue, the roadway surface is comprised of asphalt for the approximately 11' wide travel lanes, with the rest of the surface being bricked.

The land uses are predominately residential, with substantial retail and commercial land uses mixed in at several locations. The commercial/retail facilities have a high percentage of marginal uses or vacancies. There are also some public and institutional uses. Pittsburgh is beginning a TRID (Transit Revitalization Investment District) study of Beechview (SmartTRID) and is interested in determining potential new commercial and retail uses along Broadway Avenue.

Bicycle Conditions – There are no designated bicycle facilities along Broadway Avenue. The roadway is bicycle compatible according to the PennDOT *Bicycle Guidelines*, which recommends 14' shared travel lanes for this roadway type and land use context.

Pedestrian Conditions – The typical sidewalk width is 5', but the width increases in segments, particularly proximate to commercial or institutional uses. The presence of utility poles and other obstacles reduce the sidewalk clear width to 3'. Sidewalks are directly adjacent to the curb. At driveway locations, the sidewalk grade changes due to the presence of driveway cross-slopes.

Crosswalks are located on Broadway Avenue at the majority of signalized intersections. However, crosswalks are not consistently available for every controlled crossing. Crosswalks are also installed at a number of unsignalized intersections, usually where T platforms are present. At these intersections, a crosswalk is installed at only one

approach on Broadway. Virtually all of the crosswalks are standard; with the exception of one ladder striped crosswalk which traverses Broadway Avenue at its intersection with Belasco Avenue.

Most of the curb ramps along the corridor are older, without detectable warning surfaces (e.g., truncated domes). At many intersections, the curb ramps are installed only to accommodate pedestrians walking along Broadway, not across Broadway.

Transit Conditions – The PAAC T Red Line provides excellent transit service to the Beechview neighborhood. Light rail vehicle (LRT) service operates in this neighborhood with 10-minute headways during weekday peak periods and with 15-minute headways during weekday off-peak periods. The streetcar has 20-minute headways on weekends. Regular service is provided between Castle Shannon and Downtown via Beechview. Red Line service may be reduced in 2011 based on Port Authority proposed service reductions.

The Port Authority supports implementation of Complete Streets approaches so long as transit operations are not compromised. A traffic analysis is needed to better understand the impacts to rail operations and motor vehicle traffic for a Complete Streets concept on Broadway Avenue.

Complete Streets Analysis – The opportunity for the most significant reconfiguration of Broadway Avenue along complete streets principles involves bicycle facilities. Although the roadway is currently bike compatible, the design of the roadway complicates bicycle travel. As discussed above, motorists traveling along Broadway routinely depart from the major travel path (proximate to the roadway centerlines) to bypass traffic at intersections by driving to the right of passenger platforms. These motorist actions have the potential to create conflicts with bicyclists, both before and after the passenger platforms, but particularly directly between platforms and the curb, where motorist and bicyclist would be squeezing into an area of only nine feet in width. This configuration also reduces the comfort level of bicyclists.

Pittsburgh has begun investigating the potential for creating bicycle lanes on the roadway. At 52', Broadway Avenue north of Neeld Avenue is significantly wider than needed for a two lane roadway. Bike lanes can thus be



Broadway Avenue has ample capacity for accommodating bike lane

accommodated. In the process, it will be necessary to better define the vehicular travelway. The presence of the Red Line, and particularly of the platforms for passengers boarding the Red Line, presents unusual design issues.

If the passenger platforms are kept in place, one potential midblock configuration could involve 11' travel lanes, 9' parking lanes, and 6' bike lanes. The travel lane width would need to remain at about 11' to maintain the combination travel lane/light rail track on the inside of existing passenger platforms. An unusually wide parking lane (9') is recommended, as opposed to bike lanes wider than 6', in order to help discourage motorists from using the bike lane as an auxiliary travel lane.

Pittsburgh has also begun preliminary consideration of the idea of separating bike lanes from vehicular traffic, and placing the bike lanes next to the curb. This design has recently been implemented on a number of roadways in New York City and other municipalities, and has been implemented in many European cities. It is sometimes referred to as a "cycle track." Many bicyclists like this design. To be implemented on Broadway Avenue, a number of issues will need to

be addressed. It will be important to prohibit on-street parking for a sufficient distance at all intersections to permit good visibility of motorists and bicyclists of each other when they approach intersections, in order to avoid conflicts. Some bicycle advocates have also expressed general concerns that such designs have the effect of discouraging bicyclists from “taking the lane” or otherwise positioning themselves to conduct a vehicular-style left turn at intersections. In the case of Broadway, however, bicyclists will likely be less inclined to turn left from the travel lane in any event, given the constrained travel lane between the centerline and the streetcar passenger platforms. This design will require significant study; one possible configuration (from centerline to curb) would involve 11’ travel lanes; 7’ parking lanes; 3’ buffer, with pylons, curbing, or other physical means to demarcate the travelway from the bike lane; and a 5’ bike lane. The 3’ buffer will also serve to protect bicyclists from passengers who exit from their car curbside, since it will place bicyclists beyond the “door zone.”

Under a reconfiguration, whether bike lanes are placed next to the curb or next to the travel lane, the new roadway configuration will indicate that motorists will no longer be able to use the area between the platform and the curb as a bypass. Keeping vehicles in one travel lane will increase vehicular delays at intersections, but any increase should not be significant. The design will also have the effect of tempering vehicular speeds on Broadway, acting as a “traffic calming” measure.

It should be noted that the section of Broadway Avenue to the south of Neeld Avenue would not be affected by this proposed roadway reconfiguration. At a width of 20’ each for northbound and southbound lanes, including parking, the installation of bike lanes is not possible. Particularly if it is proposed to designate all of Broadway Avenue as a bike route, it is recommended to install shared lane markings (also known as “sharrows”) to indicate where bicyclists should position themselves in order to avoid being struck by opened car doors, and to alert motorists to the potential presence of bicyclists.

Other changes in roadway design will be necessary. For example, the bricked portion of Broadway Avenue will need to be replaced with asphalt or other smooth surface to improve the comfort level and safety of bicyclists. Angle parking will need to be changed to parallel parking.

Connecting Broadway Avenue bicycle facilities to the larger network will need to be investigated. At the south, Broadway terminates at Potomac Avenue. The Potomac Avenue area has recently been studied as a Transit-Oriented Development (TOD) “place”. Plans for this area are included in the South Hills TRID Study, and may be accessed at <http://www.mtlebanon.org/index.aspx?nid=338>. Potomac Avenue is a two-lane roadway of only 38’ in width, with on-street parking on both sides. No designated bicycle facility is possible here, but the appearance of this roadway – traditional commercial/retail buildings with residences above and with zero building setback, landscaped curb extensions, and well-used on-street parking – have the effect of traffic calming motorists. This is a densely settled residential neighborhood with a senior citizen high rise located at the Potomac T station. It should be noted that there are other quiet side streets in Dormont, on blocks surrounding Potomac Avenue that can accommodate bike access to Brookline’s residential and commercial retail area or to West Liberty Avenue in Dormont, another commercial/retail corridor a few blocks away from Potomac. To the north where the T crosses a narrow bridge which cannot accommodate bikes, the City has expressed interest in a bicycle route to tie into Brashear High School, and the investigation of a crossing of the Route 19 viaduct to the Mt. Washington area. These aspects will require longer range plans, which will occur during the upcoming SMART TRID study process.

To further enhance bicycle travel along Broadway Avenue, it would be desirable to improve the capacity of Red Line cars to accommodate bicycles. The T service currently accepts up to two bicycles on streetcars, but only during off-peak hours. It also accepts folding bicycles anytime. The T is very popular and filled to capacity during peak travel times. PAAC and interested parties can work together to explore alternative ways to expand bicycle storage capacity, and

expedite bicycle boarding during peak hours. A small flatcar added to the front of the T may be one avenue worth exploring, although it will likely necessitate costly adjustments to the length of the T platforms which may not be feasible at all stations/stops.

Although the most dramatic changes to Broadway Avenue would affect bicyclists, pedestrians would also benefit from the reconfiguration. Motorist movements would be better defined and controlled at intersections. The installation of bike lanes would reduce the width in which pedestrians are exposed to motorized traffic.

New pedestrian facilities should be installed along Broadway Avenue, whether or not a reconfiguration with bike lanes proceeds. High visibility crosswalks should be installed at intersections with large numbers of pedestrian crossings. More crosswalks are needed at intersections with T Red Line passenger platforms. Curb ramps are recommended where missing to facilitate crossings of Broadway, not just pedestrians along Broadway. “Yield to Pedestrians” signs mounted on stanchions are recommended to accompany installation of crosswalks. New equipment is recommended at signalized crossings, including auditory pedestrian signals.

As noted, clear sidewalk width is often reduced to 3’ by the presence of utilities and other obstacles. Driveways that run across the sidewalks result in uneven grades. Where existing right-of-way permits, sidewalks should be widened to provide greater sidewalk clearance. Because the roadway is wider than needed to accommodate traffic, the curbs could be moved away from the right-of-way line in order to widen the sidewalk. It should be noted that this would reduce the ability to install certain roadway treatments with bike lanes, and would also involve considerable expense.

Conclusion – Because the roadway capacity of Broadway Avenue is greater than necessary, a number of options are available for a complete streets approach to this roadway. A proposed reconfiguration of Broadway Avenue would improve conditions for pedestrians and bicyclists, and abet the ability of both modes to access the T Red Line on this corridor.

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CHAPTER 6. *Action for Active Transportation*

6.1 ADOPT, APPROVE AND INTEGRATE

ACTIVEALLEGHENY lays the groundwork for a complete multi-modal transportation network with successful active transportation connectivity for Allegheny County. Implementation of this plan will be by PennDOT, Allegheny County, and local municipalities. Local municipal plans should work toward consistency and this enhancement to the ALLEGHENYPLACES transportation element, pending adoption of the updates. PennDOT, Allegheny County, and local municipal support for ACTIVEALLEGHENY will enable effective implementation of this active transportation vision. Municipalities can work to plan and adopt specific active improvements consistent with the standards and guidelines in this plan. The toolboxes, model ordinance and specifications are resources for use in customizing local municipal plans. For specific projects identified in the plan, a focused team effort by all levels of government involved in each suggested improvement can lead to construction and use of the desired facilities. For regional active transportation connectivity, Allegheny County will coordinate with adjacent counties to achieve regional integration of active transportation, creating active transportation byways that connect from county-to-county.

County residents have a role to encourage leaders to implement improvements identified in the plan, and to assist in achieving active transportation facilities and usage an important component for a healthier life experience with mode choices for Allegheny County. PennDOT is a critical partner in provision of support for this vision through assistance and commitment to identifying and committing funds to construct active transportation improvements.

6.2 IMPLEMENTING AGENCY

Active transportation projects can be implemented by a variety of entities; the party implementing the improvement will help determine the funding source that can be used. The entities are:

PennDOT

PennDOT must be involved in all bicycle and pedestrian improvements that occur within state roadway rights-of-way, even if they do not necessarily provide funding for these improvements. PennDOT typically does not fund the construction of “independent bikeway projects”, or projects that are intended to construct only a bicycle facility. However, it will process federal funding (and county and state matches) for such projects if requested by sponsoring organizations for projects on state roadways. If an “incidental bikeway projects” is developed as part of a larger project; for example, on a roadway improvement, a shoulder could be modified to bicycle compatible standards. The recently opened segment of the Three Rivers Heritage Trail between the 31st Street Bridge and Millvale Riverfront Park was developed as part of the Route 28 Improvement Project. Depending on the scale of improvement, incidental bikeway projects may be incorporated into normal PennDOT funding channels. The sponsoring agency should initiate discussion and advocacy in the early stages of a PennDOT project so that incidental bikeway projects are included in the project pipeline. But more importantly, PennDOT should take the initiative to consider all modes on projects.



Pedestrian in Friendship

According to PennDOT Design Manual (DM-2, Chapter 6: Pedestrian Facilities) it is PennDOT’s policy not to use state funds for sidewalk construction. However, there are several exceptions, such as to accommodate critical pedestrian

safety needs as part of a larger roadway project, or ADA compliance for alterations as defined in the Americans with Disabilities Act. PennDOT will process federal funding (with local matching funds) for the construction of sidewalks within state rights-of-way. Municipalities can donate right-of-way, engineering, or construction funding for sidewalks as part of their local match.

Although PennDOT typically does not fund construction of independent bicycle and pedestrian projects, it will fund planning studies for such facilities (see “Pennsylvania Community and Transportation Initiative” (PCTI) in Section 6-3 below; other state departments also provide funding for active transportation projects.

Municipalities

Local municipalities can adopt their own plans and policies for developing and funding active transportation projects on locally owned roadways. These projects are then added to their capital improvements program, which are funded from local tax revenues or from state gas tax revenues. In addition to projects specifically developed as active transportation efforts, municipal officials should review projects currently on the capital improvements program, and determine where facilities that encourage pedestrian and bicycle travel, and transit use, can be added at little expense. Municipalities will also often serve as the initiating agency for projects on state roadways; they may request the installation of facilities to accommodate pedestrians and bicyclists, and secure funding offered by federal, state and county agencies, even if PennDOT ultimately performs the work. Municipalities should utilize criteria developed for active transportation safety and demand to prioritize improvements.



Need for bicycle racks

Photo: Kevin Smay

Developers

Developers may be asked to construct pedestrian and bicycle facilities along both state and local municipal roadways as part of development projects. Applications for all land developments requiring access to state highways are reviewed by PennDOT and this provides the opportunity to request needed improvements. As part of PennDOT’s Smart Transportation initiative, a 2009 revision to PennDOT’s *Transportation Impact Studies Guidelines* requires developers to state how pedestrians, bicyclists and transit users will be accommodated in projects along state roadways. For example, if a state roadway is designated as a bike route by the municipality, PennDOT should request that the shoulder be developed to bicycle compatible standards.

Likewise, municipalities should have appropriate standards for pedestrian facilities in their subdivision and land development ordinances, and request developers to install such facilities along both local and state owned roadways as part of the normal land development review. Bicycle facilities are in a different category than pedestrian facilities; a sidewalk can have utility if installed as a segment only on the developer’s property, but a bicycle facility will normally be appropriate if needed as part of a larger planned facility. The local municipality should therefore request the developer to install the appropriate bike facility if the roadway is designated as such in a local municipal bicycle plan.



Accommodations at Airside Business Park, Moon Township

6.3 FUNDING SOURCES

Following is a discussion of funding sources available to assist in the planning, design and construction of active transportation improvements.

Federal Funding

Federal transportation funding is typically authorized in 6 year cycles. In 2005, Safe, Accountable, Flexible, Efficient Transportation Equity Act – A Legacy for Users (SAFETEA-LU) guaranteed \$244.1 billion for transportation funding. According to FHWA's website, \$480 million of federal funding was spent on bicycle and pedestrian projects during the life of SAFETEA-LU (2005-2009). Due to federal budget issues and other priorities, the current Administration and Congress did not pass a new reauthorization bill in 2009 or 2010. Instead, SAFETEA-LU funding levels and programs have been extended year-to-year with continuing resolutions. After the 2010 mid-term elections, a new reauthorization bill will likely be considered in 2011. The Administration has stated that its priorities for this bill include promoting Livable Communities, transit, congestion reduction, and climate change. These priorities all offer opportunities for funding Active Transportation projects. State and local governments and Active Transportation proponents should lobby for these programs to be included in the reauthorization.

Federal funding programs for bicycle and pedestrian projects currently include:

- 1) **National Highway System (NHS)** funds if adjacent to a highway on the National Highway System.
- 2) **Surface Transportation Program (STP)** funds including Transportation Enhancement Activities and Hazard Elimination and Railway-Highway Crossing Programs.
- 3) **Congestion Mitigation and Air Quality Improvement Program (CMAQ)** funds.
- 4) **Recreational Trails Program** funds for trail projects.
- 5) **Federal Lands Highway Program** if used in conjunction with roadway improvements.
- 6) **National Scenic Byways Program** funds if adjacent to a National Scenic Byway.
- 7) **Job Access and Reverse Commute (JARC) Grants** if used to improve employment access and increase opportunities.
- 8) **High Priority Projects and Designated Transportation Enhancement (TE) Activities.**
- 9) **Federal Transit Program** to improve access to transit facilities and provide non-motorized amenities.
- 10) **State and Community Highway Safety Grants** for research, development, demonstrations and training to improve highway safety (23 USC Section 403).
- 11) **Home Town Streets/Safe Routes to School** to improve access by bicycles and pedestrians to schools and to encourage the use of alternative modes.

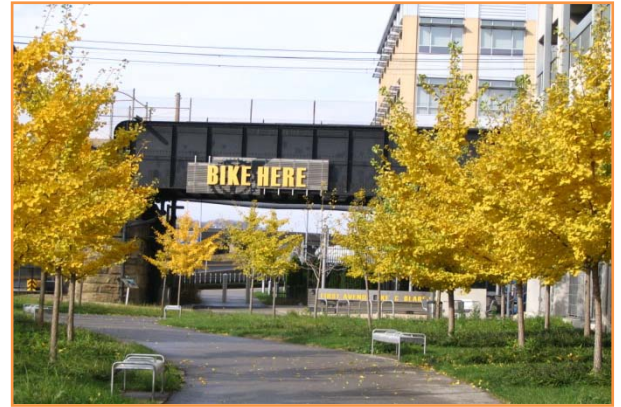
State Funding

The Pennsylvania Department of Transportation (PennDOT) initiated the **Pennsylvania Community Transportation Initiative (PCTI)** in 2008 to promote the principles of Smart Transportation throughout the Commonwealth. PCTI funds planning projects up to \$300,000 and construction projects of up to \$1 million. The program is currently in its second round of funding with applications currently under consideration. In addition to the PCTI program, PennDOT uses federally allocated money throughout the Commonwealth for bicycle and pedestrian facilities. Most commonly used sources of funding for bicycle and pedestrian facilities come from Transportation Enhancement (TE) funding and the Home Town Streets/Safe Routes to School Program, both federally funded programs administered at the state level. Not all of these programs have funding or opportunities to apply for funding at all times.

In addition to PennDOT funding, the Pennsylvania Department of Conservation and Natural Resources (DCNR) has several grant opportunities available to promote active transportation. Grant opportunities are part of the **Community Conservation Partnership Program (C2P2)** which is designed to assist communities in addressing their recreation and

conservation needs as well as supporting economically beneficial recreational tourism. The following grants are available and applicable for bicycle and pedestrian uses for development, construction, or education⁵:

- **Community Recreation and Conservation** – These grants fund municipalities and nonprofit organizations to plan for, acquire, develop and/or rehabilitate public park, recreation, open space, greenway, trail and conservation areas and facilities.
- **Pennsylvania Recreational Trails** – These grants help develop and maintain recreational trails and trail related facilities for motorized and non-motorized recreational trail use, and to purchase equipment for those purposes.
- **DCNR C2P2 Rails-to-Trails** – These grants help plan for, acquire, or develop rail-trail corridors.
- **Rivers Conservation** – These Grants help develop watershed/river-corridor conservation plans, assist with land acquisition and development projects recommended in areas with completed river conservation plans on the rivers conservation registry.
- **Technical Assistance** – Grants to develop/promote/conduct training/education programs; prepare and distribute technical assistance/education manuals/brochures/videos or otherwise provide for training and education of professionals and the general public on a local, county, regional or statewide issues concerning recreation, park, conservation, natural areas, open space, greenways, and trails.
- **Circuit Rider Projects** – These projects provide grant funds for counties, multi-municipal organizations, or COGs to hire a professional, full-time staff person. The circuit rider's purpose is to initiate new programs and services for a county and/or municipalities that individually do not have the financial resources to hire a professional staff person.
- **Peer-to-Peer** – These projects help municipalities improve their park, recreation and conservation services through a collaborative process. Projects are accomplished through contracts with experienced park, recreation and conservation professionals from nearby communities working closely with local leaders.



Eliza Furnace Trail, City of Pittsburgh

Regional Funding

MPOs

Metropolitan Planning Organizations (MPO) are federally designated transportation planning organizations located in all urbanized areas in the U.S. MPOs are charged with programming all federal transportation funding expended in an urbanized area, as well as maintaining a short-term transportation plan (Transportation Improvement Plan (TIP), 4-year horizon) and a long-term transportation and economic development plan (typically 20-year horizon). MPOs also program non-federally funded projects of regional significance. Each MPO's TIP is then integrated into the Statewide Transportation Improvement Plan (STIP), a twelve-year program to provide a broad picture of transportation projects throughout the Commonwealth. Allegheny County is one of ten counties in the Southwestern Pennsylvania Commission (SPC) region. Other SPC members are Armstrong, Beaver, Butler, Fayette, Greene, Indiana, Lawrence, Washington and Westmoreland counties, as well as the City of Pittsburgh.

TMA

Transportation Management Associations (TMA) are non-profit organizations tasked with creating a better transportation environment for residents, businesses, and visitors. TMAs are comprised of members, often businesses, who support the TMA financially. TMAs in the study area include:

⁵ all information is taken from the DCNR C2P2 Grant Website at <https://www.grants.dcnr.state.pa.us/GrantPrograms.aspx>

- The Airport Corridor Transportation Association (ACTA) serves the business community located along the Parkway West from the western portal of the Fort Pitt Tunnel through Beaver County.
- The Oakland Transportation Management Association (OTMA) serves the Oakland area.
- The Pittsburgh Downtown Partnership Transportation Management Association (PDP TMA) serves the Downtown Pittsburgh central business district.

TMA's offer transportation services (shuttles, shared-ride, etc), as well as advocate, educate, and offer technical assistance to individuals and member organizations. TMA's often provide funding for community transportation initiatives and are an excellent source of technical assistance to local municipalities for project implementation. TMA's also advocate for funding from state and federal sources and provide valuable education programs to encourage active transportation use.

Private Funding

The Bicycles Belong Coalition is sponsored by member companies of the American bicycle industry. According to their website (www.bikesbelong.org), Bikes Belong has awarded 209 grants to municipalities and grassroots groups since 1999, for a total of nearly \$1.6 million in bicycle projects. Eligibility and funding guidelines are provided on their website under "Grants."

Other local organizations provide support and funding for bicycle and pedestrian facilities. The Montour Trail Council, for example, builds, operates and maintains the Montour Trail.

Foundations are another source of private funding. There are many foundations in Allegheny County, and some have become active in transportation funding. **The Pittsburgh Foundation**, one of the largest community foundations in the United States, has provided funding for community transportation initiatives involving workforce accessibility, reverse commuting, and others. Their grant eligibility criteria include projects that "enhance access to entry-level jobs and needed services through integrated transit systems; expand strategic alliances to increase transit ridership; and promote development of healthy transportation alternatives (walking, biking, etc.)." Other foundations also provide active transportation funding. For example, the newly opened section of the Three Rivers Heritage Trail received funding from the **Laurel Foundation**, the **Richard King Mellon Foundation** and the **Heinz Foundation**.

6.4 SIGNIFICANT PROJECTS – ACTION PLAN

As noted previously, PennDOT, Allegheny County, and local municipalities should use this plan and its toolboxes and other resources as a guide in developing Active Transportation projects in their communities or as part of their transportation facilities. However, some major elements of the plan will require coordinated, multi-jurisdictional actions to move forward. These elements include:

Designated County Commuter Bicycle Routes and Recreational Route

These routes (N1-N4, W1-W4, E1-E4, S1-S4 and the Beltway Bicycle Route), will traverse a combination of state, county and municipally owned roads. Logically, Allegheny County could assume a lead role in the implementation of these routes. In this role, the County could identify and help to secure funding sources, provide engineering to determine actions and budgets needed to bring deficient route segments up to current bicycle suitability standards, prioritize the order of route implementation, establish route signing and marking standards, and identify the responsible agencies for implementing improvements (such as new trail construction, shoulder widening, signing, etc.).



Motorist and a Bicyclist

Pedestrian Corridor Improvements



Steps, Carnegie Mellon University

Similarly, the plan includes a number of pedestrian corridors identified for improvements. Although many of these corridors are within one municipality, it may be desirable for one agency, possibly the County, to serve as a “Champion” for these projects in order to identify and secure funding and coordinate in instances where they cross municipal boundaries or involve multiple jurisdictions (such as provision of sidewalks along state roads).

Complete Streets Prototype Projects

The plan includes three “Complete Streets” prototype projects. Two of them, Broadway Avenue (City of Pittsburgh and Borough of Dormont) and South Braddock Avenue (Borough of Swissvale), are each contained within one municipality. The third, Freeport Road, is within Blawnox Borough and O’Hara Township. The City of Pittsburgh has the professional expertise and resources to advance the Broadway Avenue project on its own, but the other two may require assistance from a higher level, either the County or the State. Since Freeport Road is a state road traversing two communities, PennDOT may be the logical implementation agency for this project. South Braddock Avenue in Swissvale is a municipal roadway, but Swissvale has limited professional and financial resources to undertake a project of this magnitude. In this

case, again, it may be logical for the County to act as a “Champion” for the project to at least identify and assist with securing funding and professional assistance to the Borough in moving the project forward.

INDEX

ACRONYMS

AAA	American Automobile Association
AADT	Annual Average Daily Traffic
AASHTO	American Association of State Highway and Transportation Officials
ACED	Allegheny County Economic Development
ACTA	Airport Corridor Transportation Association
ADA	Americans with Disabilities Act
APBP	Association of Pedestrian and Bicycle Professionals
BikePGH	BikePittsburgh
C2P2	Community Conservation Partnership Program
CAT	Committee for Accessible Transportation
CBD	Central Business District
CCAC	Community College of Allegheny County
CMAQ	Congestion Mitigation and Air Quality Improvement Program
COG	Council of Government
DCNR	Department of Conservation and Natural Resources
DMV	Department of Motor Vehicles
FHWA	Federal Highway Administration
GAP	Great Allegheny Passage
GHG	Greenhouse Gas Emissions
GRH	Guaranteed Ride Home Programs
HOV	High Occupancy Vehicle
IRS	Internal Revenue Service
ITE	Institute of Transportation Engineers
JARC	Job Access and Reserve Commute
MPO	Metropolitan Planning Organizations
MUTCD	Manual on Uniform Traffic Control Devices
NHI	National Highway Institute
NHS	National Highway System
NIMY	Not In My Back Yard
NJDOT	New Jersey Department of Transportation
NPLAN	National Policy and Legal Analysis Network to Prevent Childhood Obesity
OTMA	Oakland Transportation Management Association
PA	Pennsylvania
PCTI	Pennsylvania Community Transportation Initiative
PDP	Pittsburgh Downtown Partnership
PennDOT	Pennsylvania Department of Transportation
RPO	Rural Planning Organizations
RSA	Road Safety Audit
SAFETEA-LU	Safe Accountable Flexible Efficient Transportation Equity Act - A Legacy for Users

SALDO	Subdivision and Land Development Ordinance
SPC	Southwestern Pennsylvania Commission
STIP	Statewide Transportation Improvement Plan
STP	Surface Transportation Program
TE	Transportation Enhancement
TIP	Transportation Improvement Plan
TMA	Transportation Management Areas
TOD	Transit Oriented Development
TRID	Transit Revitalization Investment District
USDO	United States Department of Transportation
U.S. EPA	United States Environmental Protection Agency
VMT	Vehicle Miles of Travel
YMCA	Young Men's Christian Association

GLOSSARY

■ A

Active Transportation

A human powered transportation including bicycling, walking, kayaking and in-line skating. (ACTIVEALLEGHENY)

Alignment

The line which represents the proposed location of a new highway or transit line. (ALLEGHENYPLACES)

Annual Average Daily Traffic (AADT)

The estimate of typical daily traffic on a road segment for all seven days of the week over the period of one year. (ALLEGHENYPLACES)

■ B

Bicycle Facilities Toolbox

A resource for County and local officials, staff, residents, and stakeholders that will assist in planning and developing bicycle facilities. (ACTIVEALLEGHENY)

Bulb-out

A bulb-out, or curb extension, is a traffic calming measure which narrows the roadway width and serves multiple purposes including reducing speeds, reducing pedestrian crossing distances, improving visibility of pedestrians at crossings, and defining on-street parking,

■ C

Capacity

The maximum rate of traffic flow which can be expected to pass a certain point; usually expressed in vehicles per hour. (ALLEGHENYPLACES)

Carbon Intensity

The relative amount of carbon by emitted per unit of energy or fuels consumed. (Babylon Dictionary)

Central Business District (CBD)

The downtown retail trade area of the city or an area of very high land valuation, traffic flow, and concentration of retail business offices, theaters, hotels, and services. (ALLEGHENYPLACES)

Chicane

A traffic calming method intended to reduce vehicular speeds by creating horizontal deflection through the use S-shaped curves typically formed by curb extensions or on-street parking. Chicanes are often used on residential neighborhood streets.

Complete Streets

A complete streets policy ensures that the entire right of way is routinely designed and operated to enable safe access for all users. Pedestrians, bicyclists, motorists and transit rider of all ages and abilities must be able to safely move along and across a

complete street. (ACTIVEALLEGHENY)

Comprehensive Plan

The general, inclusive, long-range statement of the future development of a community. The plan is typically a map accompanied by description and supplemented by policy statements that direct future capital improvements in an area. (ALLEGHENYPLACES)

Congestion

The level at which transportation system performance is no longer acceptable to the traveling public due to traffic interference. The level of acceptable system performance may vary by type of transportation facility, geographic location, public tolerance, and/or time of day. (ALLEGHENYPLACES)

Congestion Mitigation and Air Quality Program (CMAQ)

A \$6 billion funding program contained in Title I of ISTEA which provides funds for projects and activities which reduce congestion and improve air quality. To be eligible for CMAQ, projects and activities must contribute to the National Ambient Air Quality Standards and must be included in a transportation improvement program (TIP). (ALLEGHENYPLACES)

Corridor

Any major transportation route that includes parallel limited access highways, major arterials or transit lines. With regard to traffic incident management, a corridor may include more distant transportation routes that can serve as viable alternatives to each other in the event of accidents. (ALLEGHENYPLACES)

Crosswalk

(a) that part of a roadway at an intersection included within the connections of the lateral lines of the sidewalks on opposite sides of the highway measured from the curbs or in the absence of curbs, from the edges of the traversable roadway, and in the absence of a sidewalk on one side of the roadway, the part of the roadway included within the extension of the lateral lines of the sidewalk at right angles to the center line; (b) any portion of a roadway at an intersection or elsewhere distinctly indicated as a pedestrian crossing by pavement marking lines on the surface, which might be supplemented by contrasting pavement texture, style, or color. (MUTCD)

■ E

Environmental

1. In a scientific context, a combination of external or extrinsic conditions present in nature. 2. In a planning context, a category of analytical studies of aesthetic values, ecological resources, cultural (historical) resources, sociological and economic conditions, etc. (ALLEGHENYPLACES)

Environmental Protection Agency (EPA)

The government agency responsible for enforcing environmental regulations such as RCRA, CERCLA, Clean Air Act, Clean Water Act. (ALLEGHENYPLACES)

■ F

Feasibility Study

Evaluation of potential remedial alternatives for their ability to meet technical, public health, environmental and cost effective programs. (ALLEGHENYPLACES)

■ G

Great Allegheny Passage (GAP)

A non-motorized shared use trail connecting Pittsburgh, PA to Washington, DC. Portions of the trail are currently under construction.

Greenway

A linear open space established along either a natural corridor, such as a river front, stream valley, or ridge line, or over land along a railroad right-of-way converted to recreational use, a canal, a scenic road, or other route;

- Any natural or landscaped course for pedestrian or bicycle passage;
- An open space connector linking parks, natural reserves, cultural features, or historic sites with each other and with populated areas; and
- Locally, certain strip or linear parks designated as a parkway or greenbelt. (ALLEGHENYPLACES)

■ I

Interchange

A system of interconnecting roadways in conjunction with one or more grade separations, providing for the movement of traffic between two or more roadways on different levels. (ALLEGHENYPLACES)

■ L

Local Government

A city, county, parish, township, municipality, borough, or other general purpose political subdivision of a state. (ALLEGHENYPLACES)

■ M

Metropolitan Planning Organization (MPO)

The agency designated by the Governor (or Governors in multi state areas) to administer the federally required transportation planning process in a metropolitan area. An MPO must be in place in an urbanized area with a population over 50,000. The MPO is responsible for the 20-year long range plan and the transportation improvement program. (ALLEGHENYPLACES)

Mitigation	The replacement of natural features which have been lost or reduced in value. (ALLEGHENYPLACES)
Montour Trail	A non-motorized shared use trail connecting Moon Township, PA to Clairton, DC. Portions of the trail are under currently under construction.
■ N	
Natural Resources	Land, fish, wildlife, drinking water supplies and other assets belonging to, maintained by, or otherwise controlled by the federal, state, or local government. (ALLEGHENYPLACES)
■ O	
Open Space	Any parcel or area of land or water essentially unimproved and set aside, dedicated, designated, or reserved for public or private use or enjoyment of owners, occupants, and their guests, of land adjoining or neighboring such open space. (ALLEGHENYPLACES)
Orange Belt	Comprised of 91.7 miles of miscellaneous county roads color coded for navigational purposes. (ACTIVEALLEGHENY)
Ordinance	A municipal ordinance regulates building setbacks, lot and building coverage, parking, and storm water management. (ALLEGHENYPLACES)
■ P	
Panhandle Trail	A rail trail extending from Collier Township in Allegheny County to Colliers in Northern West Virginia. It was built on an abandoned Conrail line once known as the Panhandle Route. (PAAC)
Park and Ride	A procedure that permits a patron to drive a private automobile to a transit station, park in the area provided for that purpose and ride the transit system to his or her destination. (ALLEGHENYPLACES)
Peak Hour	The one hour period of a typical day during which traffic volumes are the greatest. (ALLEGHENYPLACES)
Pennsylvania Department of Transportation (PennDOT)	The agency of the Pennsylvania State Government responsible for the design, construction and maintenance of state highways and bridges in Pennsylvania, no including toll highways that are under the jurisdiction of the Pennsylvania Turnpike Commission. PennDOT is funded by state and federal tax dollars. (ALLEGHENYPLACES)

Placemaking

A term that began to be used in the 1970s by architects and planners to describe the process of creating squares, plazas, parks, streets and waterfronts that will attract people because they are pleasurable or interesting. (Wikipedia)

Public Involvement

Coordination events and informational materials geared at encouraging the public to participate in project development. A successful Public Involvement Program facilitates the exchange of information among project sponsors and outside groups and the general public, and includes meetings, surveys, committees, presentations, etc. (ALLEGHENYPLACES)

■ R

Roundabout

An intersection traffic control treatment utilized throughout the world. Roundabouts form circular intersections in which entering traffic is required to yield to circulating traffic.

■ S

Safety Improvements

Roadway maintenance activities and smaller construction projects that correct conditions occurring on or alongside an existing highway. Typically involves minor widening, resurfacing, regarding roadside, hazard or obstacle elimination, guiderail installation, and miscellaneous maintenance. (ALLEGHENYPLACES)

Seeps

Location where fluids contained in the ground slowly release to the surface and often form small pools. (ALLEGHENYPLACES)

Smart Growth

An urban planning and transportation theory that concentrates growth in the center of a city to avoid urban sprawl; and advocates compact, transit-oriented, walkable, bicycle-friendly land use, including neighborhood schools, complete streets, and mixed-use development with a range of housing choices. (Wikipedia)

Smart Transportation

Partnering to build great communities for future generations of Pennsylvanians by linking transportation investments and land use planning and decision-making. (PennDOT)

Sprawl

Uncontrolled growth, usually of a low-density nature, in previously rural areas and some distance from existing development and infrastructure. (ALLEGHENYPLACES)

Subdivision

The division or re-division of lots, tracts or parcels. A municipal ordinance that regulates how this may occur, including, but not

limited to, public streets, parks, utilities and storm water management. (ALLEGHENYPLACES)

Sustainability

Of, relating to, or being a method of harvesting or using a resource so that the resource is not depleted or permanently damaged. (Merriam-Webster Dictionary)

■ T

Topography

The natural surface features of a region, including its relief; may be land or water-bottom surface. (ALLEGHENYPLACES)

Traffic Control Signal

Any highway traffic signal by which traffic is alternately directed to stop and permitted to proceed. (MUTCD)

Transit Oriented Development (TOD)

The concept of using mass transit as the focal point of new development, particularly housing, because if residents can be bought close to mass transit, people would be more inclined to use it rather than their own automobiles. (ALLEGHENYPLACES)

■ U

Urban Area

An area having a Center City population of 50,000 or more as defined by the 1990 US Census, may also include other major population concentrations where a systems planning study is deemed necessary. (ALLEGHENYPLACES)

■ V

Vehicle Miles Traveled (VMT)

The sum of distances traveled by all motor vehicles in a specified region in a given period of time. Travel demand forecasting (modeling) is used to generate the average trip lengths for a region. The average trip length measure can then be used in estimating vehicle miles of travel, which in turn is used in estimating gasoline usage or mobile source emissions of air pollutants. (ALLEGHENYPLACES)

■ W

Watershed

A region or area bounded by a water parting and draining ultimately to a particular watercourse or body of water. (ALLEGHENYPLACES)

ENDNOTES

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